PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

FOR

B.Sc./B.Sc. (Hons)/B.Sc. (Hons. with research)
Major & Minor Courses

SECOND YEAR
(SEMESTER SYSTEM)
EXAMINATIONS, 2026-27

(SEMESTER : THIRD AND FOURTH)

i.e

Third Semester: November/December, 2026
Fourth Semester: April/May, 2027

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Agriculture

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY

B.A./B.Sc., B.A./B.Sc. HONOURS and B.A./B.Sc. HONOURS WITH RESEARCH IN AGRICULTURE

(Table 2*)

Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits
III	L-200	AGR-DSC-3-Maj/Min-301	Agronomic Practices for Cultivation of Kharif Crops	3+1=4 (x3)	24
		AGR-DSC-4-Min-302	Soil Science	3+1=4	
		AGR-MDC-3-301	Statistical Techniques in Agriculture	2+1=3	
		AGR-SEC-3-301	Agricultural Extension	2+1=3	
		AEC-3-301	Eng-3/MIL-1	2	
IV	L-200	AGR-DSC-5-Maj/Min-401	Agronomic Practices for Cultivation of Rabi Crops	3+1=4 (x3)	24
		AGR-DSC-4-Maj-402	Soil Science	3+1=4 (x2)	
		AGR-VAC-3-401	-	2	
		AEC-4-401	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship					

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

B.A./B.Sc.(General) Agriculture Major/Minor

Course Title : AGRONOMIC PRACTICES FOR CULTIVATION OF *KHARIF* CROPS **Course Code: (AGR-DSC-3 Maj/Min-301)** **Semester: Third**

Learning Outcomes: To give student basic agronomic principles and practices for cultivation of *Kharif* crops.

Max. Marks: 100 **Course Credits =** 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Cereals – Rice, Basmati Rice, Maize. Bajra

UNIT – II **11 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Oilseeds- Groundnut, Sesame. Cotton (American and Desi), Sugarcane

UNIT – III **11 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :-Pulses –Moong, Mash, Arhar, Soyabean.

UNIT – IV **11 Hours**

Distribution, Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :-Green manuring Crop (Dhaincha, cowpea and sunhemp). Fodders : Maize, Sorghum, Bajra, Guara

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Crop identification based on seed and morphological characteristics.
2. Different inorganic fertilizers, fertilizers recommendation for *Kharif* crops and calculating the fertilizer requirement of various crops as per the recommendation using different combinations.
3. Practice in different methods of fertilizer application.
4. Sowing of important *kharif* crops.
5. Identification of *kharif* weeds and chemical method of weed control.
6. Visit to different farms.
7. Study of morphological characterstics of rice, maize, bajra, groundnut and cotton.

Suggested Readings:-

Reddy S.R., *Principles of Crop Husbandry,* Kalyani Publishers, Ludhiana, 2009.
Handbook of Agriculture, I.C.A.R. Publications, New Delhi, 2008.
Weeds of North India I.C.A.R. Publications, New Delhi, 2008.
Package of Practices for kharif crops, P.A.U. Publications Ludhiana, Corresponding year.

B.A./B.Sc.(General) Agriculture Minor

Course Title: SOIL SCIENCE **Course Code: (AGR-DSC-4 Min-302)**
Semester: Third

Learning Outcome: To give student elementary information on basic principles of soil sciences.

Max. Marks: 100 **Course Credits =** 3+1=4

1. Theory: 75	Written Exam: 67	Internal Assessment: 8
2. Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

Introductory knowledge of soil: definition of soil, component of soil
Definition, occurrence and classification of soil forming rocks & minerals.
Weathering of rocks and minerals.
Soil Profile, Horizons. Classification of soil based on soil taxonomy.

UNIT – II **11 Hours**

Important physical properties of soil: Soil separates/particles, their mechanical analysis and characteristics.
Soil texture, soil densities and porosity of soil.
Classification of Soil structure, factors effecting soil structure and its importance.
Soil Organic Matter origin, its nature and composition.

UNIT – III **11 Hours**

Decomposition of various organic compounds: Factors affecting decomposition, decompositions of water soluble and water insoluble organic matters.
Advantages and disadvantages of organic matter.
Concept of humus formation.
Soil reactions:- Methods of expressing and factors affecting Soil reactions, its influence on nutrient availability & plant growth.
Buffer action & buffering capacity of soils. Factors affecting buffering capacity of soils.

UNIT – IV **11 Hours**

Formation & problems of Salt effected soils: Different types of Salt effected soils, nature & sources of soluble salt in saline soil, appraisal of saline & sodic soil. Calcareous soil.
Reclamations of saline and alkali or sodic soils: Chemical reactions involving reclamations of salt affected soil. Gypsum requirement.
Plant growth concept and factors affecting plant growth.
Plant nutrition: Essential elements, classification & general function of plant nutrients, forms of nutrient elements absorbed by plants. Process of nutrient uptake.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Identification of Important rocks and minerals.
2. Examination of soil profile in the field.
3. Soil sampling and use of augers.
4. Determination of water and soil texture.
5. Study of soil colour using Munsell soil colour charts.
6. Study of soil texture and soil structure of a given sample.
7. Calculations of soil densities and porosity of soil.
8. Calculation of gypsum requirement for given sample
9. Soil survey of a farm.

Suggested Readings:

Brady, N.C. and Weil, R.R., *The Nature and Properties of Soil: 13th edn.,* Pearson education Pte. Ltd. New Delhi, 2002.
Oswal, M.C., *Soil Physics,* Oxford & IBH publishing Co.Pvt. Ltd. New Delhi, 1994.
Biswas, T.D., and Mukherjee, S.K., *Text book of soil science,* Tata McGraw Hill Publishing Co. Ltd., New Delhi, 1997.

B.A./B.Sc.(General) Agriculture Major/Minor

Course Title: Agronomic practices for Cultivation of *Rabi* Crops | **Course Code: (AGR-DSC-5 Maj/Min-401)**
Semester: Fourth

Learning Outcome: To give student basic knowledge of principles and practices for Rabi crops.

Max. Marks: 100 | **Course Credits =** 3+1=4

1. Theory: 75	Written Exam: 67	Internal Assessment: 8
2. Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Cereals – Wheat, Barley, Winter Maize.

UNIT – II **11 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of Oilseeds — Rapeseed and Mustard (Toria, Gobhi Sarson, Raya, Taramira, Linseed, and Sunflower.

UNIT – III **11 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of :- Pulses — Gram, Field Pea and Lentil, Summer moong.

UNIT – IV **11 Hours**

Climatic requirement, Soil requirements, Rotations, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of sugarcane and Fodders : Barseem, Shaftal, Lucerne, Oat and Senji.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Crop identification based on seed and morphological characteristics
2. Different inorganic fertilizers, fertilizers recommendation for *Rabi* crops and calculating the fertilizer requirement of various crops as per the recommendation using different combinations.
3. Different methods of fertilizer application used in *Rabi* crop.
4. Sowing of important *Rabi* crops
5. Identification of *Rabi* weeds and chemical method of weed control.
6. Study of morphological characteristics of important *Rabi* crops.
7. Visit to different farms.

Suggested Readings:

Reddy S.R., *Principles of Crop Husbandry,* Kalyani Publishers, Ludhiana, 2009.
Handbook of Agriculture, I.C.A.R. Publications, New Delhi, 2008.
Weeds of North India I.C.A.R. Publications, New Delhi, 2008.
Package of Practices for Rabi crops, P.A.U. Publications Ludhiana, Corresponding year.

B.A./B.Sc.(General) Agriculture Major

Course Title : SOIL SCIENCE **Course Code: (AGR-DSC-4 Maj-402)**
Semester: Fourth

Learning Outcome: To give student elementary information on basic principles of soil sciences.

Max. Marks: 100 **Course Credits =** 3+1=4

1. Theory: 75	Written Exam: 67	Internal Assessment: 8
2. Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I 12 Hours

Introductory knowledge of soil: definition of soil, component of soil
Definition, occurrence and classification of soil forming rocks & minerals.
Weathering of rocks and minerals.
Soil Profile, Horizons. Classification of soil based on soil taxonomy.

UNIT – II 11 Hours

Important physical properties of soil: Soil separates/particles, their mechanical analysis and characteristics.
Soil texture, soil densities and porosity of soil.
Classification of Soil structure, factors effecting soil structure and its importance.
Soil Organic Matter origin, its nature and composition.

UNIT – III 11 Hours

Decomposition of various organic compounds: Factors affecting decomposition, decompositions of water soluble and water insoluble organic matters.
Advantages and disadvantages of organic matter.
Concept of humus formation.
Soil reactions:- Methods of expressing and factors affecting Soil reactions, its influence on nutrient availability & plant growth.
Buffer action & buffering capacity of soils. Factors affecting buffering capacity of soils.

UNIT – IV 11 Hours

Formation & problems of Salt effected soils: Different types of Salt effected soils, nature & sources of soluble salt in saline soil, appraisal of saline & sodic soil. Calcareous soil.
Reclamations of saline and alkali or sodic soils: Chemical reactions involving reclamations of salt affected soil. Gypsum requirement.
Plant growth concept and factors affecting plant growth.
Plant nutrition: Essential elements, classification & general function of plant nutrients, forms of nutrient elements absorbed by plants. Process of nutrient uptake.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Identification of Important rocks and minerals.
2. Examination of soil profile in the field.
3. Soil sampling and use of augers.
4. Determination of water and soil texture.
5. Study of soil colour using Munsell soil colour charts.
6. Study of soil texture and soil structure of a given sample.
7. Calculations of soil densities and porosity of soil.
8. Calculation of gypsum requirement for given sample
9. Soil survey of a farm.

Suggested Readings:

Brady, N.C. and Weil, R.R., *The Nature and Properties of Soil: 13th edn.,* Pearson education Pte. Ltd. New Delhi, 2002.
Oswal, M.C., *Soil Physics,* Oxford & IBH publishing Co.Pvt. Ltd. New Delhi, 1994.
Biswas, T.D., and Mukherjee, S.K., *Text book of soil science,* Tata McGraw Hill Publishing Co. Ltd., New Delhi, 1997.

Biochemistry

AND STUDENTS OPTING FOR MULTIDICIPLINARY /INTERDICIPLINARY (TABLE 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONORS WITH RESEARCH IN BIOCHEMISTRY

NEP-2020 COLLEGES SCIENCES TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (TABLE 1*)
AND STUDENTS OPTING FOR MULTIDICIPLINARY /INTERDICIPLINARY (TABLE 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONORS WITH RESEARCH IN BIOCHEMISTRY

Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits	Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits
III	L-200	(BCH-DSC-3 Maj/Min-301)	Intermediary Metabolism	3+1=4	24	III	L-200	(BCH-DSC-3 Maj/Min-301)	Intermediary Metabolism	3+1=4 (×3)	24
		(BCH-DSC-4 Min-302)	Enzymology & Bioenergetics	3+1=4				(BCH-DSC-4 Min-302)	Enzymology & Bioenergetics	3+1=4	
		(BCH-MDC-3 -301)	Basic Cell Biology	2 + 1=3				(BCH-MDC-3 -301)	Basic Cell Biology	2 + 1=3	
		(BCH-SEC-3-301)	Analyticaltechniques in Biochemistry	0 + 3= 3				(BCH-SEC-3-301)	Analyticaltechniques in Biochemistry	0 + 3= 3	
		(AEC-3-301)	English-3/MIL-1	2				(AEC-3-301)	English-3/MIL-1	2	
IV	L-200	(BCH-DSC-5 Maj/Min-401)	Membrane Biochemistry & Neurobiology	3+1=4	24	IV	L-200	(BCH-DSC-5 Maj/Min-401)	Membrane Biochemistry & Neurobiology	3+1=4 (×3)	24
								(BCH-DSC-4 –Maj-402)	Enzymology & Bioenergetics	3+1=4(x2)	
		VAC-3-401		2				VAC-3-401		2	
		AEC-4-401	Eng-4/MIL-2	2				(AEC-4-401)	English - 4/MIL-1	2	
The student exiting the programme after two years securing 96 credits will be awarded UG diploma provided he /she secure 4 credits from summer training											

Notes:

1. In the first semester student is offered three different subjects A,B&C.
2. At the end of second semester each student as the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (upto3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc.HONOURS with research, on the condition that the Institution is a PU approved Research Centre

5. ThestudentoptingforSingleMajormustsecure50%creditsofthetotalcreditsinthe Majorsubjectandobtain24creditsfromtheMinorsubject(s).Toqualifyfor Minor in a particular subject,the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/ community work. Further,the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/ she secure 4 credits from summer internship.Thestudentsuccessfullysecuring 144 credits will beawarded BA degree.
7. Language paper chosen as MIL-1and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rdsemester ,the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04credits and write Dissertation of 08 credits in one of the two Major subjects.

Semester III

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-3 Maj/Min-301): Intermediatory Metabolism (3+1)

Marks: 75 (60+15 Int.)

Objective: To understand Metabolism of amino acids, their derivatives and nucleic acids.

Learning Outcomes:
Upon successful completion of this course, students will be able to:

- Comprehend the basic types of catabolic and anabolic pathways of carbohydrates,aminoacids,fatty acids and nucleic acids.
- Learn to perform various clinical testsrelated to metabolism.

SECTION-I

(Lectures: 10)

Introduction to Metabolic pathways. Digestion and absorption carbohydrates, Metabolism of carbohydrates, glycolysis and alcoholic fermentation, the pentose phosphate pathway, metabolism of fructose, galactose, TCA cycle and pyruvate dehydrogenase complex Glucuronate and glyoxylate pathway, gluconeogenesis, Cori cycle, glycogenolysis & glycogenesis, biosynthesis of starch.

SECTION-II

(Lectures: 10)

Digestion and absorption of lipids, beta-Oxidation of fatty acids including odd chain fatty acids. Alpha and omega oxidation of fatty acids. Formation and utilization of ketone bodies. Biosynthesis of saturated and unsaturated fatty acids Biosynthesis and degradation of phospholipids and triglycerides. Cholesterol biosynthetic pathway Biosynthesis of prostaglandins, thromboxanes, leukotrienes.

SECTION-III

(Lectures: 10)

Digestion & absorption of proteins, nitrogen balance, general reactions of amino acids: deamination,
transamination and decarboxylation. Urea cycle. Catabolism of glucogenic and ketogenic amino acid Biosynthesis of nutritionally essential amino acids: glutamate and glutamine, aspartate and asparagine, proline, alanine, cysteine & selenocysteine, tyrosine, serine, glycine. Conversion of amino acids to specialized products: catecholamine, glutathione, creatinine, serotonin.

SECTION-IV

(Lectures: 10)

Nucleic Acids: Degradation and biosynthesis of purines and pyrimidines nucleotides *Denovo* and salvage pathway, formation of deoxy ribonucleotides, biosynthesis and degradation of heme. Disorders of heme, purine & pyrimidine metabolism, anticancer drugs.

PRACTICALS:

(BCH-DSC-3 Maj/Min-301): Intermediatory Metabolism

Marks: 25 (20 + 5 Int.)

1. Estimation of blood glucose
2. Demonstration of Amylase activity
3. Estimation of total cholesterol
4. Estimation of pyruvic acid.
5. Assay of Serum Transaminases (SGPT and SGOT)
6. Estimation of Serum Urea
7. Estimation of Serum Uric Acid
8. Estimation of Serum Creatinine
9. Estimation of DNA

Suggested Books:

1. Lippincott's Illustrated Reviews: Biochemistry by P.C. Champe, R.A. Harvey & amp; D.R. Ferrier, Indian Edition, 2017.
2. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell,31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
3. Voet D and Voet JG. Biochemistry 5 th Edition, John Wiley and Sons Inc., 2021.
4. Lehninger A.L., Nelson D.L. and Cox M.M. (2021)8th edition Principles of biochemistry (W.H. Freeman, USA).
5. Stryer.L. J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman Company, New York).
6. Conn E.E.and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA).
7. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry, (Cengage Learning, US).
8. Zubay G.L., Parson W.W, Dennis E. Vance (William C. Brown, U.S.)1998 Principles of Biochemistry volume-1.

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-4 Min-302): Enzymology and Bioenergetics (3+1)

Marks: 75 (60+15 Int.)

Objective: To learn principles of thermodynamics, nature of enzymes, their mode of action and enzyme kinetics.

Learning Outcomes:

Upon successful completion of the course, students will be able to:

- Understand the concepts of enzymes, their mechanisms and kinetics of enzymatic reactions.
- Understand the mechanism of oxidative phosphorylation and ATP synthesis
- Learn to perform enzymology experiments and learn to calculate enzyme activityand kinetics.

SECTION-I

(Lectures: 10)

Introduction to enzymes. General characteristics of enzymes' system of nomenclature and classification of enzymes. Prosthetic groups. Holoenzyme, apoenzyme, cofactors, isoenzymes and their clinical significance and zymogens. Coenzymes (NAD/NADP, FMN/FAD, coenzyme A, TPP, pyridoxal phosphate, tetrahydrofolate) and their biochemical functions, assay of enzyme activity, units of enzyme activity. Active site of enzymes.

SECTION-II

(Lectures: 10)

Factors affecting velocity of enzyme catalyzed reactions: enzyme concentration, pH and temperature. Michaelis-Menten equation. Determination of Km and its significance.

Enzyme inhibition: Various types of enzyme inhibitions. Determination of Ki value. Enzyme inhibitors and their importance. Introduction to multisubstrate enzyme, Allosteric enzymes properties and dregulation.

SECTION-III

(Lectures: 10)

Enzymes as catalysts. Theories of enzymes catalysis: proximity and orientation effects, acid base catalysis, covalent catalysis. Role of metals in enzyme catalysis Monomeric enzymes, oligomeric enzymes & multienzyme complexes. Mechanism of action of chymotrypsin, abzymes and lysozyme Introduction to multisubstrate enzymes. Allosteric enzymes and enzyme regulation.

SECTION-IV

(Lectures: 10)

Concepts of bioenergetics, principles of thermodynamics & their application in Biochemistry, concept of free energy. Biological oxidations. Redox potential. ATP and High energy phosphate compounds: introduction, phosphate group transfer potential, enzymes and co-enzymes involved in oxidations and reductions. Mitochondrial electron transport chain and oxidative phosphorylation

Suggested Books:

1. T. Palmer Enzymes: Biochemistry, Biotechnology and Clinical Chemistry, 2nd Ed. (2018) Affiliated East-West press Pvt. Ltd. New Delhi.
2. L.Stevens and Nicholas C.Price Fundamentals of Enzymology: The cell and molecular Biology of catalytic Proteins (2000) 3rd edition (Oxford University Press, USA).
3. Robert A. Copeland: A practical Introduction to Structure, Mechanism and Data Analysis 2nd edition (John Wiley and Sons Inc., Publications)
4. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
5. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
6. LehningerA.L.,NelsonD.L.andCox M.M.(2021)8th edition Principles of biochemistry (W.H. Freeman, USA)
7. StryerL., J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman company, New York)
8. Conn E.E. and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA)
9. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry, (Cengage Learning, US)

PRACTICALS:

(BCH-DSC-4 Min-302): Enzymology and Bioenergetics

Marks: 25 (20 +5 Int.)

1. Assay of Enzyme Activity and Specific Activity e.g. Acid Phosphatase
2. Effect of pH on enzyme activity.
3. Effect of temperature on enzyme activity and determination of energy of activation.
4. Effect of substrate concentration on enzyme activity and determination of Km.
5. Partial Purification of Acid Phosphatase from Germinating Mung Bean
6. Effect of inhibitors on enzyme activity.
7. Demonstration of potato polyphenol oxidase activity.

Multi-Disciplinary/Inter-Disciplinary Paper

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 8 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-MDC-3-301): Basic Cell Biology (2+1) Marks: 50 (40 +10 Int.)

Objective: This course is designed to help students understand basis of the cell structure and function at cellular and molecular level.

Learning Outcomes:

Upon successful completion of this course, students are able to:

- Understand the structure and functions of the basic components of prokaryotic and eukaryotic cells, an Comprehend the structure and functions of chromosomes and phases of cell division.
- Independently gain practical hands-on expertise in basic microscopy, simple staining and isolation and different organelles.

SECTION – I

(Lectures: 8)

Cell as a unit of Life: PPLO's, bacteria, eukaryotic, microbes, plant and animal cells., The cell theory, Bio-membranes; structure and function, fluid mosaic concept, Faces of the membranes. Cell wall-structure and functions. Biochemical composition of cells (proteins, lipids, carbohydrates, nucleic acids and the metabolic pool.

SECTION- II

(Lectures: 8)

Cell Organelles: Non-membranous organelles: Structure and functions of ribosomes, centrioles and basal bodies Single membrane bound organelles: endoplasmic reticulum, golgi bodies and lysosomes, peroxisomes and glyoxysomes. Double membrane bound organelles: Mitochondria; structure and functions, semi-autonomous nature; endosymbiont hypothesis; mitochondrial DNA, Nucleus: Nuclear Envelope- structure of interphase nucleus; chromatin material, euchromatin and heterochromatin, nucleolus.

SECTION– III

(Lectures: 8)

Cell cycle: Overview of Cell cycle and its regulation, mitosis, meiosis, binary fission, amitosis, Cell senescence and death: Apoptosis and necrosis, Cell differentiation in plants and animals: totipotent, multipotent, pleuripotent cell.

SECTION-IV

(Lectures: 8)

Techniques in cell biology: Principles and applications of Light microscopy, Phase contrast microscopy, Confocal microscopy, Electron microscopy (SEM, TEM, STEM), fluorescence microscopy, Basics and uses of flow cytometry, fluorescent probes, Centrifugation :differential and density gradient, Chromatography: Paper and TLC.

Suggested Books:

1. Cell Biology by Thomas D. Pollard, William C. Earnshaw.
2. Cell Biology: Structure, Biochemistry, and Function by Phillip Sheeler, Donald E. Bianchi.
3. Molecular Biology of the Cell by Bruce Albert and Dennis Bray, 4th Ed. Garland Publishing Inc, New York and London.
4. Cell and Molecular Biology by Gerald Karp.1996. John Willey and Sons,Inc. London.

5. Cell and Molecular Biology- De-Robertis, F.D.P. and De-Robertis Jr. E.M.F. 6. Molecular Cell Biology- Lodish, H., Baltimore, D., Berk, A., Zipursky,

PRACTICALS:

(BCH-MDC-3-301): Basic Cell Biology

Marks: 25 (20 +5 Int.)

1. Microscopy: a. Use and care of Light compound microscope.
2. Study of structure of any Prokaryotic and Eukaryotic cell.
3. Isolation of chloroplasts by differential centrifugation.
4. To separate chloroplast pigments by chromatography.
6. Study of plasmolysis and de-plasmolysis.
7. Cell fractionation and determination of enzyme activity in organelles using sprouted seed or any other suitable source.
8. Preparation of Nuclear, Mitochondrial & cytoplasmic fractions.

Semester IV

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-5-Maj/Min-401): Membrane Biochemistry & Neurobiology (3+1)

Marks: 75 (60+15 Int.)

Objective: to make students aware of structural and functional aspects of membranes. Highlighting the significance of their compositional heterogeneity and to various cell functions.

Learning Outcomes:
Upon successful completion of this course, students will be able to:

- Understand the composition, structure and various properties of membrane
- Electrochemical and ionic changes that power the various membrane like RBC and neurons functions.
- They will be able to independently perform various experiments related to isolation and purification of membrane.

SECTION-I

(Lectures -10)

Introduction: membrane structure and organization: models of membranes structure. Protein- lipid interactions in membranes, fluid mosaic model of membrane structure, its merits and demerits. Constituents of biological membranes in prokaryotes and eukaryotes. Asymmetry of biomembranes. Structure of RBC membrane and ethrocyte deformities.

SECTION –II

(Lectures-10)

Membrane dynamics: Physical properties of membrane components, membrane fluidity, Phase transitions, cytological and biochemical methods to study membrane fluids. Membrane transport: Typical properties of membrane transport, diffusion and osmosis: uniport and synport, passive diffusion, facilitated diffusion, Carrier mediated active transport. Role of Na, K- ATPase.Ca-ATPase & H^+-ATPase.

SECTION –III

(Lectures-10)

Membrane analysis: isolation and characterization of cell membrane, criteria of membrane purity. Solubilization of membranes and their reconstitution. Labeling of membrane probes and determination of transbilayer distribution of membrane components. Liposomes: unilamelar and multilamelar vesicles. Methods of their preparation, characterization and their applications in targeting drugs and gene therapy. Ionophores and siderophores, structure and functions.

SECTION –IV

(Lectures-10)

Neurobiology: Central nervous system, peripheral nervous system, action potential, neurotransmitters, synapses & nerve gases. Mechanism of nerve impulse transmission: structure, classification and functions of neurons and neuroglias. The resting membrane potential & action potential. Properties of nerve fibers: excitability, conductivity, all or none law, accommodation, adaptation, summation, refractory period, indefatigability. Synapses: types, structure, synaptic transmission of the impulse, synaptic potentials, neurotransmitters, cotransmitters, neuromodulators. The neuromuscular junction: structure, transmission, end-plate potential.

Suggested Books:

1. Jain, MK. Introduction to Biological membranes, John Wiley and sons New York, 1988
2. Lodish H. Molecular Cell biology, WH Freeman and Company, 2021.
3. Bruce alberts, Molecular Biology of the cell 7th edition 2022
4. Geoffrey M. Cooper 9th edition 2023The cell: A molecular approach. Oxford University Press
5. Gerald Karp: Cell and molecular Biology 9th Edition 2019 John Wiley and sons
6. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
7. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
8. Lehninger A.L., Nelson D.L. and Cox M.M. (2021)8th edition Principles of biochemistry (W.H. Freeman, USA)
9. StryerL., J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman company, New York)

PRACTICALS:

(BCH-DSC-5-Maj/Min-401): Membrane Biochemistry & Neurobiology

Marks: 25 (20+5 Int.)

1. Demonstration of hypotonic, hypertonic and osmotic media.
2. Preparation of red cell ghost membrane.
3. Cell lysis by sonication.
4. Preparation of RBC Ghost Cell.
5. Isolation of Chloroplast from Spinach Leaves and Estimation of total Chlorophyll.
6. Extraction and Separation of Photosynthetic Pigments by Partition Chromatography (TLC)
7. Isolation of Mitochondria by cell fractionation method.

Major

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

2. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
3. All questions carry equal marks.
4. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
5. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
6. Questions should be uniformly spread over the entire syllabus.
7. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-4-Maj-402): Enzymologyand Bioenergetics (3+1)

Marks: 75 (60+15 Int.)

Objective: To learn principles of thermodynamics, nature of enzymes, their mode of action and enzyme kinetics.

Learning Outcomes:

Upon successful completion of the course, students will be able to

- Understand the concepts of enzymes, their mechanisms and kinetics of enzymatic reactions.
- Understand the mechanism of oxidative phosphorylation and ATP synthesis.
- They will learn to perform enzymology experiments and learn to calculate enzyme activityand kinetics.

SECTION-I

(Lectures: 10)

Introduction to enzymes. General characteristics of enzymes' system of nomenclature and classification of enzymes. Prosthetic groups. Holoenzyme, apoenzyme, cofactors, isoenzymes and their clinical significance and zymogens.Coenzymes (NAD/NADP, FMN/FAD, coenzyme A, TPP, pyridoxal phosphate, tetrahydrofolate) and their biochemical functions, assay of enzyme activity, units of enzyme activity. Active site of enzymes.

SECTION-II

(Lectures: 10)

Factors affecting velocity of enzyme catalyzed reactions: enzyme concentration, pH andtemperature.Michaelis-Menten equation. Determination of Km and its significance.Enzyme inhibition: Various types of enzyme inhibitions. Determination of Ki value. Enzyme inhibitorsand their importance. Introduction to multisubstrate enzyme, Allosteric enzymes properties andregulation.

SECTION-III

(Lectures: 10)

Enzymes as catalysts. Theories of enzymes catalysis: proximity and orientation effects, acid base catalysis, covalent catalysis. Role of metals in enzyme catalysis Monomeric enzymes, oligomeric enzymes &multienzyme complexes. Mechanism of action of chymotrypsin, Abzymes and Lysozyme.

SECTION-IV

(Lectures: 10)

Concepts of bioenergetics: principles of thermodynamics & their application in Biochemistry, concept of free energy. Biological oxidations. Redox potential. ATP and High energy phosphate compounds: introduction, phosphate group transfer potential, enzymes and co-enzymes involved in oxidations and reductions. Mitochondrial electron transport chain and oxidative phosphorylation. Mechanism of oxidative phosphorylation. Inhibitors and Uncouplers of Oxidative Phosphorylation

Suggested Books:

1. T. Palmer Enzymes: Biochemistry, Biotechnology and ClinicalChemistry, 2nd Ed. (2018) AffiliatedEast-West pressPvt. Ltd. New Delhi.

2. L.Stevens and Nicholas C.Price Fundamentals of Enzymology:The cell and molecular Biology of catalytic Proteins(2000) 3rd edition (Oxford University Press,USA)
3. Robert A.Copeland: A practical Introduction to Structure ,Mechanism and Data Analysis 2ndedition(John Wiley and Sons Inc.,Publications)
4. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Grannerand V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
5. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
6. LehningerA.L.,NelsonD.L.andCox M.M.(2021)8th edition Principles of biochemistry(W.H.Freeman,USA)
7. StryerL.,J.M.Berg,J.L.Tymoczko (2023)10th edition,Biochemistry(W.H.Freemancompany,New York)
8. Conn E.E.and Stumpf P.K.(2006)5th edition Outlines of Biochemistry,(John Wiley and Sons,USA)
9. R.H.Garett and C.M.Grisham (2023)7th Edition Biochemistry,(CengageLearning,US)

PRACTICALS:

(BCH-DSC-4-Maj-402): Enzymologyand Bioenergetics

Marks: 25 (20 +5 Int.)

1. Preparation of casein from milk
2. Determination of achromatic point of saliva
3. Assay of alkaline phosphatase activity.
4. Effect of pH on enzyme activity.
5. Effect of temperature on enzyme activity and determination of energy of activation.
6. Effect of substrate concentration on enzyme activity and determination of Km.
7. Effect of inhibitor on enzyme activity
8. Demonstration of potato polyphenol oxidase activity.

BIOINFORMATICS

COURSE OUTLINE & SYLLABUS FOR BIOINFORMATICS

Students opting for Single major and Single minor(Table 1*)and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)

B.Sc., B.Sc. Honours and B.Sc. Honours withResearch(1st Year Programme)

Sem	Level	Paper Code	Paper Name	Credits (Th+Pr)
I	L-100	BINF-DSC-1 (Maj/Min-101)	Bioinformatics-I	3+1=4
		BINF-MDC-1-101	Introduction to Bioinformatics-I	2+1=3
		BINF-SEC-1-101	Tools for Biological Data Analysis	0+3=3
II	L-100	BINF-DSC-2 (Maj/Min-201)	Bioinformatics-II	3+1=4
		BINF-MDC-2-201	Introduction to Bioinformatics-II	2+1=3
		BINF-SEC-2-201	Computational Tools for Small and Macromolecules	0+3=3
III	L-200	BINF-DSC-3-Min-301	Computational Methods for Protein Sequence and Structure Analysis	3+1=4
		BINF-DSC-4-Min-302	Programming Language in C	3+1=4
		BINF-MDC-3-301	Bioinformatics Methods for Biomolecule Analysis	2+1=3
		BINF-SEC-3-301	Artificial Intelligence Tools for Biological Data Analysis	0+3=3
IV	L-200	BINF-DSC-5-Min-401	Computational Methods for Gene and RNA Structure Analysis	3+1=4

Notes: -

1. MDC/IDC subject is offered by The Department of Bioinformatics for the students of allied/other streams. The students opting for Bioinformatics as DSC subject will select MDC/IDC from the list of courses offered by the allied/other departments.
2. Language paper chosen as MIL-1, MIL-2, MIL-3 and MIL-4 should be in the same subject.

SEMESTER - III

Paper: BINF-DSC-3-Min-301: COMPUTATIONAL METHODS FOR PROTEIN SEQUENCE AND STRUCTURE ANALYSIS

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Computational Methods for Protein Sequence and Structure Analysisis to equip students with computational tools and methodologies for analyzing protein sequences and predicting their structures and functions.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Motif and Domains: Basic concept of Protein family, types of motifs and domains.

Sequence patterns and profiles: Basic concept and definition of sequence patterns and profiles, various types of pattern representations viz. consensus, regular expression (Prosite-type), PHI-BLAST.
Sequence profiles: PSSM, profile-based database searches using PSI-BLAST. Applications of Pattern and Profile.

Domaindatabases: CDD, SMART, InterPro, Prosite
Tools for searching pattern and profile: InterPro Scan and Scan Prosite and MEME.

UNIT – II

Secondary Structure Prediction methods:
Ab-initio based: Chou Fasman and GOR, Homology based and Neural Network (PSIPRED).

Tertiary Structure Prediction Methods: Homology Modeling, Threading/Fold Recognition, Ab-initio Methods, AI-Based Tertiary Structure Prediction (AlphaFold)

Structural Validation and Quality Assessment Tools: PROCHECK and Verify3D.

Recommended Books:

1. David W. Mount (2004). Bioinformatics: Sequence and Genome Analysis, 2nd Edition, Cold Spring Harbor Laboratory Press.
2. Richard Durbin, Sean R. Eddy, Anders Krogh, and Graeme Mitchison (1998). Biological Sequence Analysis: Probabilistic Models of Proteins and Nucleic Acids, Cambridge University Press.
3. Carl Branden and John Tooze (1999). Introduction to Protein Structure, 2nd Edition, Garland Science.
4. Jin Xiong (2006). Essential Bioinformatics, Cambridge University Press.
5. BharathRamsundar, Peter Eastman, Patrick Walters, and Vijay Pande (2019). Deep Learning for the Life Sciences: Applying Deep Learning to Genomics, Microscopy, Drug Discovery, and More, O'Reilly Media.
6. Tiago Antao (2018). Bioinformatics with Python Cookbook, 2nd Edition, PacktPublishing.

BINF-DSC-3-Min-301: COMPUTATIONAL METHODS FOR PROTEIN SEQUENCE AND STRUCTURE ANALYSIS – (PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. MEME for motif identification and Prosite for motif analysis.
2. Sequence patterns and profiles by PSI-BLAST and PHI-BLAST
3. Use InterPro and SMART to annotate protein domains.
4. Compare CHOU-FASMAN, GOR, and PSIPRED for a given protein sequence.
5. Perform homology modeling using SWISS-MODEL.
6. Structure prediction by Phyre2, I-TASSER and AlphaFold
7. Validate modeled structures using PROCHECK and Verify3D

BINF-DSC-4-Min-302: PROGRAMMING LANGUAGE IN C

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

To introduce basic concepts of "C" Language, which is required to do programming and solve problems related to Bioinformatics.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Introduction to 'C' Language: Character set, variable and Identifiers, Built-in data types, Variable Definition, Arithmetic operations and Expressions, Constants and Literals, Simple assignment statement. Basic input/output statement.

Conditional statement and Loops: Decision making, Conditions, Relational Operators. Logical connectives If Statement, If-else statement Loops, while loop, do while loop, for loop, Nested loops. Switch statement, structural programming Programs related to decision making statements, loops.

Array: One dimensional arrays, Array manipulation; Searching, Insertion, Deletion of an element from an array. Finding the largest /smallest element in an array; Two dimensional arrays, multi-dimensional array, Addition / Multiplication of two matrices, Transpose of a square matrix, Strings, operation on strings.

UNIT - II

Functions: Top-down approach and Bottom-up approach of problem solving, Modular programming and functions, Standard library of C functions, Prototype of a function: Formal parameter list, Return Type function Call, Block Structure, Passing arguments to a function: Call by reference, call by value, Recursive functions, Arrays as function arguments.

Structure and Unions: Structure variables, initialization, structure assignment, nested structure, structures and functions, structures and arrays, arrays of structures, structures containing arrays, unions.

Pointers: Address Operators, Pointer type declaration, pointer assignment, pointerinitialization, pointer arithmetic, functions and pointers, Arrays and Pointers, Pointer arrays, Dynamic memory allocation malloc() and calloc() and free().

File Processing: Concept of Files, Files operating in various modes and closing of a file, reading from a file, writing to a file.

Recommended books:

1. Byron Gottfried "Programming wit 'C' Second addition. Tata McGrawhill, (2000).
2. RG Dromey "How to solve it by computer", Prentice Hall of India, 7th Edition, (2001).
3. E. Balaguruswami "Programming with Ansi-C", Pearson Education, 1st Edition, (2002).
4. Kamthane "Programming with ANSI and Turbo C", Pearson Education, First Edition, (2002).
5. Venugopal and Prasad " Programming with C" Tata McGrawhill, 1st Edition, (1997).
6. BW Kemighan and DM Ritchie "The C Programming Language", Prentice Hall of India, Second edition, (2001).

BINF-DSC-4-Min-302: PROGRAMMING LANGUAGE IN C(PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Writeaprogramtocalculateaprimenumber is evenandoddnumbers
2. Writeafunctiontoreadamatrixofsizemxnfromthekeyboard and rotate the array clock-wise.
3. Writeaprogramtoaddandsubtractatwo-threematrixsimultaneously
4. Writeaprogramtofind factorial of number using recursion.
5. ProgrammingproblemsonArray,PointerandFiles
6. Write a program to translate protein sequence using DNA.
7. Write a program to convert protein or DNA sequence into FASTA FORMAT
8. Write a program to cut a DNA using restriction enzyme like EcoRII.

SEMESTER - IV

BINF-DSC-5-Min-401: COMPUTATIONAL METHODS FOR GENE AND RNA STRUCTURE ANALYSIS

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Computational Methods for Gene and RNA Structure Analysis is to equip students with computational tools and methodologies for analyzing gene structure and predicting their structures and functions.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Basic Concepts of Gene Structure: Exons, introns, promoters, enhancers, and untranslated regions (UTRs). Differences between prokaryotic and eukaryotic gene structures.

Nucleic acid sequence analysis: Reading frames, Concept of ORF, Translational andtranscriptional signals.

Computational methods for gene prediction: Ab initio methods: Markov Model, HMM, Artificial Neural Networks and Discriminant Analysis. Homology-based approaches, Consensus Based Programs. Performance Evaluation: Real vs Predicted gene, Sensitivity, Specificity.
Prediction tools: Genscan, GenMark and Grail.

UNIT – II

Promoter analysis: Promoter and Regulatory Elements in Prokaryotes and Eukaryotes. Predictions of promoters in prokaryotic and eukaryotic Genomes: Ab initio based and Phylogenetic Footprinting.

Basics of RNA structure: Primary, secondary, and tertiary structures of RNA.

RNA secondary structure prediction: RNA secondary structure thermodynamics. Ab-initio and Comparative methods.

Tools of RNA secondary structure prediction: Mfold, Sfold and Vienna RNA package.

Post transcriptional Gene regulation: miRNA and siRNA.

Recommended Books:

1. **Jin Xiong (2006).** Essential Bioinformatics, Cambridge University Press.
2. Bioinformatics: Sequences, structure and databanks by Des Higgins and WillieTaylor, Oxford University Press, (2000).
3. Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
4. Bioinformatics: A practical guide to the analysis of genes and proteins. Ed. By Baxevanis, 3
5. Pevsner, Jonathan. ***Bioinformatics and Functional Genomics***. 3rd ed. Hoboken, NJ: Wiley-Blackwell, 2015

BINF-DSC-5-Min-401: COMPUTATIONAL METHODS FOR GENE AND RNA STRUCTURE ANALYSIS (PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Gene Prediction Tools: GenMark, GenScan and GENEID.
2. Promoter 2.0 prediction server.
3. Tools of RNA secondary structure: Mfold
4. Transcription factor databases like Transfac.
5. miRDB.

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B.Sc./B.Sc. (Hons)/B.Sc. (Hons. with research) Major & Minor Courses
Second Year (Semester System), Examinations, 2026-27

Biotechnology

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						Table 2*					
Sem	Level	Course Code	Course Name	Credits(Th+Pr+Tut)	Total Credits	Sem	Level	Course Code	Course Name	Credi (Th+Pr+Tut)	Total Credits
III	L-200	BIOT-DSC-3-(MAJ/MIN)-301	Bio chemical and Bio physical techniques	3+1	24	III	L-200	BIOT-DSC-3(MAJ/MIN)-301	Bio chemical and Bio physical techniques	3+1	24
		BIOT-DSC-4-(MAJ/MIN)-302	Genetics	3 +1				BIOT-DSC-4(MAJ/MIN)-302	Genetics	3+1	
		MIC-DSC-3-MIN-301	Bacteriology and Virology	3+1				Major		3+1	
		MIC-DSC-4-MIN-302	Molecular Biology	3 +1				Major		3+1	
		BIOT-MDC-3-301	Bio analytical Techniques	2+1				MDC-3-301	MDC	2+1	
		**BIOT-SEC-3-301	Molecular Diagnostics	2+1				SEC-3-301	SEC	2+1	
		AEC-3-301	Eng-3/MIL-1	2				AEC-3-301	Eng-3/MIL-1	2	
IV	L-200	BIOT-DSC-5-(MAJ/MIN)-401	Genetic Engineering	3 +1	24	IV	L-200	BIOT-DSC-5(MAJ/MIN)-401	GeneticEngineering	3+1	24
		BIOT-DSC-6-(MAJ)-402	Environment Biotechnology	3 +1				Major		3+1	
		BIOT-DSC-7-(MAJ)-403	Bio safety& Bioethics	3+1				Major		3+1	
		BIOT-DSC-8-(MAJ)-404	Plant Tissue Culture	3+1							
		*MIN-B-5/MIN-C-5	Minor	3+1				Major		3+1	
		BIOT-VAC-3-401	Value Added Course	2				Major		3+1	
		AEC-4-401	Eng-4/MIL-2	2				VAC-3-401	ValueAdded Course	2	
								AEC-4-401	Eng-4/MIL-2	2	

The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/ she secures 4 credits from summer internship
*Minor subjects to be selected from the three streams (Biochemistry, Microbiology & Bioinformatics (Annexure1)which are approved by BOS Biotechnology PU colleges being offered to the students of B.Sc. Hons. Biotechnology or other Science faculty.
**Candidate has to select all the SEC s offered by the Department of Biotechnology in colleges as proposed and approved by BOS Biotechnology PU colleges.

Semester III

BIOT-DSC-3-(MAJ/MIN)-301:BIOCHEMICALANDBIOPHYSICALTECHNIQUES

Learning Outcomes: Students will be able to understand, select, and apply core biophysical and biochemical techniques by explaining the principles and instrumentation of spectroscopic, microscopic, electrophoretic, chromatographic, crystallographic, radioisotopic, and mass spectrometric methods, and by interpreting experimental data for the qualitative and quantitative analysis of biological molecules and systems in research and diagnosticcontexts.

CourseDuration

45 hours of lectures forTheory+ 30 hours ofpractical (1 credit = 15 hours of theory learning based on1hr/week)
(1 credit = 30 hours of practical based on 2 hrs./week)

(1Semester=15 weeks)

Credits:3+1credits(Each credit is equal to 25 marks)

ExternalTheory	InternalTheory	Externalpractical	Internalpractical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No.1(COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks.Rest all questions will be of 14marks each

UNIT I

Principle,working and applications of Spectrophotometry: UV-visible,spectroflourimetry, Atomic absorption spectrophotometry, Infrared and Ramanspectroscopy.

ORD and circulardichroism,Nuclear MagneticResonance and Electron Spin Resonance spectroscopy, Magnetic Resonance Imaging.
Electro phoresis: Principle, types and applications (Agarose and PAGE).

UNIT II

Microscopy: Principle working and applications of Bright field Microscopy, Darkfield Microscopy, Phase contrast, Fluorescent and Confocal microscopy.

Electron microscopy: Principle, working and applications of Scanning and Transmission Electron Microscopy.

Centrifugation: Principle, types & applications.

UNIT III

Crystallography: Physical basis of crystallization; formation of crystals; Mountingof crystals X-ray diffraction: Braggs law, Diffraction of x-rays by crystals

Over view of column chromatography: Principle, working and applications of Gelfiltration, Ion exchange, Gas- chromatography and HPLC.

UNIT IV

Radio isotope techniques: Radiotracers; GM counter,proportional and scintillation counters, autoradiography.

Massspectrometry: Principle, instrumentation and Applications

Hyphenated Techniques: Principle, instrumentation and Applications GC-MSandLC-MS.

ReferenceBooks:

1. KWilson,JWalker(2000)Principlesandtechniquesofbiochemistryand MolecularBiology Seventh Edition, (Cambridge University Press).
2. SRMikkelsen,ECortin(2004).BioanalyticalChemistry(2004)(JohnWiley &Sons)
3. DavidSheehan(2009).PhysicalBiochemistry:Principlesand Applications, Second Edition. (John Wiley & Sons)
4. P.Carmona,R.Navarro,A,Hernanz(1997)Spectroscopyofbiological molecules: Modern trends(Illustrated, Publisher Springer)
5. BernardValeur(2001).Moleculafluorescence:Principlesandapplication (Wiley-VCH). Plenum USA.
6. PlummerD.T.(2004).Anintroductiontopracticalbiochemistry(Tata McGraw Hill Publishers Co. Ltd., New Delhi).
7. Bansal,D.D.,KHardori,R&Gupta,M.M.(1985).Practicalbiochemistry (Standard Publication, Chandigarh).

BIOT-DSC-3-(MAJ/MIN)-301:BIOCHEMICALANDBIOPHYSICALTECHNIQUES (PRACTICALS)

1. Determination of DNA/RNApurityusingUV-Visspectrophotometer.
2. AgarosegelelectrophoresisofDNAsample.
3. Separationofaminoacids/plantpigments.
4. StudyofGMcounter/scintillationcounter.
5. Interpretationofmassspectraofbiomolecules.
6. Measurementofcellsizeusingmicroscope.

BIOT-DSC-4-(MAJ/MIN)-302:GENETICS

Learning Outcomes: Students will be able to understand, analyze, and apply fundamental principles of genetics by explaining patterns of inheritance and gene interactions; interpreting linkage, recombination, gene mapping, and chromosomal behavior; evaluating mutations, DNA repair, and chromosomal aberrations; and applying microbial and population genetics concepts to solve genetic problems, analyze experimental data, and interpret hereditaryvariation in organisms.

CourseDuration

45hoursof lecturesforTheory+30hoursofpractical

(1 credit = 15 hours of theorylearningbased on 1hr/week) (1 credit = 30 hours of practical based on 2 hrs./week)

(1Semester=15weeks)

Credits:3+1credits(Eachcreditisequalto25 marks)

ExternalTheory	InternalTheory	Externalpractical	Internalpractical
68	07	22	03

TheoryPaper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit,including Question No. 1(COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks.Rest all questions will be of 14 marks each

UNIT I

Mendelian lawsofinheritance, Gene interactions, Sex determination in drosophila, plants and animals, Sex linked inheritance, Non-disjunction as a proof of chromosomal theory of inheritance.

Extra chromosoma linheritance: mito chondrial and chloroplastgenetic systems.

UNITII

Crossing over: molecular mechanism and cytological proof, Recombination, linkage, gene mapping, Three point test cross, interference, coincidence, recombination frequencies, Tetrad analysis, somatic cell hybridization for gene linkage studies, Hereditary defects.

UNIT III

Mutation: Spontaneous versus induced mutations, types of mutations,mutagenic agents:Physical,chemicalandradiation,molecularbasisofmutations, mechanisms of DNA repair, mutations frequency, correlation between mutagenicity and carcinogenicity. Numerical chromosome aberrations: polyploidy, aneuploidy, Chromosomal aberrations: Deletion, duplications, inversions, translocations, position effects.

UNIT IV

Basic microbial genetics: Conjugation, transduction, transformation, isolation of auxotrophs, replica plating techniques, analysis of mutations inbiochemical pathway,one gene – one enzyme hypothesis.

Population genetics: Hardy-Weinberg equilibrium, gene and genotypicfrequencies, Chi-square test, probability, pedigree analysis.

ReferenceBooks:

1. Birge,E.A.(2006).Bacterial andbacteriophagegenetics(Springer).
2. DaleJ.,ParkS.(2005)Moleculargeneticsofbacteria(JohnWileyandSonsInc).
3. Freifelder,D.(2005).Molecularbiology(JonesandBartlett).
4. Synder,L.,ChampnessW.(2007).Moleculargeneticsofbacteria(ASMPress).
5. TurnN.J,TrempyJ.E.(2006).Fundamentalbacterialgenetics(WileyBlackwell).
6. T.A.Baker,J.D.Watson,S.Bell.(2004).Molecularbiologyofthegene(Benjamin B.Cummings).
7. A.J.F.Griffiths,S.R.Wessler,R.C.Lewontin,S.B.Carroll(2007).Introductiontogeneticanalysis (W.H. Freeman & Co Ltd).
8. StrickbergerM.W.(2006)3rdedition.Genetics(Macmillan).
9. VermaP.S(2012)Genetics(S.Chand&CompanyLtd.)
10. SandersM.F.andBowmanJ.L.(2012)GeneticAnalysis(Pearsons)

BIOT-DSC-4-(MAJ/MIN)-302:GENETICS(PRACTICALS)

1. Examination of permanent slides of various stages of mitosis and meiosis and different types of chromosomes.
2. Demonstration of law of segregation and independent as sortment (use of driedpeas, colored peas, *etc*.).
3. Numericals for segregation and independent as sortment.
4. UseofChi-squareforpredictionofphenotype/genotypefrequenciesofparentsfromprogeny and vice-versa, epistasis.
5. Detectionofbloodgroups.
6. Calculationofvarianceinrespectofpodlengthandnumberofseeds/pod.
7. Calculationofgenefrequenciesandrandommating(coloredbeads,capsules).
8. Dermatographics:Palmprinttakingandfingertippatterns

Semester IV

BIOT-DSC-5-(MAJ/MIN)-401:GENETIC ENGINEERING

Learning Outcomes: Students will be able to understand the principles and applications of genetic engineering, use of recombinant DNA technology, cloning vectors, and nucleic acid manipulation techniques.

They will gain practical skills in gene cloning, PCR, DNA analysis, Southern blotting, transformation, and recombinant protein production, enabling them to design, perform, and analyze experiments for gene identification, mutagenesis, and molecular characterization of DNA.

CourseDuration

45hoursoflecturesfor Theory+30hoursof practical

(1credit=15hoursoftheorylearningbasedon1hr/week)(1 credit = 30 hours of practical based on 2 hrs./week)

(1Semester=15 weeks)

Credits:3+1credits(Eachcreditisequalto25 marks)

ExternalTheory	InternalTheory	Externalpractical	Internalpractical
68	07	22	03

TheoryPaper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1(COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks.Rest all questions will be of 14marks each

UNITI

Introduction to Genetic Engineering: Basic Principles, Significance, and Overview of Gene Cloning.

Tools in Recombinant DNA Technology: Restriction and modifying enzymes, TypeI, Type*II and TypeIII*

enzymes and their characteristic features; restrictions equences, isoschizomers, rarecutting enzymes, enzyme cutting similar sequence in different manner. DNAModifyingenzymes:Characteristicsand applicationsofNucleases:DNaseand RNase,DNA-PolymeraseI,Klenow fragment,T4DNApolymerase,T7 DNApolymerase, T4Polynucleotidekinase,Phosphatase,Reversetranscriptase,Taq polymeraseandLigase.

Terminal deoxy ribonucleotidyl transferase.

Polymerase Chain Reaction: Principle,types and applications

UNIT II

Basic biology of plasmids and Phage vectors: Basic features of plasmids, plasmid classification, Bacteriophage λ, lytic & lysogeny, Linear and circular forms oflambda vector, DNA cloning

With single stranded DNA vectors.

DNAcloningvectors: Cloning vectors for*E.coli*(Nomenclature,pBR322,pBR327,pUC18, pGEM3Z), Insertion and replacement vectors, Vectors based on M13.

Methods of identification of recombinants: Insertion alin activation, blue/ white selection.

Cloning vectors for yeast:YEp,YIp,YRp.

Advanced Vectors: cosmid, phagemid, Bacterial Artificial Chromo somes (BACs), shuttle vectors, yeast artificial chromo somes.

UNIT III

Preparation of genomic and cDNA library: Partial digests, Choice of vectors, Construction and Evaluationofa genomiclibrary.cDNAlibrary:mRNAenrichment,cDNAsynthesis,Random,arrayed and Ordered library.

Finding The Right Clone: Gene identification, Nucleic acid hybridization, screening Procedure,Probeselection,immunoscreening,functionalcomplementation.Southernblotting,northern blotting.

Preparation of nucleic acid probes: DNA and RNA labeling techniques, nick translation, random priming, end labelling, radioactive and non- radioactive labels.

UNIT IV

Site directed mut agenesis: cassette, primer extension, RT, realtime, multiplex, inverse,

DNA sequencing: Maxam-Gilbert,Sanger,pyrosequencing.

Production of Protein from Cloned Genes: Special vectors for expression of foreign genes in *E. coli,* General problems with the production of recombinant protein in *E. coli.* Production of recombinant protein by yeast cells.

Referencebooks:

1. RecombinantDNAbyWatson,J.D.et.al.,1993,Scientific AmericanBooks,NewYork.
2. PrinciplesofGeneManipulationbyOld.,R.W.andPrimrose,S.B.,2004,BlackwellPublishing,UK.
3. MethodsinGeneTechnologybyDale,J.N.,1994,JAIPressLtd.London,England.
4. RecombinantPrinciplesofgenemanipulationandgenomics,primrose&Twyman 2006,Blackwell,U.K.
5. GenecloningandDNAanalysis:AnintroductionbyT.A.Brown(5thEd.)2006,Blackwell Science Ltd.
6. Molecularcloning:AlaboratorymanualVol.I-IIISambrookJ.andRussel,DavidN.2001Cold Spring Harbor,

BIOT-DSC-5-(MAJ/MIN)-401:GENETICENGINEERING (PRACTICALS)

1. Toperformpolymerasechainreaction.
2. DemonstrationofSouthernblottingtechnique.
3. Preparationofcompetentcellsforgenetictransformation.
4. Transformationofcompetentcellsandblue/whitescreening.
5. Restrictiondigestionusingdifferenttypesofrestrictionendonucleases.
6. AnalysisandinterpretationofDNAsequencingdata.

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BOTANY

MAJORAND SINGLE MINOR(Table 1*) and STUDENTS OPTING FORMULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)B.Sc.,B.Sc.HONOURSANDB.Sc.HONOURSWITHRESEARCHINBOTANY

Table1*						Table2*					
Sem	**Level**	**Course Code**	**CourseName**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**CourseName**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(BOT-DSC-3Maj/Min-301)	Morphologyof Angiosperms	3+1=4(x2)	24	III	L-200	(BOT-DSC-3 Maj/Min-301)	Morphologyof Angiosperms	3+1=4(x3)	24
		(BOT-DSC-4Maj/Min-302)	Cytogenetics	3+1=4(x2)				(BOT-DSC-4Min-302)	Cytogenetics	3+1=4	
		(BOT-MDC-3-301)	Natural ResourceManagement	3				(BOT-MDC-3-301)	Natural ResourceManagement	3	
		(BOT-SEC-3-301)	Organic Farming	2+1=3				(BOT-SEC-3-301)	Organic Farming	2+1=3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(BOT-DSC-5Maj/Min-401)	Reproductive Biology ofAngiosperms	3+1=4(x2)	24	IV	L-200	(BOT-DSC-5 Maj/Min-401)	Reproductive Biology ofAngiosperms	3+1=4(x3)	24
		(BOT-DSC-6Maj-402)	Phycology	3+1=4				(BOT-DSC-4 Maj-402)	Cytogenetics	3+1=4(x2)	
		(BOT-DSC-7Maj-403)	Mycology	3+1=4							
		(BOT-DSC-8Maj-404)	CellBiology	3+1=4							
		(BOT-VAC-3-401)	Plantsand HumanWelfare	2				(BOT-VAC-3-401)	Plantsand HumanWelfare	2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
Thestudentexitingtheprogrammeaftertwoyearssecuring96creditswillbeawardedUGDiplomaprovidedhe/shesecures4creditsfrom summerinternship											

***Note:ThecandidatewilloptforeitherTemplate1or2fromSemesterIIIonwards.**

Notes:

1. In the first semester student is offered three different subjects A, B&C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjectsand complete his/her three-year degree by securing 36 credits each in the chosen two subjectsasMajorand24creditsinMinorsubject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% creditsof the total credits in theMajor subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attainat least 12 credits in that subject(s).
6. The student exiting the programme after one years ecuring 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/communitywork.Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship.The student successfully securing 144 credits will be awarded BA degree.
   Language paper chosen as MIL-1andMIL-2 should be in the same subject.
7. Once Major and Minor papers are chosen in 3rdsemester, the same will continue as Major and Minor subject in the subsequent semesters.
8. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

BOTANY

SEMESTER III

CourseTitle:MORPHOLOGY OF ANGIOSPERMS Course Code:BOT-DSC-3

Maj/Min-301

Learning Outcome:This paper deals with the morphology and reproductive structures in flowering plant forms. The course work of this paper covers vegetative and reproductive morphology of these plants and will familiarize the students withplants bearing the enclosed seeds.

Max.Marks:100			**CourseCredits=**	3+1=4
1. Theory:75		WrittenExam:67	InternalAssessment:	8
2. Practical:25		WrittenExam:22	InternalAssessment:	3

Periodsperweek: 1.Theory-Fourof45minutesduration(Threehours)
2.Practical-Oneoftwohoursduration(Twohours)

Instructions for Paper Setters:

Question Paper will have four sections.Students will attempt five questions out of total nine comprising of one COMPULSORYquestion consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids:Teaching methodology includes series of lectures making use of charts, transparencies, LCD, Models, slides ,practical demonstrations ,extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT–I 12Hours

General characters of angiosperms, Diversity in plant forms:annuals ,biennia lsand perennials. Diversity on the basis of size, habit and habitat. Monocots and dicots.

UNIT-II 11Hours

Root:Characteristics and functions.Taprootandadventitiousrootsystemandtheirvarious types; structural modifications for storage, respiration and reproduction

UNIT–III 11Hours

Stem:Characteristics and functions, branching pattern. Modifications of aerial, subaerial and underground stem.

UNIT-IV **11Hours**

Leaf:

PartsofLeaf,Venation,phyllotaxy,simpleandcompoundleaves,functionsand modifications of leaves.

Flower:Partsofflower(excludingterminology).Solitaryflowerandinflorescence (racemose, cymose, compound and special inflorescence)

SuggestedReadings:

1. Bhojwani, S.S. and Bhatnagar, S.P. The Embryology of Angiosperms, 5th edition. Vikas Publishing House Pvt. Ltd., Delhi, 2009
2. Goyal J.P. and Saini, A.Angiosperms: Structure Development and Reproduction, Trueman Book Co. Jalandhar & New Delhi, 2006.
3. Eames,Al.;1961:MorphologyofAngiosperms,Mc-GrawHill,NewYork.
4. DuttaA.C.2016.BotanyforDegreeStudents.OxfordUniversityPress.

Suggestedlaboratoryexercises:

1. TostudytheModificationsofroot.
2. Tostudythemodificationsofstem.
3. Tostudythemodificationsofleaf.
4. Tostudyphyllotaxyofleaf.
5. Tostudyinflorescencetypes.
6. Topreparephotoalbumonroot/stem/leaf.

GUIDELINESFORPRACTICALEXAMINATION

Time:3Hrs **Max.Marks:25**

Practical:22 **Int.Assessment:3**

1. Identify,writeillustratedmorphologicalnoteonSpecimenAfromroot.	(4)
2. Identify,writeillustratedmorphologicalnoteonspecimenBfromstem.	(4)
3. Identify,writeillustratedmorphologicalnoteonspecimenCfromleaf.	(3)
4. Identify,writeillustratednoteonspecimenDfrominflorescence.	(3)
5. Practicalnote-book	(3)
6. Viva-voce	(3)
7. Photoalbum	(2)

CourseTitle:CYTOGENETICS **CourseCode:BOT-DSC-4**

Maj/Min-302/
Maj-402

Learning Outcome: This paper deals with the basic structural unit of life i.e. Cell & its organelles. It also provides an insight into structure of chromosomes, cell division, Mendelism and genetic code.

Max.Marks:100		**CourseCredits=**	3+1=4
1. Theory:75	WrittenExam:67	InternalAssessment:	8
2. Practical:25	WrittenExam:22	InternalAssessment:	3

Periodsperweek: **1.Theory-Fourof45minutesduration(Threehours)**
2.Practical-Oneoftwohoursduration(Twohours)

InstructionsforPaperSetters:

Question Paper will have four sections.Students will attempt five questions out of total nine comprising of one COMPULSORYquestion consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering thewhole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

InstructionAids:Teaching methodology includes seriesoflectures,making useof charts,transparencies,LCD,Models,slides,practicaldemonstrations,extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT–I 12Hours

Cell:Ultrastructure and functions of a typical plant cell and its organelles: Nucleus, Mitochondrion, Plastids, Ribosome, Endoplasmic reticulum, Golgi apparatus, Lysosomes; Structure and functions of cell wall and plasma membrane: fluid mosaic model only.

UNIT–II 11Hours

Chromosomes: Physical structure of chromosome; Giant chromosomes: Polytene and Lampbrush chromosomes.
Celldivision:mitosisandmeiosisanditssignificance.

UNIT–III 11Hours

Mendelism: Mendel's experiments and results, Mendel's Laws of Dominance, Segregationand Independent assortment; Linkage: complete and incomplete linkage, linkage groups,linkage maps, importance of linkage.

UNIT–IV 11Hours

Genetic Code: Characteristics, exceptions, Wobble hypothesis; DNA: Structure (Watson and Crick model) and Replication. RNA: Structure and types, transcription, translation.

SuggestedReadings:

1. AggarwalSK(2009)FoundationCourseinBiology,2ndEdition,AneBooksPvt.Ltd
2. Gardner, E.J., Simmons, M.J., Snustad, D.P. (1991). Principles of Genetics, John Wiley& sons, India. 8th e
3. Snustad, D.P. and Simmons, M.J. (2010). Principles of Genetics, John Wiley & Sons Inc., India.5th edition.
4. Karp,G.(2010).CellBiology,JohnWiley&Sons,U.S.A.6thedition.
5. Cooper, G.M. and Hausman, R.E. (2009) The Cell: A Molecular Approach. 5th edition. ASM Press & Sunderland, Washington, D.C.; Sinauer Associates, MA.17
6. Becker, W.M., Kleinsmith, L.J., Hardin. J. and Bertoni, G. P. (2009) The World of the Cell. 7th edition. Pearson Benjamin Cummings Publishing, San Francisco
7. Gupta,P.K.(2010).Cytogenetics.RastogiPublications,Meerut,India.
8. Majumdar, R., Sisodia, R. (2019). Laboratory Manual of Cell Biology, with reference to Plant Cells. New Delhi, Delhi: Prestige Publication

Suggestedlaboratoryexercises:

1. Studyofplantcellstructurewiththehelpofepidermalpeelmountof*Allium/Rhoeo*.
2. Preparation of temporary slides to show different stages of mitosis from root tips of *Allium cepa/A. sativum*.
3. Preparation of temporary slides to show different stages of meiosis from floral buds of *Allium/Brassica*.
4. Studyofcellanditsorganelleswiththehelpofcharts/Photographs/slides.

GUIDELINESFORPRACTICALEXAMINATION

Time:3Hrs **Max.Marks:25**

Practical:22 **Int.Assessment:3**

1. Prepare a squash mount of specimen A to show the stage of cell division visible in the slide andshow it to the examiner. Identify it giving at least one reason. Draw the stage of

cell division and show it to the examiner. (06)

2. TopreparesliideofpeelmountspecimenBanddrawitsdiagram. (04)
3. Identify the photograph/slide of specimen C and D (from cell organelles) and write a note on it along with well labelled diagram. (06)
4. Viva-voce (03)
5. PracticalNote-book. (03)

BOTANY **CourseCode:BOT-DSC-5**
SEMESTER IV **Maj/Min-401**

Course Title: REPRODUCTIVE BIOLOGY OF ANGIOSPERMS

Learning Outcome: This paper deals with the structure, development and reproduction in flowering plants- the most fascinating group of plants on earth. The studentswill learn about reproduction and developmental processes in angiosperms.

Max.Marks:100 **CourseCredits**=3+1=4

1. Theory:75 WrittenExam:67 InternalAssessment:8
2. Practical:25 WrittenExam:22 InternalAssessment:3

Periodsperweek: **1.Theory-Fourof45minutesduration(Threehours)**
2.Practical-Oneoftwohoursduration(Twohours)

InstructionsforPaperSetters:

Question Paper will have four sections.Students will attempt five questions out of total nine comprising of one COMPULSORYquestion consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering thewhole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

InstructionAids:Teaching methodology includes seriesoflectures,making useof charts,transparencies,LCD,Models,slides,practicaldemonstrations,extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT–I 12Hours

Structure of anther and pistil, Microsporogenesis and development of male gametophyte, Megasporogenesis and development of female gametophyte (*Polygonum* type).

UNIT–II 11Hours

Typesofpollination,advantagesanddisadvantagesofselfandcross-pollination, Contrivancesforselfandcross-pollination,Variousagenciestobringaboutcross pollination, characters of flowers pollinated by various agencies.

UNIT-III 11Hours

Types of ovules, Types of embryo sac (monsporic, bisporic and tetrasporic), process of fertilization, double fertilization and its significance.

UNIT–IV 11Hours

Typesoftypesofendosperm(nuclear,cellularandhelobial)anditsimportance, Development of monocot embryo (*Sagittaria sagittifolia*), Development of dicot embryo (*Capsella bursa-pastoris*), Structure of Dicot seed and monocot Seed,endospermic and non-endospermic, Seed germination (Hypogeal and Epigeal).

1. Bhojwani, S.S. and Bhatnagar, S.P. The Embryology of Angiosperms, 5th edition. Vikas Publishing House Pvt. Ltd., Delhi, 2009
2. Goyal J.P. and Saini, A.Angiosperms: Structure Development and Reproduction, Trueman Book Co. Jalandhar & New Delhi, 2006.
3. Singh, V., Pande, P. C. and Jain, D. K., (1997). Embryology of Angiosperms, Rastogi Publications, Meerut.
4. Johri, B. M., Srivastava, P. S., & Singh, N. (2001). Reproductive Biology of Angiosperms. Reproductive Biology of Plants, 237–272
5. P. Maheshwari, (2004). An introduction to the embryology of Angiosperms. Tata McGraw-Hill Edition, ISBN: 0-07-099434-X.
6. Johri,B.M. (1984). Embryology of Angiosperms. Netherlands: Springer-Verlag. ISBN: 13:978-3-642-69304-5
7. Raghavan, V. (2000). Developmental Biology of Flowering plants. Netherlands: Springer. ISBN: 978-1-4612-7054-6.
8. Mangla,Y., Khanduri, P., Gupta, C.K. 2022. Reproductive Biology of Angiosperms: Concepts and Methods.Cambridge University Press ISBN 978-1-009-16040-7.

Suggestedlaboratoryexercises:

1. Tostudyvarioustypesofovules(slides/photographs/charts).
2. Tostudythedicotandmonocotseeds.
3. Tostudythevarioustypesofembryosacs(charts/photographs).
4. Tostudyseedgermination(hypogeal/epigeal).
5. Topreparereportonpollinationtypes.

GUIDELINESFORPRACTICALEXAMINATION

Time:3Hrs Max.Marks:25

Practical:22 Int.Assessment:3

1. Identify,writeillustratedmorphologicalnoteonspecimenA(Dicot/Monocotseed). (4)
2. Identify,writeillustratednoteonspecimenB(ovule). (4)

3. Write a morphological note onspecimenC(embryosac). (3)
4. Write illustrated note on seed germination(epigeal/hypogeal) (3)
5. Practica lnote-book (3)
6. Viva-voce (3)
7. Project report (2)

Course Title: PHYCOLOGY **Course Code:BOT-DSC-6**
Maj-402

Learning Outcome: To impart knowledge about Algae on various aspects like Classification, Life cycles, Thallus organization and Reproduction in various groups; Evolution, Habitats and Economic importance of algae.

Max.Marks:100 **CourseCredits=**3+1=4

1.	Theory:75	WrittenExam:67	InternalAssessment:8
2.	Practical:25	WrittenExam:22	InternalAssessment:3

Periodsperweek: **1.Theory-Fourof45minutesduration(Threehours)**
2.Practical-Oneoftwohoursduration(Twohours)

InstructionsforPaperSetters:

Question Paper will have four sections.Students will attempt five questions out of total nine comprising of one COMPULSORYquestion consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering thewhole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

InstructionAids:Teaching methodology includes seriesoflectures,making useof charts,transparencies,LCD,Models,slides,practicaldemonstrations,extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT-I **12Hours**

Criteria for algal classification (pigments,reserve food, flagella, chloroplasts,pyrenoids,eye spots, etc.)
Salientfeatures:Cell structure,thallus organization ,reproduction and broad classification of i)Cyanophytaii)Xanthophytaiii)Bacillariophytaiv)Dinophyta.

UNIT-II **11Hours**

Salientfeatures:Cell structure,thallus organization, reproduction and broad classificationof v)Chlorophytavi)Phaeophytavii)Cryptophytaandviii)Rhodophyta.

UNIT-III **11Hours**

Diversity in algal habitats (terrestrial, freshwater and marine). Thallus organization in algae.
Reproduction(vegetative,asexualandsexual),lifecycles.

UNIT-IV **11Hours**

Rhythm sand biolumine scence in dinoflagellates.
Economic importance of algae (algal biofertilizers, algal blooms, algae as food and feed, usesin industries etc.)
AlgaeinBiotechnology

SUGGESTEDREADINGS

1. Ahluwalia, A.S., (Ed.) Phycology: Principles, Processes and Applications, Daya Publishing House, New Delhi, 2003
2. Kumar,H.D.,IntroductoryPhycology,EastWestPress,NewDelhi,1999
3. Lee,R.E.Phycology.4thEd.CambridgeUniversityPress,London.2008.

Suggestedlaboratoryexercises:

1. Morpholgy and microscopic observation of following genera (through permanent slides/photographs/ Fresh material):
   Cyanophyta: Nostoc,
   Chlorophyta: Spirogyra
   Xanthophyta: Vaucheria.
   Phaeophyta: Dictyota
   Rhodophyta: Polysiphonia
2. Projectreportoneconomicimportanceofalgae.

GUIDELINESFORPRACTICALEXAMINATION

Time:3Hrs **Max.Marks:25**

Practical:22 **Int.Assessment:3**

1.	Identifyandwriteillustratedmorphologicalnoteonspecimen AandB.	(8)
2.	Identify the slides / photographs of specimen C and D giving at least two reasons for each.	(6)
3.	Projectreport	(2)
4.	Vivavoce	(3)
5.	PracticalNoteBook.	(3)

Course Title: MYCOLOGY **Course Code:BOT-DSC-7**
Maj-403

Learning Outcome:The objective of the paper is to impart knowledge about Classification, Mycelium, and Reproduction in various groups of Fungi.

Max.Marks:100 **CourseCredits**=3+1=4

1. Theory:75 WrittenExam:67 InternalAssessment:8
2. Practical:25 WrittenExam:22 InternalAssessment:3

Periodsperweek: 1.Theory-Fourof45minutesduration(Threehours)
2.Practical-Oneoftwohoursduration(Twohours)

InstructionsforPaperSetters:

Question Paper will have four sections.Students will attempt five questions out of total nine comprising of one COMPULSORYquestion consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering thewhole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

InstructionAids:Teaching methodology includes seriesoflectures,making useof charts,transparencies,LCD,Models,slides,practicaldemonstrations,extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT–I 10Hours

IntroductiontoFungi: CharacteristicsofFungi;habitats;hyphalstructureandtheir modifications; mode of nutrition; reproduction types and life cycle in Fungi. ClassificationofFungi(comparativestatement)withrecenttrends

UNIT–II 11Hours

General account (mainly structure, reproduction and life cycle) of the following: Oomycota – *Phytopthora, Plasmopara*
Chytidriomycota- *Synchytrium endobioticum*; Zygomycota - *Mucor*
Ascomycota- *Claviceps* and *Morchella*

UNIT–III 11Hours

General account (mainly structure, reproduction and life cycle) of the following: Basidiomycota- *Puccinia* and *Agaricus*
Anamorphic fungi- *Alternaria* and *Cercospora*
Myxomycota-True slime moulds

UNIT–IV **13Hours**

Mycorrhiza: General characters, types and mode of nutrition, mycorrhizal application in agriculture and plant growth; economic importance to humans, ecosystem and environment; negative aspects of Fungi.

SuggestedReadings:

1. Alexopolous, C.J., C.W. Mims and M. Blackwell. Introductory Mycology. 4 th Ed. John Wiley & Sons, New York. 880 pp. 2007.
2. Webster, J. and Webes R. Introduction to Fungi. Cambridge University Press, Cambridge 3rd Edition, 2007.
3. Bos,L.IntroductiontoPlantVirology.Longman,NewYork.160pp.1992
4. 4.Bilgrami,K.S. and R.N.Verma. Physiology of Fungi.2 nd Ed. Vikas Publication House, New Delhi
5. Pelczar M.J. Jr., E.C.S.Chan and N.R.Krieg. Microbiology. 5th Ed. Tata McGraw Hill, New Delhi. 2007.
6. Kendrick,B.TheFifthKingdom.FocusPublishing,Newburyport,M.A.U.S.A,2000

Suggestedlaboratoryexercises:

1. Study of morphological and reproductive structures of following genera :
   Oomycota – *Phytopthora, Plasmopara*
   Chytidriomycota- *Synchytrium endobioticum*
   Zygomycota - *Mucor*
   Ascomycota- *Claviceps, Morchella*
   Basidiomycota- *Puccinia, Agaricus*
   Anamorphic fungi- *Alternaria,Cercospora*

2. ProjectreportofMycorrhizalapplicationinagricultureandplantgrowth

GUIDELINESFORPRACTICALEXAMINATION

Time:3Hrs **Max.Marks:25**

Practical:22 **Int.Assessment:3**

1. Identifyandwriteillustratedmorphologicalnoteonspecimen AandB. (08)
2. IdentifythephotographorslideofspecimenCandDgivingatleasttworeasons. (06)
3. Viva-voce (03)
4. PracticalNote-book. (03)
5. Projectreport (02)

Course Title: CELL BIOLOGY **Course Code:BOT-DSC-8**
Maj-404

Learning Outcome: This paper deals with the prokaryotic and eukaryotic cell, basicstructure of bacterial cell. It provides an insight into chromosomal alterations, non- allelic interactions and it also deals with various aspects of sex linked inheritance.

Max.Marks:100		**CourseCredits=**	3+1=4
1. Theory:75	WrittenExam:67	InternalAssessment:	8
2. Practical:25	WrittenExam:22	InternalAssessment:	3

Periodsperweek: **1.Theory-Fourof45minutesduration(Threehours)**
2.Practical-Oneoftwohoursduration(Twohours)

InstructionsforPaperSetters:

Question Paper will have four sections.Students will attempt five questions out of total nine comprising of one COMPULSORYquestion consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering thewhole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

InstructionAids:Teaching methodology includes seriesoflectures,making useof charts,transparencies,LCD,Models,slides,practicaldemonstrations,extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT–I **12Hours**

Cell: Introduction to cell theory, Structure of prokaryotic and eukaryotic cell, structure of bacterial cell.
Structure and biological roles of carbohydrates, lipids, proteins and nucleic acids, ATP an energy currency molecule.

UNIT–II **11Hours**

Chromosomal alterations: Deletion, Duplication, Inversion, Translocation and their importance. Variations in chromosome number (aneuploidy and polyploidy) and their importance.

UNIT–III **11Hours**

Non allelic gene interactions: Dominant and recessive epitasis, supplementary genes, complementary genes, quantitative or polygenic inheritance, duplicate genes. Allelic gene interactions: Incomplete dominance, co dominance, multiple alleles, pleiotropic genes.

UNIT–IV **11Hours**

Sex linked Inheritance: Characteristics and examples (haemophilia, color blindness), cytoplasmic or extra nuclear inheritance:Mitochondrialand plastidDNA: Plastid Inheritancein *Mirabilis*, Mitochondrial inheritance in Yeast.

SuggestedReadings:

1. Karp, G. (2009). Cell and molecular biology: concepts and experiments. John Wiley& Sons.
2. Singh, S. P., & Tomar, B. S. (2008). Cell biology. Rastogi Publications, Meerut, India.
3. Renu Gupta, Seema Makhija and Ravi Toteja (2018). Cell Biology Practical Manual,Prestige Publishers, New Delhi
4. Iwasa, J, Marshall, W. (2020). Karps's Cell Biology, 9th edition, New Jersey, U.S.A.: John Wiley & Son
5. Majumdar, R., Sisodia, R. (2019). Laboratory Manual of Cell Biology, with referenceto Plant Cells. New Delhi, Delhi: Prestige Publication.
6. Nelson, D.L., Cox, M.M. (2021). Lehninger Principles of Biochemistry, 8thedition. New York, NY: W.H. Freeman and Company.
7. Reven,F.H.,Evert,R.F.,Eichhorn,S.E.(1992).BiologyofPlants.NewYork,NY: **W.H.FreemanandCompany.**

Suggestedlaboratoryexercises:

1. Studyofbacterialcellstructurewiththehelpofslide/photograph/chart.
2. Studyofprokaryoticcellandeukaryoticcellwiththehelpofslide/photograph/chart.
3. ProblemsrelatedtoGeneInteractions.
4. Microchemicaltestsforcarbohydrates(reducing,non-reducingsugarsandstarch).
5. Projectreportonsex-linkedinheritance

GUIDELINESFORPRACTICALEXAMINATION

Time:3Hrs **Max.Marks:25**

Practical:22 **Int.Assessment:3**

1. Problemrelatedtogeneinteraction(tobeannouncedbytheexaminer). (06)
2. Identify the photograph or slide of specimen A (bacterial/ prokaryotic/ eukaryotic cell) and write a note on it along with a well-labelled diagram. (04)
3. Writeprocedureofmicrochemicaltests(tobeannouncedbytheexaminer). (04)
4. Viva-voce (03)
5. PracticalNote-book. (03)
6. Projectreport (02)

CHEMISTRY

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MAJOR SUBJECT AS PER TABLE 1 AND TABLE 2

Table 1						Table 2					
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**	**Total Credits**
III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	24	III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	24
		(CHM-DSC-4 MAJ-302)	BASIC CONCEPTS IN CHEMISTRY - 1	4				Major - 3	To be chosen by the student	4	
		Minor – 3	To be chosen by the student	4				Minor – 3	To be chosen by the student	4	
		Minor - 4	To be chosen by the student	4				Minor - 4	To be chosen by the student	4	
		IDC-3	To be chosen from the available options	3				IDC-3	To be chosen from the available options	3	
		(CHM-SEC-3-301)	Hands on chemistry Lab skills-3/ to be chosen from available options	3				(CHM-SEC-3-301)	Hands on chemistry Lab skills-3/ to be chosen from available options	3	
		ENG-3	Compulsory	2				ENG-3	Compulsory	2	
IV	L-200	(CHM-DSC-5 MAJ-401/MIN-302)	GENERAL CHEMISTRY – 4	4	24	IV	L-200	(CHM-DSC-5 MAJ-401/MIN-302)	GENERAL CHEMISTRY – 4	4	24
		(CHM-DSC-6 MAJ-402/MIN-401)	GENERAL CHEMISTRY – 5	4				(CHM-DSC-6 MAJ-402/MIN-401)	GENERAL CHEMISTRY – 5	4	

		(CHM-DSC-7 MAJ-403)	BASIC CONCEPTS IN CHEMISTRY - 2	4				Major - 5	To be chosen by the student	4	
		(CHM-DSC-8 MAJ-404)	BASIC CONCEPTS IN CHEMISTRY - 3	4				Major - 6	To be chosen by the student	4	
		Minor - 5	To be chosen by the student	4				Minor - 5	To be chosen by the student	4	
		ENG-4	Compulsory	2				ENG-4	Compulsory	2	
		VAC-3	To be chosen from the common basket	2				VAC-3	To be chosen from the common basket	2	

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MINOR SUBJECT AS PER TABLE 1 AND TABLE 2

Table 1					Table 2				
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**
III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4
	L-200	(CHM-DSC-5-MIN-302/MAJ-401)	GENERAL CHEMISTRY – 4	4		L-200	(CHM-DSC-5-MIN-302/MAJ-401)	GENERAL CHEMISTRY – 4	4
IV	L-200	(CHM-DSC-6 MIN-401/MAJ-402)	GENERAL CHEMISTRY – 5	4	IV	L-200	(CHM-DSC-6 MIN-401/MAJ-402)	GENERAL CHEMISTRY – 5	4

Note:-

1. In the first semester, studentsare offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awardeda BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3rd semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

Semester	**III**
Code of Paper	**CHM-DSC-3-MAJ/MIN-301**
Title of Paper	**General Chemistry-3**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1 or Table 2
2. Those who have opted for chemistry as a Minor subject in Table 1 or Table 2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of s-block elements and noble gases
2. Chemistry of compounds of alkali metals, alkaline earth metals and noble gases
3. Important aspects of Alkenes, Cycloalkenes, alkynes and dienes
4. Chemical reactions of alkenes, alkynes, cycloalkanes, and dienes along with their mechansims.

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 3
Syllabus for Theory Paper
CHM-DSC-3-MAJ/MIN-301
General Chemistry-3

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Chemistry of s-Block Elements (7 Hrs.)

Comparative study, diagonal relationships, salient features of hydrides, solvation and complexation tendencies, including their function in biosystems, and an introduction to alkyls and aryls.

Chemistry of Noble Gases (8 Hrs.)

Occurrence and uses, rationalisation of inertness of noble gases, Clathrates; preparation and properties of XeF_2, XeF_4 and XeF_6; Nature of bonding in noble gas compounds (Valence bond treatment and MO treatment for XeF_2), Molecular shapes of noble gas compounds (VSEPR theory).

UNIT II: ORGANIC CHEMISTRY(15 Hrs.)

Alkenes andCycloalkenes (8 Hrs.)

Nomenclature of alkenes, methods of formation, mechanisms of dehydration of alcohols and dehydrohalogenation of alkyl halides, regioselectivity in alcohol dehydration. Saytzeff's Rule, Hofmann elimination, physical properties and relative stabilities of alkenes, Chemical reactions of alkenes–mechanisms involved in hydrogenation, electrophilic and free radical additions, Markownikoff's rule, hydroboration-oxidation, oxymercuration-reduction. Epoxidation, ozonolysis, hydration, hydroxylation and oxidation with $KMnO_4$, Polymerisation of alkenes, Substitution at the allylic and vinylic positions of alkenes, Methods of formation, conformation and chemical reactions of cycloalkenes.

Dienes andAlkynes (7 Hrs.)

Nomenclature and classification of dienes: Isolated, conjugated and cumulated dienes. Structure of allenes and butadiene, methods of formation, and polymerisation. Chemical reactions – 1,2 and 1,4 additions, Diels-Alder reaction.Nomenclature, structure and bonding in alkynes. Methods of formation.Chemical reactions of alkynes, acidity of alkynes. Mechanism of electrophilic and nucleophilic addition reactions, hydroboration-oxidation, metal-ammonia reductions, oxidation and polymerisation.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Thermochemistry (7 Hrs.)

Standard state, standard enthalpy of formation-Hess's Law of constant Heat Summation and its applications. Heat of reaction at constant pressure and at constant volume.Enthalpy of neutralisation.Bond dissociation energy and its calculation from thermochemical data, temperature dependence of enthalpy.Kirchhoff's equation.

Thermodynamics-II (8 Hrs.)

Second Law of Thermodynamics: Need for the law, different statements of the law, Carnot cycle and its efficiency, Carnot theorem. Thermodynamic scale of temperature, Concept of Entropy: Entropy as a state function, entropy as a function of V & T, entropy as a function of P & T, entropy change in physical change, Clausius inequality, entropy as a criterion of spontaneity and equilibrium. Entropy change in ideal gases and mixing of gases.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.

18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:PearsonEducation,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising of **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. Theuse of scientific calculators is not allowed.

Semester – 3
Syllabus for Practical Paper
CHM-DSC-3-MAJ/MIN-301
General Chemistry-3

End semester = 22 **Credit - 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

I. INORGANIC CHEMISTRY

1. Gravimetric analysis
    a. Ni^{2+} as Ni-DMG complex
    b. Cu^{2+} as copper(I)-thiocyanate
    c. Ba^{2+} as barium(II) sulphate

II. ORGANIC CHEMISTRY

2. Qualitative Analysis:
    a. Investigation and characterisation of simple organic compounds for the detection of elements (N, S and halogens) present in the given organic compound. (Alcohols, carboxylic acids, carbonyl compounds, alkanes, amines, nitro, thio- and halo-compounds) and Identification of an unknown organic compound.

III. PHYSICAL CHEMISTRY

3. Thermochemistry
    a. To determine the solubility of benzoic acid at different temperatures and to determine the ph of the dissolution process.
    b. To determine the enthalpy of neutralisation of a weak acid/weak base versus strong base/strong acid, and determine the enthalpy of ionisation of the weak acid/weak base.

Books/Literature suggested

1. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
2. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
4. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
5. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

Semester	**III**
Code of Paper	**CHM-DSC-4-MAJ-302**
Title of Paper	**Basic Concepts in Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table-2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects and principles of acids and bases
2. Chemistry of carbohydrates
3. Basic concepts of quantum chemistry

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 3
Syllabus for Theory Paper
CHM-DSC-4-MAJ-302
Basic Concepts in Chemistry-1

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Non-aqueousSolvents(5 Hrs.)

Introduction to non-aqueous solvents, their classification, Physical properties of a solvent, types of solvents and their general characteristics, Solvent system concept and its failure, reactions in non-aqueous solvents with reference to liquid NH_3, liquid SO_2, HF and H_2SO_4.

AcidsandBases (5 Hrs.)

Introduction, Arrhenius concept,Bronsted-Lowry,theLux-Flood,solventsystemandLewisconceptofacidsandbases.

HardandSoftAcidsandBases(HSAB)(5 **Hrs.)**

Classification of acids and bases as hard and soft, Pearson's HSAB concept, acid-base strength and hardness and softness, Symbiosis, theoretical basis of hardness and softness, electronegativity and hardness and softness, Limitation of HSAB principle

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Carbohydrates(9 Hrs.)

Carbohydrate Introduction: brief descriptions (structure and properties) of: (i) Monosaccharides: glucose, fructose, and galactose, (ii) Disaccharides: sucrose, lactose, and maltose(iii) Polysaccharides: starch, cellulose, and glycogen. Monosaccharides,mechanismofosazoneformation,interconversionofglucoseand fructose, chain lengthening and chain shortening of aldoses. Configuration of monosaccharides.Erythro and threodiastereomers.Conversionofglucoseintomannose.Formationofglycosides,ethersandesters.D eterminationof ring size of monosaccharides. Structure of D (+)-glucose.Mechanism of mutarotation.

Structureofriboseanddeoxyribose (6 Hrs.)

Concept of reducing and non-reducing sugars, analysis of carbohydrates (Molisch test, Benedict's test, and Barfoed's test); carbohydrate metabolism, glycemic index, common diseases related to carbohydrate metabolism (diabetes, galactosemia, lactose intolerance, etc.).

UNIT II: PHYSICAL CHEMISTRY (15 Hrs.)

Elementary Quantum Mechanics-I (15 Hrs.)

Black-body radiation, Planck's radiation law, photoelectric effect, heat capacity of solids, Bohr's model of the hydrogen atom (no derivation) and its defects, Compton effect.De Broglie hypothesis, the Heisenberg's uncertainty principle, Sinusoidal wave equation, Hamiltonian operator, Schrodinger wave equation and its importance, physical interpretation of the wave function, postulates of quantum mechanics, particle in a one dimensional box,Schrodinger wave equation for H-atom, separation into three equations (without derivation), quantum numbers and their importance, hydrogen like wave functions, radial wave functions, angular wave functions.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWiley and Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6th edition, Pubs: Prentice-Hall, 1992.
18. Solomons, T.W., Fryhle, C.B., Organic Chemistry; 9th edition, Pubs: Wiley India, 2007.
19. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6th edition, Pubs: Pearson Education, 2008.
20. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985.
21. Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry Part B: Reactions and Synthesis; 5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 3
Syllabus for Practical Paper
CHM-DSC-4-MAJ-302
Basic Concepts in Chemistry-1

End semester = 22 **Credit - 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

I. INORGANIC CHEMISTRY

1. Preparation of the following compounds
   a. Cuprous Chloride,Cu_2Cl_2
   b. Manganese(III) phosphate, $MnPO_4.H_2O$
   c. Turnbull's Blue and Prussian Blue

II.ORGANIC CHEMISTRY

2. Qualitative Analysis:

   a. Investigation, characterisation and identification of simple carbohydrates. Analysis of carbohydrates (Molisch test, Benedict's test, Fehling's test and Barfoed's test).

   b. Separation of a mixture of 2 sugars by ascending paper chromatography

III. PHYSICAL CHEMISTRY

3. pH metry

   a. Preparation of buffer solutions of different pH: i. Sodium acetate-acetic acid, ii. Ammonium chloride-ammonium hydroxide

   b. pH metric titration of (i) strong acid vs. strong base, (ii) weak acid vs. strong base.

   c. Determination of the dissociation constant of a weak acid.

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2nd edition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

Semester	**III**
Code of Paper	**CHM-DSC-5-MIN-302/MAJ-401**
Title of Paper	**General Chemistry-4**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. Those who have opted for chemistry as a Minor subject in Table 2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects group 13 and 14 elements
2. Chemistry of compounds of B and C and their group elements
3. Chemistry of arenes, and alkyl ad aryl halides
4. Concept of aromaticity
5. Concepts of thermodynamics and chemical equilibrium

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 3
Syllabus for Theory Paper
CHM-DSC-5-MIN-302/MAJ-401
General Chemistry-4

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Group 13 Elements (8 Hrs)

Comparative study (including diagonal relationship) of groups 13 elements, Study of the following compounds with emphasis on structure, bonding, preparation, properties and uses: Boric acid and borates, boron nitrides, boron hydrides (diborane), boron halides, alumina.

Group 14 Elements (7 Hrs)

Comparative study (including diagonal relationship) of groups 14 elements, , allotropic forms of carbon, graphitic compounds, compounds like hydrides, oxides, oxyacids and halides of groups 14, fullerenes, carbides, fluorocarbons.

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Aromaticity and Arenes (8 Hrs.)

Aromaticity: The Huckel's rule, aromatic ions.

Nomenclature of benzene derivatives, aryl group, aromatic nucleus and side chain, Structure of benzene: Molecular formula and Kekule structure. Stability and carbon-carbon bond lengths of benzene, resonance structure, and MO picture.

Aromatic electrophilic substitution-General pattern of the mechanism, role of σ and π-complexes. Mechanism of nitration, halogenation, sulphonation, mercuration and Friedel-Crafts reaction.Energy profile diagrams. Activating and deactivating substituents, orientation and ortho/para ratio.Side chain reactions of benzene derivatives. Methods of formation and chemical reactions of alkylbenzenes, alkynylbenzenes and biphenyl.

Alkyl and Arylhalides(7 Hrs.)

Nomenclatureandclassesofalkylhalides,methodsofformation,chemicalreactions.Mechanism sofnucleophilicsubstitutionreactionsofalkylhalides,SN2andSN1reactionswithenergyprofiled iagrams.

Methodsofformationofarylhalides,nuclearandsidechainreactions.Theaddition-eliminationandtheelimination-additionmechanismsofnucleophilicaromaticsubstitutionreactions.

Relativereactivitiesofalkylhalidesvs.allyl,vinylandarylhalides.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Thermodynamics-III (8 Hrs.)

Third Law of Thermodynamics: Nernst heat theorem, statement and concept of residual entropy, evaluation of absolute entropy from heat capacity data. Gibbs and Helmholtz functions; Gibbs function (G) and Helmholtz functions (A) as thermodynamic quantities, A & G as criteria for thermodynamic equilibrium and spontaneity, and their advantage over entropy change. Variation of G and A with P, V and T.

Chemical Equilibrium(7 Hrs.)

Equilibrium constant and free energy, Thermodynamic derivation of law of mass action, Le-Chatelier's principle, Reaction isotherm and Reaction isochore-Clapeyron equation and Clausius–Clapeyron equation, applications.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
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14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.

19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:Pearson Education,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester – 3
Syllabus for Practical Paper
CHM-DSC-5-MIN-302/MAJ-401
General Chemistry-4

End semester = 22 **Credit - 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

I. INORGANIC CHEMISTRY

1. Iodometry and Iodimetry
    a. Determination of Cu^{2+}
    b. Determination of vitamin C
    c. Determination of chlorine in a sample of bleaching powder

II. ORGANIC CHEMISTRY

2. Preparation of the following compounds via Aromatic electrophilic substitution:-
    a. 1,3-dinitrobenzene from nitrobenzene
    b. 4-nitroacetanilide from acetanilide
    c. 4-iodoaniline from aniline

III. PHYSICAL CHEMISTRY

3. Refractive indices

    a. Determine the Refractive indices of given liquids (water, acetone, methanol, ethylacetate,cyclohexane) by Abbe's refractometer and calculate their specific refractions.

4. Molecular Weight Determination

    a. Determination of molecular weight of a non–volatile solute by the Rast method.

    b. Determinationoftheapparentdegreeofdissociationofanelectrolyte(e.g.NaCl) inaqueous solution of the substance.

Books/Literature suggested

1. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
2. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
4. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.

Semester	**IV**
Code of Paper	**CHM-DSC-5-MAJ-401/MIN-302**
Title of Paper	**General Chemistry-4**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1 or Table 2
2. Those who have opted for chemistry as a Minor subject in Table 1 or Table 2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects group 13 and 14 elements
2. Chemistry of compounds of B and C and their group elements
3. Chemistry of arenes, and alkyl ad aryl halides
4. Concept of aromaticity
5. Concepts of thermodynamics and chemical equilibrium

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 4
Syllabus for Theory Paper
CHM-DSC-5-MAJ-401/MIN-302
General Chemistry-4

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Group 13 Elements (8 Hrs)

Comparative study (including diagonal relationship) of groups 13 elements, Study of the following compounds with emphasis on structure, bonding, preparation, properties and uses: Boric acid and borates, boron nitrides, boron hydrides (diborane), boron halides, alumina.

Group 14 Elements (7 Hrs)

Comparative study (including diagonal relationship) of groups 14 elements, , allotropic forms of carbon, graphitic compounds, compounds like hydrides, oxides, oxyacids and halides of groups 14, fullerenes, carbides, fluorocarbons.

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Aromaticity and Arenes (8 Hrs.)

Aromaticity: The Huckel's rule, aromatic ions.

Nomenclature of benzene derivatives, aryl group, aromatic nucleus and side chain, Structure of benzene: Molecular formula and Kekule structure. Stability and carbon-carbon bond lengths of benzene, resonance structure, and MO picture.

Aromatic electrophilic substitution-General pattern of the mechanism, role of σ and π-complexes. Mechanism of nitration, halogenation, sulphonation, mercuration and Friedel-Crafts reaction.Energy profile diagrams. Activating and deactivating substituents, orientation and ortho/para ratio.Side chain reactions of benzene derivatives. Methods of formation and chemical reactions of alkylbenzenes, alkynylbenzenes and biphenyl.

Alkyl and Arylhalides(7 Hrs.)

Nomenclatureandclassesofalkylhalides,methodsofformation,chemicalreactions.Mechanism sofnucleophilicsubstitutionreactionsofalkylhalides,SN2andSN1reactionswithenergyprofiled iagrams.

Methodsofformationofarylhalides,nuclearandsidechainreactions.Theaddition-eliminationandtheelimination-additionmechanismsofnucleophilicaromaticsubstitutionreactions.

Relativereactivitiesofalkylhalidesvs.allyl,vinylandarylhalides.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Thermodynamics-III (8 Hrs.)

Third Law of Thermodynamics: Nernst heat theorem, statement and concept of residual entropy, evaluation of absolute entropy from heat capacity data. Gibbs and Helmholtz functions; Gibbs function (G) and Helmholtz functions (A) as thermodynamic quantities, A & G as criteria for thermodynamic equilibrium and spontaneity, and their advantage over entropy change. Variation of G and A with P, V and T.

Chemical Equilibrium(7 Hrs.)

Equilibrium constant and free energy, Thermodynamic derivation of law of mass action, Le-Chatelier's principle, Reaction isotherm and Reaction isochore-Clapeyron equation and Clausius–Clapeyron equation, applications.

Books/Literature suggested

22. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
23. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
24. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
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26. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
27. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
28. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
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31. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
32. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
33. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
34. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
35. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
36. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
37. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
38. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
39. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
40. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:PearsonEducation,2008.
41. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
42. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th edition, Pubs: Springer, 2007

Instructions for the examiner

4. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
5. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
6. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

3. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
4. The use of scientific calculators is not allowed.

Semester – 4
Syllabus for Practical Paper
CHM-DSC-5-MAJ-401/MIN-302
General Chemistry-4

End semester = 22 **Credit - 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

II. INORGANIC CHEMISTRY

2. Iodometry and Iodimetry
    a. Determination of Cu^{2+}
    b. Determination of vitamin C
    c. Determination of chlorine in a sample of bleaching powder

II. ORGANIC CHEMISTRY

5. Preparation of the following compounds via Aromatic electrophilic substitution:-
    a. 1,3-dinitrobenzene from nitrobenzene
    b. 4-nitroacetanilide from acetanilide
    c. 4-iodoaniline from aniline

III. PHYSICAL CHEMISTRY

6. **Refractive indices**

    a. Determine the Refractive indices of given liquids (water, acetone, methanol, ethylacetate,cyclohexane) by Abbe's refractometer and calculate their specific refractions.

7. **Molecular Weight Determination**

    a. Determination of molecular weight of a non–volatile solute by the Rast method.

    b. Determinationoftheapparentdegreeofdissociationofanelectrolyte(e.g.NaCl) inaqueous solution of the substance.

Books/Literature suggested

5. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
6. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
7. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
8. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.

Semester	**IV**
Code of Paper	**CHM-DSC-6-MAJ-402/MIN-401**
Title of Paper	**General Chemistry-5**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1 or Table 2
2. Those who have opted for chemistry as a Minor subject in Table 1 or Table 2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects group 15, 16 and 17 elements
2. Chemistry of compounds of N, As, Sb, O, S, Se, F, Cl, and Br
3. Chemistry of alcohols, phenols, ethers and epoxides
4. Concepts of solid state chemistry

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 4
Syllabus for Theory Paper
CHM-DSC-6-MAJ-402/MIN-401
General Chemistry-5

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Group 15 Elements (5 Hrs.)

Types of covalence in nitrogen, stereochemistry, chemical reactivity, dinitrogen complexes (basic idea only), hydrides of nitrogen, nitrogen halides, oxides and oxoacids, Phosphorus, As, Sb & Bi: Stereochemistry of their compounds, production of elemental P and its allotropic forms, hydrides, halides, oxides and oxy-acids

Group 16 Elements (5 Hrs.)

Chemical Reactivity, group trends and stereochemistry of Oxygen, S, Se and Te Family, dioxygen as a ligand (basic idea only), structure of O_3 and H_2O_2, clathrate hydrates allotropic forms of S and Se, structures of halides, oxides and oxyacids of S, Se and Te, S-N compounds (neutral) Polyatomic cations of S, Se and Te.

Group 17 Elements (5 Hrs.)

Comparative study of group 17 elements, basic properties of halogens, Oxides and oxoacids of Chlorine, interhalogen compounds, polyhalide ions, and pseudohalogens.

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Alcohols and Phenols(9 Hrs.)

Classification and nomenclature, Monohydric alcohols-Nomenclature, methods of formation by reduction of aldehydes, ketones, carboxylic acids and esters. Hydrogen bonding, Acidic nature, Reactions of alcohols, Dihydric and Trihydric alcohols- Nomenclature, methods of formation, chemical reactions of vicinal glycols and glycerol,Preparation of phenols, physical properties and acidic character, Comparative acidic strengths of alcohols and phenols, resonance stabilization of phenoxide ion, Reactions of phenols-electrophilic aromatic substitution, acylation and carboxylation, Mechanisms of Fries rearrangement, Claisen rearrangement, Gattermann synthesis, and Reimer-Tiemann reaction.

Ethers and Epoxides(6 Hrs.)

Nomenclature of ether and methods of their formation, physical properties, Chemical reaction-cleavage and autoxidation, Ziesel's method,Synthesis of epoxides, Acid and base-catalyzed ring opening of epoxides, orientation of epoxide ring opening, reactions of Grignard and organolithium reagents with epoxides.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Solid State(15 Hrs.)

Definition of space lattice, unit cell and Miller Indices

Laws of Crystallography – (i) Law of Constancy of Interfacial Angles, (ii) Law of Rationality of Indices, (iii) Law of Symmetry.Symmetry elements in crystals.
X-ray diffraction by crystals. Derivation of the Bragg equation.Determination of the crystal structure of NaCl, KCl and CsCl (Laue's method and powder method).
Applications of Powder diffraction for structure determination, Thermal and photochemical reaction in the solid state

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:PearsonEducation,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:ReactionsandSynthesis;5th editio n, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester – 4

Syllabus for Practical Paper

CHM-DSC-6-MAJ-402/MIN-401

General Chemistry-5

End semester = 22 **Credit - 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

I. INORGANIC CHEMISTRY

1. Preparation of the crystals of the following compounds
   a. Potash alum
   b. Chrome alum
   c. Mohr's salt

II. ORGANIC CHEMISTRY

2. Williamson ether synthesis using cresol and chloroacetic acid.
3. Determination of the amount of phenol in the given sample by acetylation and bromination methods, and perform acid-base and iodimetric titrations.
4. To observe the solubility of alcohols relative to their chemical structure
5. Tests to distinguish primary, secondary and tertiary alcohols and a colour test for phenol

III. PHYSICAL CHEMISTRY

a) Calibration of Thermometer: 80-82° (Naphthalene), 113-114° (acetanilide), 132.5-133^{o}(Urea), 100° (distilled Water)

b) Determination of Critical Micelle Concentration (CMC) of a Cationic Surfactant by Conductivity Measurements

c) Determination of the Critical Micelle Concentration (CMC) of an Anionic Surfactant by Conductivity Measurements

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2ndedition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.

Semester	**IV**
Code of Paper	**CHM-DSC-7-MAJ-403**
Title of Paper	**Basic Concepts in Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects oxidation and reduction
2. Chemistry methods and techniques used in metallurgy
3. Chemistry of organic polymers
4. Concepts of liquid state crystals and colloids

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 4
Syllabus for Theory Paper
CHM-DSC-7-MAJ-403
Basic Concepts in Chemistry-2

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

OxidationandReduction (7 Hrs)

Introduction, oxidation-reduction reactions (electronic and oxidation number concept), Balancing of redox reactions, Use of redox potential data – analysis of redox cycle, redox stability in water – Frost, Latimer and Pourbaixdiagrams.

Metallurgy (8 Hrs)

General Principles of Metallurgy: Chief modes of occurrence of metals based on standard electrode potentials. Ellingham diagrams for the reduction of metal oxides using carbon and carbon monoxide as reducing agents. Electrolytic Reduction, Hydrometallurgy. Methods of purification of metals: Electrolytic process, Parting process, van Arkel-de Boer process and Mond's process, Zone refining

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Polymer Chemistry(15 Hrs.)

Basic characteristics of polymers, melting point of polymers, glass transition temperature, molecular weight and molecular weight distribution.

Organic polymers: Synthetic polymers and their syntheses via free radical polymerisation, polymerisation by ionic initiators, Ziegler-Natta type polymerisation, atom transfer radical polymerisation, ring opening polymerisation, olefin metathesis and polymerisation, group transfer polymerisation, condensation polymerisation.

Polyesters,polyamides,phenolformaldehyderesins,ureaformaldehyde resins, epoxy resins and polyurethanes.

Naturalandsyntheticrubbers.

UNIT II: PHYSICAL CHEMISTRY (15 Hrs.)

Liquid state and liquid crystals(7 Hrs.)
Introduction, Intermolecular forces, Structure of Liquids, Theories of Liquid state, Structural differences between Solids, Liquids and Gases.Liquid crystals, classification and structure of thermotropic phases (Nematic, Smectic and Cholesteric phases), applications of liquid crystals.
Colloidal State (8Hrs.)
Definition of colloids, classification of colloids, Solids in liquids (sols): Properties –kinetic, optical and electrical; stability of colloids, protective action, Hardy-Schulze rules, gold number, Liquids in liquids (emulsions): Types of emulsions, preparation. Emulsifier, Liquids in solids (gels): Classification, preparation and properties, inhibition, general applications of colloids.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

3. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
4. The use of scientific calculators is not allowed.

Semester – 4

Syllabus for Practical Paper

CHM-DSC-7-MAJ-403

Basic Concepts in Chemistry-2

End semester = 22 **Credit - 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

I. INORGANIC CHEMISTRY

1. Precipitation Titration: Volhard's method (for Cl^- ions)
2. Gravimetric analysis
   a. Ag^+ as AgCl
   b. Fe^{2+} as Iron(III) oxide
   c. Ba^{2+} as barium(II) sulphate

II. ORGANIC CHEMISTRY

3. Investigate and identify a variety of polymers used in everyday materials by testing their density.
4. Study of properties of cross-linked polymers.
5. To study the effect of varying concentrations of the poly(vinyl alcohol) and borax solution on slime

III. PHYSICAL CHEMISTRY

Determination of boiling points; Ethanol 78°, Cyclohexane 81.4^o, Toluene 110.6^o, Benzene 80°.

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2ndedition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

7.

Semester	**IV**
Code of Paper	**CHM-DSC-8-MAJ-404**
Title of Paper	**Basic Concepts in Chemistry-3**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects chain like inorganic compounds 2. Chemistry of inorganic polymers 3. Chemistry of amino acids 4. Chemistry of peptides and proteins 5. Chemistry of nucleic acids	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Syllabus for Theory Paper
CHM-DSC-8-MAJ-404
Basic Concepts in Chemistry-3

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Inorganic Chains (8 Hrs.)

Chains: Catenation including heterocatenation, intercalation compounds, silicate minerals, aluminate minerals, one-dimensional sulphur-nitrogen conductor.

SiliconesandPhosphazenes(7 Hrs.)

Structure and bonding in siloxanes, triphosphazenes, phosphazenes, Synthesis, properties and applications of polysiloxanes (Silicones), polysilanes and polyphosphazenes

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Chemistry of AminoAcids (5 Hrs.)

Classification, structure and stereochemistry of amino acids, acid-base behaviour, isoelectric point and electrophoresis, Preparation and reactions of α-amino acids.

Chemistry of Peptides and Proteins (5 Hrs.)

Structure and nomenclature of peptides and proteins, Classification of proteins, Peptide structure determination, end group analysis, selective hydrolysis of peptides, Classical peptide synthesis, solid–phase peptide synthesis, Structures of peptides and proteins, Levels of protein structure, Protein denaturation/renaturation.

Chemistry of Nucleic Acids (5 Hrs.)

Introduction, Constituents of nucleic acids, Ribonucleosides and ribonucleotides, the double helical Structure of DNA.

UNIT III: PHYSICAL CHEMISTRY (15 Hrs.)

Elementary Quantum Mechanics-II (15 Hrs.)

Molecular orbital theory, basic ideas – criteria for forming M.O. from A.O., construction of M.O.'s by LCAO – H_2^+ ion. Calculation of energy levels from wave functions, physical picture of bonding and antibonding wave functions, concept of σ, σ∗, π, π* orbitals and their characteristics. Hybrid orbitals – sp, sp^2, sp^3, calculation of coefficients of A.O.'s used in these hybrid orbitals. Introduction to the valence bond model of H_2, comparison of M.O. and V.B. models.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-

WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rdedition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30thedition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Syllabus for Practical Paper
CHM-DSC-8-MAJ-404
Basic Concepts in Chemistry-3

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1.Preparation of crystals of ferrous oxalate and estimation of iron and oxalate.

2. Preparation of double salt $CoSO_4(NH_4)_2SO_4.6H_2O$ and estimation of copper and sulphate.

3. Preparation of tris (thiourea) Copper (I) sulphate (estimation of copper and sulphate.

II.ORGANIC CHEMISTRY

4. Qualitative Analysis:
    a. Separation of a mixture of 2 amino acids by radial &ascending paper chromatography and to determine the R_f values.
    b. Titration curve of an amino acid & determination of its pK_a and isoelectric point.
    c. To determine the acid-base behaviour of amino acids by the volumetric method.

III. PHYSICAL CHEMISTRY

5. Chemical Kinetics
    a. Study the kinetics of the following reactions.
        i. Acid hydrolysis of methyl acetate with hydrochloric acid.
        ii. Saponification of ethyl acetate.

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2nd edition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

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COMPUTER SCIENCE AND APPLICATION

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE/ DOUBLE MINOR (Table 1*)
and STUDENTS
OPTING FOR MULTI- DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)
BSc Computer Science subjects; For session 2026-27

Table 1*							Table 2*						
Sem	Level	Course Code	Course Name	Credits (Th+Pr)		Total Credits	Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)		Total Credits
I	L-100	CSA-DSC-1(Maj/Min)-101	Computer Fundamentals and PC Software	2+2	4(x3)	24	I	L-100	CSA-DSC-1(Maj/Min)-101	Computer Fundamentals and PC Software	2+2	4(x3)	24
		CSA-MDC-101/201/301	E-Commerce	3+0	3				CSA-MDC-101/201/301	E-Commerce	3+0	3	
		CSA-MDC-102/202/302	Management Information System	3+0					CSA-MDC-102/202/302	Management Information System	3+0		
		CSA-MDC-103/203/303	Web Development/ Designing	0+3					CSA-MDC-103/203/303	Web Development/ Designing	0+3		
		CSA-MDC-104/204/304	Introduction to Information Technology	0+3					CSA-MDC-104/204/304	Introduction to Information Technology	0+3		
		CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0					CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0		
		CSA-MDC-106/206/306	Data Analysis using Excel	0+3					CSA-MDC-106/206/306	Data Analysis using Excel	0+3		
		CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0	2				CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0	2	
		CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0					CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0		
		CSA-VAC-103/203/303/403	Digital Empowerment	2+0					CSA-VAC-103/203/303/403	Digital Empowerment	2+0		
		CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0					CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0		
		CSA-SEC-101/201/301	Computer Fundamentals and Personal	0+3	3				CSA-SEC-101/201/301	Computer Fundamentals and	0+3	3	

			Computing Software							Personal Computing Software			
									CSA-SEC-102/202/302	Desktop Publishing using GIMP	0+3		
									CSA-SEC-103/203/303	Backend Web Development	0+3		
		CSA-SEC-102/202/302	Desktop Publishing using GIMP	0+3					CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3		
		CSA-SEC-103/203/303	Backend Web Development	0+3					CSA-SEC-105/205/305	Digital Marketing	0+3		
		CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3					CSA-SEC-106/206/306	Computer Literacy	0+3		
									CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3		
									CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3		
		CSA-SEC-105/205/305	Digital Marketing	0+3					CSA-SEC-109/209/309	Application Software for Office Automation	0+3		
		CSA-SEC-106/206/306	Computer Literacy	0+3									
		CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3									
		CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3					CSA-SEC-110/210/310	Python Programming	0+3		
		CSA-SEC-109/209/309	Application Software for Office Automation	0+3									
		CSA-SEC-110/210/310	Python Programming	0+3									
		(AEC-1-101)	Eng-1	2					(AEC-1-101)	Eng-1	2		
		(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2					(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2		
II	L-100	CSA-DSC-2(Maj/Min)-201	Programming in C	2+2	4 (x3)	24	II	L-100	CSA-DSC-2(Maj/Min)-201	Programming in C	2+2	4 (x3)	2
		CSA-MDC-101/201/301	E-Commerce	3+0	3				CSA-MDC-101/201/301	E-Commerce	3+0	3	
		CSA-MDC-102/202/302	Management Information System	3+0					CSA-MDC-102/202/302	Management Information System	3+0		
		CSA-MDC-103/203/303	Web Development/ Designing	0+3					CSA-MDC-103/203/303	Web Development/ Designing	0+3		
		CSA-MDC-104/204/304	Introduction to Information Technology	0+3					CSA-MDC-104/204/304	Introduction to	0+3		

		CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0						Information Technology			
		CSA-MDC-106/206/306	Data Analysis using Excel	0+3					CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0		
									CSA-MDC-106/206/306	Data Analysis using Excel	0+3		
		CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0					CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0		
		CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0	2				CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0		
		CSA-VAC-103/203/303/403	Digital Empowerment	2+0					CSA-VAC-103/203/303/403	Digital Empowerment	2+0	2	
		CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0					CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0		
		CSA-SEC-1101/201/301	Computer Fundamentals and Personal Computing Software	0+3					CSA-SEC-101/201/301	Computer Fundamentals and Personal Computing Software	0+3		
		CSA-SEC-102/202/302	Desktop Publishing using GIMP	0+3					CSA-SEC-102/202/302	Desktop Publishing using GIMP	0+3		
		CSA-SEC-103/203/303	Backend Web Development	0+3	3				CSA-SEC-103/203/303	Backend Web Development	0+3		
		CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3					CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3	3	
		CSA-SEC-105/205/305	Digital Marketing	0+3					CSA-SEC-105/205/305	Digital Marketing	0+3		
		CSA-SEC-106/206/306	Computer Literacy	0+3					CSA-SEC-106/206/306	Computer Literacy	0+3		
		CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3					CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3		
		CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3					CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3		
									CSA-SEC-	Application	0+3		

		CSA-SEC-109/209/309	Application Software for Office Automation	0+3					109/209/309	Software for Office Automation			
		CSA-SEC-110/210/310	Python Programming	0+3					CSA-SEC-110/210/310	Python Programming	0+3		
		(AEC-2-201)	Eng-2	2					(AEC-2-201)	Eng-2	2		
		(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2					(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2		
The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II													
The studentexitingtheprogramme after oneyear, securing48credits,willbeawardedUG Certificateprovidedhe/shesecures 4creditsfromsummer internship/community work.													

Note: Student need to select one MDC, VAC, SEC out of given options, respectively. NO MDC, SEC, VAC will be repeated. All MDC, SEC and VAC are Level-100

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.C, SEC and VAC are Level-100**

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Paper Title: Programming in C++
Paper Code:CSA-DSC-3(Maj/Min)-301

Th	Pr	Cr	TheoryExternalMarks:45
2	2	4	TheoryInternalMarks:5
Level: L- 200			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical External hours: 60

Time Duration: 3 Hrs.

Objectives: By the end of the course, students will be able to write C++ programs using the more esoteric language features, utilize Object Oriented techniques to design C++ programs, use the standard C++ library, and explore advanced C++ techniques.

Learning Outcome:

On successful completion of the course, students will be able to:

- Understand the features and importance of object oriented programming concepts.
- Use the concepts like inheritance, polymorphism, and dynamic binding to develop codes.
- Gets better understanding of developing reusable components using C++ programming.
- Develop of large software that needs bottom up and better modular approach.

Note :

- The Question Paper will consist of Four Units.
- Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
- The students are required to attempt ONE question from each Unit and the Compulsory question.
- All questions carry equal marks unless specified.

UNIT - I

Principles of Object-Oriented Programming (OOP): Introduction to OOP, Difference between OOP and Procedure Oriented Programming; Concepts: Object, Class, Encapsulation, Abstraction, Polymorphism and Inheritance, Applications of OOP. Special operators: scope resolution operator, Member Dereferencing operators, Memory management operators, Manipulators and Type cast operator.

Structure of a C++ Program and Classes and Objects: Class Declaration: Data Members, Member Functions, Private and Public members, Creating Objects, accessing class data members, Accessing member functions; Class Function Definition: Member Function definition inside the class declaration and outside the class declaration.

UNIT - II

Friend function, inline function, Static data members, Function Overloading, Arrays within a class. Arrays of Objects; Objects as function arguments: Pass by value, Pass by reference, Pointers to Objects.

Constructors: Declaration and Definition, Types of Constructors, (Default, Parameterized, Copy Constructors). Destructors: Definition and use.

Operator Overloading: Concept of Operator Overloading, Overloading unary and binary operators.

UNIT - III

Inheritance: Extending Classes Concept of inheritance, Base class, defining derived classes, Visibility modes: Public, Private, Protected; Types of Inheritance: Single inheritance:
Privately derived, publicly derived; Making a protected member inheritable, multilevel inheritance, multiple Inheritance and ambiguity of multiple inheritance, Hierarchal Inheritance, Hybrid, Nesting of classes.
Polymorphism: Definition, Application and demonstration of Data Abstraction, Encapsulation and Polymorphism, Static and Dynamic Polymorphism, Virtual Functions, pure virtual functions.

UNIT - IV

Exception Handling: Definition, Exception Handling Mechanism: Throwing mechanism and Catching Mechanism, Rethrowing an Exception
File Processing: Opening and closing of file, Binary file operations, structures and file operations, classes and file operations, Random file processing.

Suggested Readings :

Essential :

1. E. Balaguruswamy, 2008 : Object Oriented Programming with C++, TMH.

Further Reading :

2. Bjarne Stroustrup, 2009 : The C++ Programming Language, Addison-Wesley Publishing Company.
3. Robert Lafore, 2003 : Object Oriented Programming in Turbo C++, Galgotia Pub.
4. Salaria, R. S. : Object Oriented Programming Using C++, Khanna Book Publishing Co. (P.) Ltd., New Delhi.

Paper Code: CSA-DSC-3(Maj/Min)-301P
(Practical based on CSA-DSC-3(Maj/Min)-301)

Time Duration: 3 Hrs.

Note : Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Paper Title : Web Designing
Paper Code :CSA-DSC-3(Maj/Min)-302A

Th	Pr	Cr	
2	2	4	TheoryExternalMarks:45
			TheoryInternalMarks:5
Level: L- 200			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical External hours: 60

Time Duration: 3 Hrs.
Objective: This course enables students to create web pages using HTML, CSS, Java script and dream weaver.

Learning Outcome:
After completing this course students will be able to:

i. Get familiar with the terms related to World Wide Web.
ii. Use the front end web site developing technologies like HTML and CSS
iii. Know about the basics of JavaScript.
iv. Use Dreamweaver for creating Web Pages

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT-I

Web Terminology: Web Server; Web Client /Browser, Understanding how a Browser communicates with a Web Server, Internet, Intranet, Extranet, WWW, URL

Introduction to HTML : Structure of an HTML program, Paragraph Breaks, Line Breaks; Emphasizing Material in a Web Page (Heading Styles, Drawing Lines); Text Styles (Bold, Italics, Underline); Other Text Effects (Centering (Text, Images etc.) ;Lists: Unordered List, Ordered Lists, Definition lists; Adding Images: Img element using Border, Width, Height, Align, ALT Attributes ;Tables: Caption Tag, Width, Border, Cell padding, Cell spacing, BGCOLOR, COLSPAN and ROWSPAN Attribute

UNIT-II

Linking Documents: Anchor tag, External Document References, Internal Document References and Image Maps

Frames: understanding frames, creating frames, Targeting Named Frames

Cascading style sheets (CSS): Style tag, Link tag, Types of CSS: In-Line, Internal, External Forms: Attributes of Form element, Input element: Text Element, Password, Button, Submit Button, Reset Button, The Checkbox, Radio, Text Area, Select and Option

UNIT-III

Java Script: Features, tokens, data types, variables, operations, control constructs, strings arrays, functions, core language objects, client side objects, event handling. Applications related to client side form validation

Other Built-In Objects in Java Script: The String Object, The Math Object, and The Date Object; User Defined Objects: Creating a User Defined Object, Instances, Objects within Objects

UNIT-IV

Web Development Tools: Overview of Web Development Environment, Code Editors and IDEs, Features of Modern Editors, popular open-source tools for web development, creating a website using IDE, Incorporating HTML, CSS and JavaScript in Web Pages, benefits of using built-in templates, Steps for deploying a website.

Suggested Reading:

Essential :

1. Wanger&Wyke : Java Script Unleased, Pearson, Latest reprint New Delhi.
2. Bayross, Ivan : HTML, DHTML, JAVA SCRIPT by BPB, Latest reprint

Further Reading :

3. Joseph Lowery : Adobe Dream weaver CS6 Bible by Wiley India
4. Kogent Learning Solutions Inc DreamweaverCS5inSimple Steps by Wiley India

Paper Code: CSA-DSC-3(Maj/Min)-302P
(Practical based on paper CSA-DSC-3(Maj/Min)-302A)

Time Duration: 3 Hrs.

Note : Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Paper Title: Software Engineering
Paper Code: CSA-DSC-4(Maj/Min)-401

Th	Pr	Cr	TheoryExternalMarks:90
4	-	4	TheoryInternalMarks:10
Level: L- 200			NumberofTheoryHours:60

Time Duration: 3 Hrs.
Objectives: This course enables students to understand Software Engineering Tools and Techniques.

Learning Outcome:
After the completion of this paper, student will be able to: -

(i) Use principles, concepts, methods, and techniques of the software engineering approach to produce quality software.
(ii) Apply software engineering principles and practices in the planning and development of an actual software product.

Note:
The Question Paper will consist of Four Units.

(i) Examiner will set total of **NINE** questions comprising **TWO** questions from each Unit and **ONE** compulsory question of short answer type covering whole syllabi.
(ii) The students are required to attempt **ONE** question from each Unit and the Compulsory question.
(iii) All questions carry equal marks unless specified.

UNIT – I

Introduction to Software Engineering: Definition, Software Engineering goals, Characteristics of well-engineered software.
Software Process Models: Waterfall Model, Prototyping Model, Spiral Model.
Software Requirement Specification (SRS): Software Requirements, Definition of SRS, Characteristics of SRS, Components of SRS, Designing of SRS.

System Analysis: Principles of Structures Analysis, DFDs, E-R Diagrams, Data Dictionary.

UNIT – II

Software Design: Design Objectives, Design Principles, Concepts, Design Process, Design Methodologies: Structured Design, Modular Design, Object Oriented Design, User Interface Design and its elements and its Characteristics.

Software Project Planning & Scheduling: Objectives, Decomposition techniques, Planning and Scheduling Tools: GANTT Chart, PERT Chart, Critical Path Method and Work Breakdown Structure; Cost estimation, Cost estimation Models: Single Variable Model, COCOMO Model; Software Risks, Risk Assessment and management.

UNIT – III

Software Metrics: Role of Metrics and Measurement, Types of Software Metrics: Product Metrics, Software Size Metrics: LOC and Function Points, Process Metrics, People Metrics.

System Maintenance and Reliability: Software Quality and features, Maintenance and its types; Factors Affecting Software Reliability, Software Reliability vs Hardware Reliability, Software Reliability Metrics.

UNIT-IV

Software Testing Techniques: Introduction to Software Testing Process, Objectives of Software Testing, BBT & its Techniques, WBT & its Techniques. Integration Testing, Functional Testing, Object Oriented Testing, Alpha and Beta Testing.

Hardware and Software Selection: Types of Software, Procedure for Hardware/Software selection: Major phases in selection, Evaluation and Validation, Vendor Selection, Post – Installation Review. Software selection- Criteria for Software Selection, the evaluation process.

REFERENCES

1. Roger S. Pressman and Bruce R. Maxim: Software Engineering, Edition 2023, Tata-McGraw Hill Publishing House
2. Ali Behforooz and Frederick J. Hudson: Software Engineering Fundamentals, Edition 2021, Oxford University Press
3. Rajib Mall: Fundamentals of Software Engineering, Edition 2018, PHI Learning Pvt. Limited
4. Sommerville, I: Software Engineering, Edition 2017, Pearson Education.

Paper Title: Database Management System
Paper Code: CSA-DSC-4(Maj)-402

Th	Pr	Cr	
2	2	4	TheoryExternalMarks:45
			TheoryInternalMarks:5
Level: L- 200			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical External hours: 60

TimeDuration : 3 Hrs

Objective: Thiscourseaimsatgivingthestudentstheinsightoftheunderlyingconceptsof databasemanagementsystemandimplementthemusing Databasesoftware.

Learning Outcome:

Onsuccessfulcompletionofthecourse,students willbeableto:

(i) Gain a solid grasp of fundamental database concepts, Including database models and schemas.
(ii) Understand the basic elements of a relational database management system
(iii) Master the basics of SQL and construct queries using SQL
(iv) Basic programming concepts using PL/SQL

Note:

- TheQuestionPaper willconsistofFourUnits.
- Examiner willset totalof**NINE**questionscomprising**TWO**questionsfromeachUnitand**ONE**
- **compulsoryquestionofshortanswer typecoveringwholesyllabi.**
- Thestudentsarerequiredtoattempt**ONE**questionfromeachUnitandtheCompulsoryquestion.
- Allquestionscarryequal marksunlessspecified.

UNIT – I

Database Concepts: Introduction to databases, need, database architecture, Terminology: Tuple, Degree, attributes, Domain, Primary key, foreign keys, candidate keys; The 12 Rules (Codd's Rule) for an RDBMS, Normalization: First, second and third Normal Form, Boyce Codd Normal Form.

UNIT – II

Understanding SQL-1: Data Types, Creating Tables, Creating a Table with data from Another table, Inserting Values into a Table, Updating Column(s) of a Table, Deleting Row(s) from a Table, Dropping a Column, Querying database tables, Conditional retrieval of rows, Working with Null Values, Matching a pattern from a table, ordering the result of a Query Aggregate Functions, Grouping the Result of a Query, creation and deletion of Views

UNIT – III

Understanding SQL-II: Managing privileges with Grant and Revoke Command, COMMIT and ROLLBACK, Functions: Arithmetic Functions, Character Functions, Date Functions, Group Functions, Querying Multiple Tables using Equi-Joins, Cartesian Joins, Outer Joins, Self-Joins, SET Operators: Union, Intersect, Minus**.**

UNIT – IV

Procedural SQL: Creating, Executing and deleting procedures and functions, VariablesandConstants, Scope and Visibility of a Variable, Assignmentsand Expressions, OperatorPrecedence, Conditional and Iterative Control, CursorManagement:Implicit/explicit Cursor Attributes; Exception Handling: Predefined Exceptions,User Defined Exceptions; Database Trigger: types of triggers, dropping triggers**.**

SuggestedReadings:

1. Elmasri&Navathe:FundamentalsofDatabasesystems, 2017, 7thEdition,AddisonWesley,NewDelhi.
2. IvanBayross:SQL,PL/SQL, 2020, Fourth Edition, BPBPublication,NewDelhi.
3. C.J.Date:AnIntroductiontoDatabasesSystems 8th Edition, 2003, Addison Wesley,NewDelhi.
4. Korth&Silberschatz: DatabaseSystemConcept,2010, 6thEdition,McGrawHillInternationalEdition
5. BipinC.Desai:AnIntroductiontoDatabaseSystem, 2015, Galgotia Publication,NewDelhi
6. Raghu Ramakrishnan&JohannesGehrke: Database ManagementSystems, 2002, 3rd edition,McgrawHillInternationalEdition.
7. Paul DuBois, "MySQL Developer's Library", Fourth Edition, Addison Wesley.

Paper Code: CSA-DSC-4(Maj)-402P

(Practical Based on paper CSA-DSC-4(Maj)-402)

Time Duration: 3 Hrs.

List ofPracticals:

1. Define tables with columns of different data types and apply primary key.
2. Insert records into a table, update existing records, and delete data based on conditions.
3. Update records from table based on a condition.
4. Delete records from table based on condition.
5. Write queries to display all records, select specific columns, and filter results using `WHERE` conditions.
6. Retrieve data and sort it in ascending or descending order.
7. Use the `ALTER TABLE` statement to add, modify, rename and drop columns.
8. Create a foreign key in a child table that references the primary key of a parent table.
9. Insert data into a child table with a valid foreign key reference.
10. Calculate aggregates on columns such as finding the total sum, average, or number of records.
11. Use various **date and time functions** to manipulate and retrieve date and time values
12. Use `GROUP BY` to group records and then apply aggregate functions like `SUM()`, `AVG()`, etc.
13. Retrieve data from multiple tables by joining them using joins.
14. Use subqueries to retrieve data based on conditions from other queries.
15. Combine results from two or more queries using UNION or INTERSECTION. Ensure unique results using `UNION`.
16. Create a simple view to display a selected set of data from a table.
17. Use functions like `CONCAT()`, `LENGTH()`, `NOW()`, `ROUND()`, and others to manipulate and format data.
18. Create a function that takes two integers as input parameters and returns their sum.
19. Create a function that takes an integer input and returns 'Even' or 'Odd' based on whether the number is even or odd.
20. Create a function that calculates the average salary of employees from the `employees` table.
21. Create and call a stored procedure that prints the names of all students from the `student` table.
22. Create a stored procedure that takes multiple parameters and returns a result.
23. Write triggers for `INSERT`, `UPDATE`, and `DELETE` operations. Automatically execute actions before or after a data modification operation
24. Create a simple cursor to retrieve data from a table.
25. Create and use a cursor with a `SELECT` statement that retrieves multiple columns.

Paper Title: Data Structure
Paper Code: CSA-DSC-4(Maj)-403

Th	Pr	Cr	
2	2	4	TheoryExternalMarks:45
			TheoryInternalMarks:5
Level: L- 200			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical External hours: 60

Time Duration: 3 Hrs.

Objectives: The objective of this course is to familiarize students with Basic Data Structure and its implementation and to develop skills to apply appropriate Data Structure in Problem Solving.

Learning Outcome:
On successful completion of the course, students will be able to:
(i) Understand the Basic types of Data Structure, implementation, Applications.
(ii) Implement Linear and Non-Linear Data Structure.
(iii) Analyze and apply various searching and sorting techniques.
(iv) Develop Programming skills which helps in problem solving.

Note:
- TheQuestionPaper willconsistofFourUnits.
- Examiner willset totalof**NINE**questionscomprising**TWO**questionsfromeachUnitand**ONE**
- **compulsoryquestionofshortanswer typecoveringwholesyllabi.**
- Thestudentsarerequiredtoattempt**ONE**questionfromeachUnitandtheCompulsoryquestion.
- Allquestionscarryequal marksunlessspecified.

UNIT – I

Data Structure: Introduction and Classification of Data Structure, Data Structure operations. Applications of Data Structure, Basic data Structures;
Arrays: Introduction, Types of Arrays, Memory representation, Applications and operations.

UNIT – II

Stacks: Introduction, representing stack in memory, Applications and operations.
Linked List: Introduction, Operations: -traversing, searching, inserting, deleting, Types of Linked List, Memory representation.

UNIT – III

Queue: Introduction, Types, Memory Representation.
Trees: Definition and Basic concepts, Representation in Contiguous Storage, Binary Tree, Binary Tree Traversal, Binary Search tree.

UNIT – IV

Graphs: Introduction, Memory Representation, Breadth First Search, Depth First Search.
Searching: Binary and Linear Search; **Sorting**: Bubble Sort, Merge Sort, Quick Sort.

Suggested Readings:

1. Lipschuitz L. Seymour, 2007: Data Structure, Schaum Outline Series, TMH, New Delhi.
2. Tannenbaum, Aaron M., 2015: Data Structure Using C & C++ 2e, Pearson.
3. Jean-Paul Tremblay & Paul G. Sorenson,2017: An Introduction to Data Structures with Applications, Publisher-Tata McGraw Hill.

Paper Code: CSA-DSC-4(Maj)-403P

(Practical Based on paper CSA-DSC-4(Maj)-403)

Time Duration: 3 Hrs.

Implement Data Structures Operations using C / C++.

LIST OF PRACTICALS

1. Write a program to check whether a string is palindrome or not
2. To read n numbers and display it.
3. To demonstrate the concept of one-dimensional array.
4. Implementation of Program to perform the following operations on an Array
   - Traversal
   - Insertion
   - Deletion
5. Program to Implement Stack
   - Push an Element
   - Pop an Element
6. Write a program to implement linear and binary search algorithm.
7. Write a program to create a linked list and perform following operations:
   - Insertion
   - Deletion
   - Traversals
   - Searching
8. Program to implement Linear Queue using an array
9. Program to Implement Circular Queue using an array
10. Program to implement Linear Linked List
11. Program to implement Doubly Linked List
12. Program to perform Insertion in a Binary tree
13. Program to perform Deletion in a Binary tree
14. Program to perform Searching in a Binary tree
15. Program to perform Traversal in Graph
16. Write a program to sort a given list of number using following algorithm.
    - Bubble sort
    - Merge sort
    - Quick sort

Paper Title: Introduction to Artificial Intelligence
Paper Code: CSA-DSC-4(Maj)-404

Th	Pr	Cr	TheoryExternalMarks:90
2	2	4	TheoryInternalMarks:10
Level: L- 200			NumberofTheoryHours:60

Time Duration: 3 Hrs.

Objective: The course aims at introducing basic concepts of artificial intelligence, understanding of search techniques. To learn working of expert system and natural language processing.

Learning Outcome:

Onsuccessfulcompletionofthecourse,students willbeableto:

(i) Understanding of artificial intelligence, intelligent agents and state-space representation.
(ii) Understanding of working of searching techniques and its algorithms.
(iii) Understand basics of game playing, knowledge representation and expert systems.
(iv) Obtain insight about natural language processing and its components.

Note :

- TheQuestionPaper willconsistofFourUnits.
- Examiner willset totalof**NINE**questionscomprising**TWO**questionsfromeachUnitand**ONE**
- compulsoryquestionofshortanswer typecoveringwholesyllabi.
- Thestudentsarerequiredtoattempt**ONE**questionfromeachUnitandtheCompulsoryquestion.
- Allquestionscarryequal marksunlessspecified.

UNIT-I

Introduction: What is AI?, Intelligent Agents: Agents and environment, the concept of Rationality, the nature of the environment, the structure of agents, Introduction to AI technique, Turing test, History and developments in AI, Applications of AI, State space representation, Production systems and problem characteristics.

UNIT-II

Searching techniques: Breadth First Search, Depth First Search, Generate and test, Hill climbing, Best First Search, A* and AO* algorithms, Constraint Satisfaction Problem, Means-End Analysis.

UNIT-III

Game Playing, Knowledge Representation and Expert Systems: Introduction to Game Playing, Minimax Search, Alpha-Beta pruning. Propositional Logic, First Order Predicate Logic, Resolution Principle, Architecture and role of expert systems, atleast two case studies of Expert Systems.

UNIT-IV

Natural Language Processing: Introduction to Natural Language Processing, Components of Natural Language Processing, Tokenisation and Sentence splitting, Lexical Analysis, Syntactic Analysis, Semantic Analysis, Discourse Integration and Pragmatic Analysis.

Suggested Readings:

1. Elaine Rich and Kevin Knight, "Artificial Intelligence," Tata McGraw Hill, 2018
2. Patterson, Dan W.: Introduction to Artificial Intelligence and Expert systems, Pearson Education, 2015
3. Sharma, A., Essentials of AI and Soft Computing, PHI Learning, 2024
4. Russell, Stuart, J. and Norvig, Peter, Artificial Intelligence - A Modern Approach, Pearson, 4th edition, 2020

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ENVIRONMENT CONSERVATION

SEMESTER-III

MINOR COURSE

EMERGING ISSUES ON ENVIRONMENT

CourseCode: MINC3Credits=4(T=3+ P=1)

Total Hours = 60 (Theory=45+Practical=15)
Total Marks = 100 (Theory=75 [Final =60; Internal Assessment =15] +Practical=25)

CourseOutcomes

1. *Thestudentswillbeabletounderstandaboutnaturalresources,theirtypesandrealizethe importance of resources.*
2. *Thestudentswillbeabletoacquiredeepknowledgeaboutlandandforestresources.*
3. *Thestudentswillbeabletodescribeenergyresources.*
4. *ThestudentswillbeabletoanalyseIndianaswellasglobalstatusofnaturalresources.*
5. *Thestudentswillbeabletostudyconservationmethodsofnaturalresources.*

THEORY

Unit-I(15Hrs)

Ozone Depletion and Its causes

1.1.Ozone Layer depletion
1.2.Science of Ozone destruction
1.3.Ozone hole
1.4.Impacts of ozone depletion on living things
1.5.Impact of ozone depletion on materials

Unit-II(15Hrs)

Climate Change

2.1 Greenhouse gGases: sources and trends
2.2 Greenhous effect and Global warming
2.3 Cause of climate change
2.4 Effects of climate change (Sea level rise, on human health)
2.5 Mitigation strategies

UnitIII(15Hrs)

Solid Waste Management

3.1 Concept of Solid Waste and Its types
3.2 Rising Solid Waste; Reasons and Issues
3.3 Impact of Solid Waste Accumulation on Environment
3.4 Solid Waste Management and Treatment
3.5 Concept of Vermicomposting and Vermiculture
3.6 Legislatives Measures on Solid Waste

UNIT IV (15 Hrs)

Environmental Disasters

4.1 Concept: Definition, hazard and vulnerability
4.2 Types of disasters: Natural
4.3 Types of disasters: Man-made disasters
4.4 Disaster management: Mitigation and Preparedness, Response and Recovery

PRACTICALS(15Hrs)

1. TodoaHousehold Waste Audit and give recommendations
2. Tostudy and compare different composting methods
3. To draw analysis of types of Solid waste and their management techniques
4. To visit Solid Waste Handling Site / Biomedical Incinerator/ landfills in the surrounding area.
5. Project report on Ozone depletion and Climate change, and Disaster mitigations

Instructions for paper setter for Theory Paper:

There will be nine questions in all. Question # 1 shall be compulsory comprising of four short answer questions covering whole of the syllabus. There shall be two questions each from Unit I to IV. The candidate shall be required to attempt compulsory question, and one question each from Unit I to IV. Each question shall be of 12 marks.

SEMESTER-IV
MINOR COURSE

ENVIRONMENTAL IMPACT ASSESSMENT AND ENVIRONMENT AUDITING

Course Code: MINC4 Credits=4(T=3+ P=1)

Total Hours = 60 (Theory=45+Practical=15)
Total Marks = 100 (Theory=75 [Final =60; Internal Assessment =15] +Practical=25)

CourseOutcomes

1. *The students will be able to understand about the importance of EIA.*
2. *The student will be able to understand the impacts of different projects on environment.*
3. *The students will be acquiring knowledge about environment auditing.*
4. *The students will get knowledge about the career prospects in the field of EIA and Environment Auditing.*

THEORY

Unit-I (15Hrs)
Introduction to EIA

1.1 EIA origin, development, purpose and aims
1.2 Principles of Environment Impact Assessment
1.3 EIA Methodology
1.4 Environmental Management Plan
1.5 Environmental components of EIA.

Unit-II(15Hrs)
EIA Process and Procedures

2.1 EIA processes: Project screening, scoping, base-line data, impact identification; prediction, evaluation, valuation of environmental impacts, mitigation.
2.2 Public participation, presentation, review and decision making, monitoring and auditing.
2.3 Environmental Appraisal procedures in India

Unit-III(15Hrs)
Environmental Impacts and EIA

3.1 Impact identification methods.
3.2 Environmental impacts of mining industry; nuclear power plant, textile industry; pulp and paper industry; petroleum refining; pesticide manufacturing industry; fertilizer industry.
3.3 Environmental impacts of mining industry; nuclear power plant, textile industry; pulp and paper industry; petroleum refining; pesticide manufacturing industry; fertilizer industry.

Unit-IV (15Hrs)
Environmental Auditing

4.1 Types of environmental audits: Assessment and compliance audit
4.2 Occupation health and safety
4.2 Energy audits. Green audit, Water audit
4.4 Environmental Management systems in India
4.5 Environmental audit and EIA, Vocational prospects in the field of EIA

PRACTICALS (15Hrs)

1. To prepare a EIA report
2. To prepare an energy audit report of college campus
3. To survey and prepare an Green audit report of college campus
4. To prepare a water audit report of college campus
5. Internship with some organization in the field of EIA or Environment Auditing

Instructions for paper setter for Theory Paper:

There will be nine questions in all. Question # 1 shall be compulsory comprising of four short answer questions covering whole of the syllabus. There shall be two questions each from Unit I to IV. The candidate shall be required to attempt compulsory question, and one question each from Unit I to IV. Each question shall be of 12 marks.

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INDUSTRIAL CHEMISTRY

CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MAJOR** SUBJECT AS PER TABLE 2

Table 2

III	L-200	(ICH-DSC-3 MAJ/MIN-301)	Industrial Chemistry and Environmental Sustainability	4	24
		Major - 3	To be chosen by the student	4	
		Minor – 3	To be chosen by the student	4	
		Minor - 4	To be chosen by the student	4	
		IDC-3	To be chosen from the available options	3	
		SEC-3	To be chosen from available options	3	
		ENG-3	Compulsory	2	
IV	L-200	(ICH-DSC-5 MAJ-401/MIN-302)	Chemical & Industrial Elemental Process	4	24
		(ICH-DSC-6 MAJ-402/MIN-401)	Heat and Mass Transfer Processes	4	
		Major - 5	To be chosen by the student	4	
		Major - 6	To be chosen by the student	4	
		Minor - 5	To be chosen by the student	4	
		ENG-4	Compulsory	2	
		VAC-3	To be chosen from the common basket	2	

CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MINOR**SUBJECT AS PER TABLE 1 AND TABLE 2

Table 2

Sem	**Level**	**Course Code**	**Course Name**	**Credits**
III	L-200	(ICH-DSC-3 MIN/MAJ-301)	Industrial Chemistry and Environmental Sustainability	4
	L-200	(ICH-DSC-5-MIN-302/MAJ-401)	Chemical & Industrial Elemental Process	4
IV	L-200	(ICH-DSC-6 MIN-401/MAJ-402)	Heat and Mass Transfer Processes	4

Note:-

1. In the first semester, students are offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awarded a BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3rd semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

Semester	**III**
Code of Paper	**ICH-DSC-3-MAJ/MIN-301**
Title of Paper	**Industrial Chemistry and Environmental Sustainability**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67 Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22 Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week Practical: 3 Hrs. per week**

Applicable to the students

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this applied course, the students will be able to:

- Explain the different sources and causes of air pollution, their control measures and equipment used for pollution control.
- The student shall be able to give the details of water pollutants, the source, characterization and methods used in water purification.
- Explain and discuss various aspects of waste management and strategies required for its minimization.
- Explain the heavy metal ions in the environment, its monitoring, assessment and remediation techniques.
- Emphasis is given to developing practical skills that will aid the student to understand various test involved in testing water samples, industrial and domestic.

GENERIC OUTCOMES

The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 3
Syllabus for Theory Paper
ICH-DSC-3-MAJ/MIN-301
Industrial Chemistry and Environmental Sustainability

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT-1 (15Hrs.)

Air Pollution: Introduction, Sources of air pollution,Causes of air pollution – types & sources from chemical industries, Classification of air pollutants: based on the origin, chemical composition and state of matter, Important air pollutants: oxides of Sulphur, CO, NOx, hydrocarbons, Suspended particulate matter, Effect of air pollution in human health, Effect of air pollution on animals, Effect of air pollution on plants, Effects of air pollution on aquatic life, Different types of plume behaviour: looping, coning, fanning, lofting, fumigating, and trapping plumes, Key factors influencing plume behavior, Importance of plumes for industries.

Air Pollution Control Measures & Equipment: Control of Particulate Emission – Control of Gaseous Emission: Absorption, adsorption, Combustion or incineration and condensation, Removal of particulate matter, Removal of Sulphur dioxide, Removal of oxides of nitrogen, Control of CO_2 and CO, Construction, working and principle of Spray tower, packed tower, Gravitational settling chambers, Cyclone separators, Fabric filters, Electrostatic precipitators, Wet scrubbers, Advantages and disadvantages.

UNIT-2 (15Hrs.)

Water pollution: Definition of water pollution, Categorization of pollutants by source (e.g., industrial, domestic, agricultural) and by type (e.g., physical, chemical, biological), Sources of water pollution, Analysis of the impacts on human health, ecosystems, and aquatic life, Techniques for measuring water pollution, Water purification methods. Effluent treatment plants (primary, secondary and tertiary treatment).

Water Quality Parameters for waste water, industrial water and domestic water: Physical Parameters: Turbidity, temperature, color, and total dissolved solids (TDS), Chemical Parameters: pH, acidity, alkalinity, hardness, chlorides, salinity, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), nitrates, phosphates, heavy metals, Biological Parameters, Water sampling procedures, Sample containers, Sample preservation and preparation, physicochemical analysis of water: pH, acidity, alkalinity and hardness.

UNIT-3 (15Hrs.)

Solid Waste Management: Introduction, Sources of solid waste, Types of solid waste, Composition of solid waste and its determination, Waste characterization, Classification of hazardous waste, Impacts of solid waste on human health and the environment, Methods of solid waste management: Sanitary landfill: Site selection, Landfilling methods, Design and operation of landfills, Incineration: Types of incinerators, Factor affecting incineration, Methods of incineration and Concept of recycling, Strategies for minimizing waste generation at the source, including waste reduction, reuse, and recycling.

Heavy metal ions in the environment: Sources of heavy metal pollution, Heavy metal behavior in the environment, Toxicity of specific heavy metals (Mercury, Copper, Chromium, Cadmium, Cobalt, lead, Arsenic), Mechanism of toxicity, Health effects on humans, Effects on ecosystems: impact on plant and animal life, Remediation techniques, Monitoring and assessment of heavy metal contamination: sampling techniques, analytical methods, and risk assessment.

Books Recommended

1. Rao, C. S.: Environmental Pollution Control Engineering, New Age Int. (P) Ltd. Publishers, 2006.
2. Rao, M.N. and Rao, H.V.N.: Air Pollution, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2003.
3. Zajic, J.E.: Water pollution, Marcel Dekker, Inc. New York, 1971.
4. Davis, M.L. and Cornwell, D.A.: Introduction to Environmental Engg. McGraw-Hill International Editions, 1991.
5. Hazardous Waste minimization by Harry Freeman (1990).

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-3
Syllabus for Practical Paper
ICH-DSC-3-MAJ/MIN-301
Industrial Chemistry and Environmental Sustainability

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

Industrial Chemistry Lab

1. To determine the pH of given samples using (a) pH paper and (b) digital pH meter.
2. To determine the acidity of the given sample of water.
3. To determine the alkalinity of the given sample of water.
4. To determine the hardness of a water sample by measuring the amount of calcium present.
5. To perform the Limit test for Sulphate.
6. To perform the Limit test for chloride.
7. Calibration of conductivity meter and determination of conductivity of given water sample.
8. Determination of heavy metal ion impurities from given waste water sample using titration method.
9. Power point presentations.

Books Recommended

1. Vogel: Practical Organic Chemistry, 3rd Edition, ELBS, 1989.
2. Vogel: Textbook of quantitative analysis, 5th Edition, ELBS, 1989.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**III**
Code of Paper	**ICH-DSC-5-MIN-302**
Title of Paper	**Chemical & Industrial Elemental Process**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 3 Hrs. per week**

Applicable to the students

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this applied course, the students will be able to:

- Explain that out of thedifferent sources of organic compounds, how petroleum, natural gas and coal play a substantial role in providing an array of organic compounds.
- The student shall be able to give the details of different physiochemical processes required for extraction of metals.
- Explain and discuss various types of inorganic material of industrial importance such as Alumina, Silicates and Zeolites.
- Emphasis is given to developing practical skills that will aid the student to understand various test involved in adsorption of heavy metals. And how to estimate the different components of coal.

GENERIC OUTCOMES

The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester-3
Syllabus for Theory Paper
ICH-DSC-5-MIN-302
Chemical & Industrial Elemental Process

End semester = 67 **Credits - 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT-1 (15Hrs.)

Raw materials for organic compounds: Sources of organic chemicals, Petroleum: crude oil, refining, petrochemicals, Fractionation of crude oilcomposition using fractional distillation,Understanding the properties and uses of different fractions like refinery gases, gasoline, kerosene, diesel, fuel oil, and lubricating oils, purpose and mechanisms of cracking, types of cracking, Reforming processes, hydroforming, isomerization.
Natural gas:Hydrocarbon gases (Methane, Ethane, Propane, Butane, and Pentane), Non-hydrocarbon gases (Carbon dioxide, Nitrogen, Hydrogen sulfide, Water Vapor).
Coal: Types (Anthracite, Bituminous, Subbituminous, Lignite), Coal structure and composition, Properties of coal, distillation of Coal, advantages and disadvantages of coal.

UNIT-2 (15Hrs.)

Surface chemistry and Interfacial phenomena: Introduction to surface and interfacial phenomena, Surface and interfacial tension, Absorption and adsorption, Adsorption isotherm: Freundlich and Langmuir adsorption isotherms.
Catalysis: Introduction to Catalysis, Fundamental Concepts, Characteristics of Catalysts, Specificity of Catalyst, Types of Catalysis-Homogeneous and Heterogeneous, Catalytic Steps in Heterogeneous Catalysis, Activation Energy, their mechanism of action and their applications, Catalytic poisoning, Catalyst Promoter, Catalyst properties, Activity of a catalyst, Autocatalysis, Negative catalysis.
Basic metallurgical operations: Pulverizations, Calcinations, Roasting, Refining, Types of refining: Fire Refining, Electrolytic Refining, Zone Refining.

UNIT-3 (15Hrs.)

Physicochemical principles of extraction of Metals: Fundamental principles and methods used to extract metals from their ores, Occurrence of Metals, Extraction Methods, Ore Concentration, Extraction of specific metals like – Iron: (Concentration of the Ore, Blast Furnace, Reduction and Collection), Lead: (Ore Concentration, Roasting, Smelting, Refining), Silver:(leaching), Aluminum: (Bayer Process).
Inorganic materials of industrial importance - Availability, Forms, Structure, and Modifications of Key Inorganic Materials, Characteristics of materials like Alumina: Definition and Formation, Structure and Properties, Industrial Applications, Silicates, Zeolites: structure, properties, and applications of zeolites.

Books Recommended

1. Admson, A.W., Physical Chemistry of Surfaces, 4th edition, Pubs: John Wiely& Sons, New York, 1982.
2. Austin H. T., Shreve's Chemical Process Industries, Pubs: McGraw Hill Book Company,New York (1984).
3. Kent James A. (ed.), Reigel's Handbook of Industrial Chemistry, Pubs: Van Nostrand inhold Company, London (1983).
4. Pandey G.N., Text Book of Chemical Technology, Vol. I & II, Pubs: Vikas Publishing House, Pvt. Ltd., New Delhi (1999).
5. Buchner V., Sohliebs P., Winter G. & Buchel K.H., Industrial Inorganic Chemistry, Pubs:V. Ch. Publishers, New York (1989).
6. Badger W.L. and Bancher J.T., Introduction to Chemical Engineering, Pubs: McGraw-HillCo., U.S.A,(1986).
7. McCabe W.L. and Smith J.C., Unit Operations in Chemical Engineering, Pubs:McGraw-HillBook Company, New York,(1984).
8. Perry J.H., Chemical Engineering Handbook, Pubs: McGraw-Hill Book Company, NewYork, (1993).

Instructions for the examiner

4. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
5. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
6. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-3
Syllabus for Practical Paper
ICH-DSC-5-MIN-302
Chemical & Industrial Elemental Process

End semester = 22 **Credit - 01**
Internal Marks = 03 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

Industrial facets of Chemistry Lab

1. Adsorption of heavy metal ions from given waste water sample using various adsorbents:
   (a) silica (b) nanoparticles
2. Study of Freundlich and Langmuir adsorption isotherm models and determination of isotherm constants.
3. Determination of surface tension of NaCl/Sugar at different concentrations.
4. Limit tests for heavy metals
   (a) lead (b) Iron
5. To determine the kinematic viscosity and absolute viscosity of the given oil at different temperatures using Redwood Viscometer.
6. Determination of volatile content in coal.
7. Determination of ash content in coal.
8. Power point presentations.

Books Recommended

1. Vogel: Practical Organic Chemistry, 3rd Edition, ELBS, 1989.
2. Vogel: Textbook of quantitative analysis, 5th Edition, ELBS, 1989.
3. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11thedition.
4. Pubs: R. Chand & Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw HillPublishing Co. Ltd., 1998.
6. Levitt, B.P., Findlays Practical Physical Chemistry; 8th edition, Pubs: Longman GroupLtd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**IV**
Code of Paper	**ICH-DSC-5-MAJ-401**
Title of Paper	**Chemical & Industrial Elemental Process**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 3 Hrs. per week**

Applicable to the students

2. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
3. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this applied course, the students will be able to:

- Explain that out of thedifferent sources of organic compounds, how petroleum, natural gas and coal play a substantial role in providing an array of organic compounds.
- The student shall be able to give the details of different physiochemical processes required for extraction of metals.
- Explain and discuss various types of inorganic material of industrial importance such as Alumina, Silicates and Zeolites.

Emphasis is given to developing practical skills that will aid the student to understand various test involved in adsorption of heavy metals. And how to estimate the different components of coal.

GENERIC OUTCOMES

The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

4th semester: TRAINING/WORKSHOP/INDUSTRIAL VISITS

1.Visit to a local industry during the year and students to submit a report on the same.
2. Students will also submit a project report on water-effluent analysis/soil analysis/ food adulteration etc.
3. Student to submit a report on one small scale unit: Safety matches, Naphthalene balls, Wax candles, Shoe Polish, Gum Paste, Pen Ink, Chalk, Plaster of Paris
4. Each Student to present a seminar relevant to the syllabus.

Semester – 4
Syllabus for Theory Paper
ICH-DSC-5-MAJ-401
Chemical & Industrial Elemental Process

End semester = 67
Credits - 03
Internal Marks = 08
Contact Hours – 3/week
Examination Time allowed 3 Hours

UNIT-1 (15Hrs.)

Raw materials for organic compounds: Sources of organic chemicals, Petroleum: crude oil, refining, petrochemicals, Fractionation of crude oilcomposition using fractional distillation,Understanding the properties and uses of different fractions like refinery gases, gasoline, kerosene, diesel, fuel oil, and lubricating oils, purpose and mechanisms of cracking, types of cracking, Reforming processes, hydroforming, isomerization.
Natural gas:Hydrocarbon gases (Methane, Ethane, Propane, Butane, and Pentane), Non-hydrocarbon gases (Carbon dioxide, Nitrogen, Hydrogen sulfide, Water Vapor).
Coal: Types (Anthracite, Bituminous, Subbituminous, Lignite), Coal structure and composition, Properties of coal, distillation of Coal, Advantages and disadvantages of coal.

UNIT-2 (15Hrs.)

Surface chemistry and Interfacial phenomena: Introduction to surface and interfacial phenomena, Surface and interfacial tension, Absorption and adsorption, Adsorption isotherm: Freundlich and Langmuir adsorption isotherms.
Catalysis: Introduction to Catalysis, Fundamental Concepts, Characteristics of Catalysts, Specificity of Catalyst, Types of Catalysis-Homogeneous and Heterogeneous, Catalytic Steps in Heterogeneous Catalysis, Activation Energy, their mechanism of action and their applications, Catalytic poisoning, Catalyst Promoter, Catalyst properties, Activity of a catalyst,Autocatalysis, Negative catalysis.
Basic metallurgical operations: Pulverizations, Calcinations, Roasting, Refining, Types of refining: Fire Refining, Electrolytic Refining, Zone Refining.

UNIT-3 (15Hrs.)

Physicochemical principles of extraction of Metals: Fundamental principles and methods used to extract metals from their ores, Occurrence of Metals, Extraction Methods, Ore Concentration, Extraction of specific metals like – Iron: (Concentration of the Ore, Blast Furnace, Reduction and Collection), Lead: (Ore Concentration, Roasting, Smelting, Refining), Silver:(leaching), Aluminum: (Bayer Process).
Inorganic materials of industrial importance - Availability, Forms, Structure, and Modifications of Key Inorganic Materials, Characteristics of materials like Alumina: Definition and Formation, Structure and Properties, Industrial Applications, Silicates, Zeolites: structure, properties, and applications of zeolites.

Books Recommended

1. Admson, A.W., Physical Chemistry of Surfaces, 4th edition, Pubs: John Wiely& Sons, New York, 1982.
2. Austin H. T., Shreve's Chemical Process Industries, Pubs: McGraw Hill Book Company,New York (1984).
3. Kent James A. (ed.), Reigel's Handbook of Industrial Chemistry, Pubs: Van Nostrand inhold Company, London (1983).
4. Pandey G.N., Text Book of Chemical Technology, Vol. I & II, Pubs: Vikas Publishing House, Pvt. Ltd., New Delhi (1999).
5. Buchner V., Sohliebs P., Winter G. & Buchel K.H., Industrial Inorganic Chemistry, Pubs:V. Ch. Publishers, New York (1989).
6. Badger W.L. and Bancher J.T., Introduction to Chemical Engineering, Pubs: McGraw-HillCo., U.S.A,(1986).
7. McCabe W.L. and Smith J.C., Unit Operations in Chemical Engineering, Pubs:McGraw-HillBook Company, New York,(1984).
8. Perry J.H., Chemical Engineering Handbook, Pubs: McGraw-Hill Book Company, NewYork, (1993).

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-4
Syllabus for Practical Paper
ICH-DSC-5-MAJ-401
Chemical & Industrial Elemental Process

End semester = 22
Credit - 01
Internal Marks = 03
Contact Hours – 3/week
Examination Time allowed 3 Hours

Industrial facets of Chemistry Lab

5. Adsorption of heavy metal ions from given waste water sample using various adsorbents:
   (b) silica (b) nanoparticles
6. Study of Freundlich and Langmuir adsorption isotherm models and determination of isotherm constants.
7. Determination of surface tension of NaCl/Sugar at different concentrations.
8. Limit tests for heavy metals
   (b) lead (b) Iron

5. To determine the kinematic viscosity and absolute viscosity of the given oil at different temperatures using Redwood Viscometer.
6. Determination of volatile content in coal.
7. Determination of ash content in coal.
8. Power point presentations.

Books Recommended

7. Vogel: Practical Organic Chemistry, 3rd Edition, ELBS, 1989.
8. Vogel: Textbook of quantitative analysis, 5th Edition, ELBS, 1989.
9. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11thedition.
10. Pubs: R. Chand & Co., New Delhi, 2002.
11. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw HillPublishing Co. Ltd., 1998.
12. Levitt, B.P., Findlays Practical Physical Chemistry; 8th edition, Pubs: Longman GroupLtd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**IV**
Code of Paper	**ICH-DSC-6-MAJ-402/MIN-401**
Title of Paper	**Heat and Mass Transfer Processes**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 3 Hrs. per week**

Applicable to the students

3. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
4. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this applied course, the students will be able to:

- Explain the different modes of heat transfer from convection to conduction. And types of heat exchangers.
- The student shall be able to give the details of different types of distillation and drying techniques.
- Explain and discuss various types of evaporation and filtration techniques, their advantages and disadvantages.
- Emphasis is given to developing practical skills that will aid the student to understand various test involved in heat transfer methods. Rate of evaporation and filtrations.

GENERIC OUTCOMES

The second year of the course is designed to deepen the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They work is supported by both theoretic and practical skill development.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 4
Syllabus for Theory Paper
ICH-DSC-6-MAJ-402/MIN-401
Heat and Mass Transfer Processes

End semester = 67
Credits - 03
Internal Marks = 08
Contact Hours – 3/week
Examination Time allowed 3 Hours

UNIT-1 (15Hrs.)

Heat transfer through Conduction and Convection: Introduction to Heat Transfer, Modes of Heat Transfer: Conduction, Convection, and Radiation, Fundamentals: Heat transfer mechanisms, definitions, and units, Heat Transfer Mechanisms, Conduction Heat Transfer: Steady state conduction, conduction through a plain wall, composite wall, cylindrical wall, Convective Heat Transfer: Natural convection and forced convection.

Heat exchangers: Types of Heat exchangers- Construction Types: 1,1 and 1,2 shell and Tube type; Difference between single pass and multipass shell and tube heat exchanger, finned tube heat exchangers, plate heat exchangers, Double pipe heat exchanger, Fluid Flow Configurations: Parallel Flow, Crossflow, Counterflow, Applications of Heat exchangers in various chemical processes.

UNIT-2 (15Hrs.)

Distillation-Introduction, Applications, Vapour liquid equilibrium, Boiling point diagram, Ideal solutions, Raoult's law, Dalton's law, Henry's law, Volatility, Relative volatility, Batch and continuous distillation, Classification of distillation methods: working and construction of Simple distillation or differential distillation, Rayleigh equation, working and construction of Flash distillation, Minimum boiling azeotropic solutions, Maximum boiling azeotropic distillation, working and construction of Azeotropic distillation and Extractive distillation with examples, working and construction of Steam distillation, Advantages of steam distillation, Rectifying section and stripping sections of distillation column.

Drying- Introduction, Definition and Importance of Drying, Objectives of Drying, Applications of drying operation, Theory of drying, Bound water, Unbound water, Mechanism of drying process, Diffusion theory, capillarity theory, Pressure gradient theory, Equilibrium moisture content (EMC), Free moisture content (FMC), Percentage loss on drying, Percentage moisture content, Drying rate, Analysis of factors influencing the drying rate, including air velocity, temperature, and moisture content. Graphical representation of drying rate vs moisture content using FMC curve, Classification of drying equipment: Principle, working, construction and uses of tray dryer, rotary dryer, flash dryer, fluidized bed dryer, drum dryer, spray dryer, Advantages and Disadvantages.

UNIT-3 (15Hrs.)

Evaporation- Introduction, Definition, Importance, Distinguishing feature of evaporation with other heat processes, Applications, Factors Affecting Evaporation: Temperature, vapour pressure, surface area, moisture content of the feed, types of product required, time of evaporation, film and deposits, economic factor, Classification of Evaporators, Principle, working, construction and uses of Horizontal tube evaporator, Vertical tube evaporator, short tube (standard) Evaporator, forced circulation evaporators, falling film evaporators, climbing film (upward flow) evaporators or rising film evaporator, Advantages and Disadvantages, Economy of a single effect evaporator.

Filtration-Introduction, Definition, Process of filtration, Clarification, Applications, Driving forces, Mechanism of filtration: Straining, Impingement, Entanglement, Attractive forces, Types

of filtration, Surface filtration (Screen filtration), Depth filtration, Theories of filtration, Rate of filtration, Factors influencing filtration process, filter media and filter aids, Classification of filtration equipment, Principle, working, construction and uses of Plate and frame filter press, rotary drum filter, bag filter, Advantages and Disadvantages.

Books Recommended

1. McCabe, W.L., Smith, J.C. and Harriot H.P.: Unit Operations of Chemical Engineering, 7th Edition, McGraw Hill,2005.
2. Holman, J.P.: Heat Transfer, McGraw Hill Publication Co.Ltd., New Delhi, 9th Edition, 2008.
3. Chapman, A.J. : Heat Transfer, McMillan Publishing Co., 4th Edition, 1984
4. Kern, D.Q. : Process heat Transfer, Tata McGraw Hill Publishing Co., New Delhi, Edition 1997, 2006
5. Kreith, F.: Principles of Heat Transfer, Harper & Row Pub., London.
6. Geankoplis, C.J.: Transport Processes and Unit Operations, Prentice Hall of India Pvt. Ltd., 3rd Edition, 1999.
7. Treybal, Robert E.: Mass Transfer Operations, 3rd Edition, McGraw-Hill, 1981.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester – 4
Syllabus for Practical Paper
ICH-DSC-6-MAJ-402/MIN-401
Heat and Mass Transfer Processes

End semester = 22
Credit - 01
Internal Marks = 03
Contact Hours – 3/week
Examination Time allowed 3 Hours

Heat and Mass Transfer Operation Lab

1. Determination of heat transfer coefficient under forced convection.
2. Determination of heat transfer coefficient under natural convection.
3. To separate a mixture of water and alcohol using simple distillation.
4. To study the factors affecting the rate of evaporation.
5. To do the effect of thickness of filter medium and pressure difference across the filter medium using (a) Simple Filtration (b) Vacuum Filtration
6. To determine the porosity of the given packing material for packed column.
7. To determine the % loss on drying, % moisture content and drying rate of given sample.
8. Power point presentations.

Books Recommended

1. McCabe, W.L., Smith, J.C. and Harriot H.P.: Unit Operations of Chemical Engineering, 7th Edition, McGraw Hill,2005.
2. Holman, J.P.: Heat Transfer, McGraw Hill Publication Co.Ltd., New Delhi, 9th Edition, 2008.
3. Chapman, A.J.: Heat Transfer, McMillan Publishing Co., 4th Edition, 1984.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

MATHEMATICS

AS PER NEP (2020)

Syllabus of **Mathematics** for UG Classes

Table-1(SCIENCE TEMPLET)

SINGLE MAJOR, SINGLE / DOUBLE MINOR.**BSc, BSc HONOURS AND BSc HONOURS WITH RESEARCH**

Sem.	Level	MAJOR (4Cr)MINOR (4 Cr)		MDC(3 Cr) Any paper in each of the first three semesters	VAC(2Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC(3Cr) Required three papers, one for each of the first three semesters	AEC (2Cr)	PBI / HCP (2Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101:BasicMathematics • MAT-MDC-102:Calculus • MAT-MDC-103:Algebra	• MAT-VAC-101:Vedic Mathematics • MAT-VAC-102:Vedic Algebra • MAT-VAC-103:Theory of Numbers • MAT-VAC-104:Theory of Numbers with html	• MAT-SEC-101:Maths Based Practicals • MAT-SEC-102:Maths Based Programming AndPracticals • MAT-SEC-103:Latex And Html • MAT-SEC-104:Quantitative Aptitude • MAT-SEC-105:Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Discrete Mathematics and Graph Theory • MAT-DSC4 (MAJ)-403: Coordinate Geometry-I • MAT-DSC4 (MAJ)-404: Numerical Analysis	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers • MAT-DSC5 (MAJ)-503: Differential Equations-II • MAT-DSC5 (MAJ)-504: Solid Geometry • MAT-DSC5 (MAJ)-505: Statics			Internship / community work			
VI	300	• MAT-DSC6 (MAJ/MIN)-602: Analysis • MAT-DSC6 (MAJ)-603: Linear Programming • MAT-DSC6 (MAJ)-604: Probability Theory • MAT-DSC6 (MAJ)-605: Dynamics • MAT-DSC6 (MAJ)-606: Modern Algebra	MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry					

Table-2(SCIENCE TEMPLETE)

MULTI-DISCIPLINARY / INTER-DISCIPLINARY**. BSc,BScHONOURSANDBSc HONOURS WITH RESEARCH**

Sem.	Level	MAJOR (4 Cr) A	MAJOR (4 Cr) B	MINOR (4 Cr) C	MDC(3 Cr) Any paper in each of the first three semesters	VAC(2Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC(3Cr) Required three papers, one for each of the first three semesters	AEC (2Cr)	PBI / HCP (2Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101:Basic Mathematics • MAT-MDC-102:Calculus • MAT-MDC-103:Algebra	• MAT-VAC-101:Vedic Mathematics • MAT-VAC-102:Vedic Algebra • MAT-VAC-103:Theory of Numbers • MAT-VAC-104:Theory of Numbers with html	• MAT-SEC-101:Maths Based Practicals • MAT-SEC-102:Maths Based Programming And Practicals • MAT-SEC-103:Latex And Html • MAT-SEC-104:Quantitative Aptitude • MAT-SEC-105:Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Differential Equations-I	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II MAT-DSC4 (MAJ)-402: Differential Equations-I	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers • MAT-DSC5 (MAJ)-503: Differential Equations-II	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra		Internship / community work			

VI	300	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis • MAT-DSC6 (MAJ)-603: Modern Algebra	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis					
		•	•	•					

Notes.

1. Members of the Board of Studies unanimously decided not to offer mathematics in Table-3 of Arts stream.
2. Practicals (wherever included in the syllabi) may be conducted into small groups(say 15 – 20 students)to facilitate the smooth conduct of practical sessions.
3. Scientific Calculator is allowed in the Numerical Analysis Paper.
   IMPORTANT NOTE: All papers in VAC/SEC/MDC would be floated in each of the first three semesters. However, the students are not allowed to repeat any paper.

Semester - III
MAT-DSC3(MAJ/MIN)-301: THEORY OF EQUATIONS

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 6 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to provide a comprehensive understanding of polynomials and polynomial equations. Students will explore methods for solving these equations, the relationship between roots and coefficients, and various techniques for transforming polynomial equations. Additionally, they will discover key theorems and interesting properties related to polynomials, deepening their problem-solving skills in this area.

Learning Outcomes: After attending this course students will be able to
(i) Analyze the relationship between the roots and coefficients of a polynomial,
(ii) Solve polynomial equations up to degree 4 by learning certain classical methods.
(iii) Apply transformations to polynomial equations and understand their implications.
(iv) Solve complex problems involving polynomial equations and transformations.

Unit-I

Polynomials (definition and examples), Euclid's algorithm, synthetic division, common divisors, G.C.D. of polynomials. Roots of a polynomial equation, repeated roots and their multiplicity, common roots. Fundamental theorem of Algebra, Factor theorem, Complex roots of real polynomials occur inconjugate pairs with same multiplicity. Irrational roots of polynomials over rationals occur inconjugate pairs with same multiplicity.

Unit-II

Relation between roots and coefficients, Vieta's formulae, symmetric functions, diminishing roots of a polynomial equation by *h* and its application, solution of a cubic when its roots are in A.P./G.P., solution of a biquadratic when its roots are in A.P. (respect. in G.P.)and sum (resp. product) of two roots is given,Descartes'Rule ofSigns, Newton's method of divisors for integral roots.

Unit-III

Transformation of equations: Transform the given polynomial equation into another such that signs of the roots changed, roots multiplied by a constant, roots are symmetric functions of the roots of the original equation. Solutions of cubic and bi-quadratic equations when their roots are in H.P.

Unit-IV

Cardan's method and trigonometric methods for solvingcubic equations. Discriminant and nature of roots, of a real cubic equation. Descartes'andFerrari'smethodof solving a bi-quadratic equation.

References:

1. **S.R.KnightandH.S.Hall** : HigherAlgebra,H.M.Publications,1994.
2. **ChandrikaPrasad**: TextBookonAlgebraandTheoryofEquations,Pothishala PrivateLtd.,Allahabad.
3. **William S Burnside and A W Panton**: The theory of equations, Dublin, Hodges, Figgis and co. Dublin University Press Series (1881).

Semester - III
*MAT-DSC3(MAJ/MIN)-302: DIFFERENTIAL EQUATION-I

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to equip students with essential techniques for solving different types of differential equations. It also introduces the fundamentals of calculus involving two and three variables, which are crucial for understanding and solving differential equations effectively.

Learning Outcomes:By the end of this course, students will be able to:
1. Apply appropriate methods to solve various types of differential equations.
2. To understand calculus concepts for functions of two and three variables.

Unit-I

Limit and continuity of functions of two and three variables. Partial differentiation up to second order. Total differential, differentiation of composite and implicit functions. Euler's Theorem on homogeneous functions, differentiability of real-valued functions of two and three variables. Schwarz and Young's theorems (without proof).

Unit-II

Exact differential equations; Necessary and sufficient condition for a differential equation of the type M(x, y) dx + N(x ,y) dy = 0 to be exact. Integrating factor of a D.E. and methods to find it. First order and higher degree differential equations solvable for x, y, $p = \frac{dy}{dx}$. Clairaut's form and equations reducible to Clairaut's form. Singular solution as an envelope of general solutions.

Unit-III

Geometrical meaning of a differential equation, orthogonal trajectories. Homogeneous linear differential equations with constant coefficients and its solutions.

Theorems for finding particular integrals $\frac{1}{\mathrm{D}-\mathrm{a}}f(x)$, $\frac{1}{f(D)}e^{ax}$, $\frac{1}{f(D)}e^{ax}\mathrm{V}(x)$, $\frac{1}{f(D^2)}\sin x$, $\frac{1}{f(D^2)}\cos x$, $\frac{1}{f(D)}x\,V(x)$, $\frac{1}{f(x)}x^n$; n **is non negative integer.** **Non-homogeneous linear differential equations with constant coefficients and its solutions.**

*Not for those students who opt mathematics as a major subject in table-2.

Unit-IV

Solution of Homogenous and non Homogeneous Linear differential equations with variable coefficients-Cauchy and Legendre Equations. Exactness of Linear differential equations of nth order and solution of linear differential equations that can be made exact using integrating factor.

References

1. **Erwin Kreyszig**: Advanced Engineering Mathematics, John Wiley &Sons Inc., New York, 1999.
2. **D.A. Murray**:Introductory Courseon Differential Equations, Orient Longmen, (India), 1967.
3. **A.R. Forsyth**: A Treatiseon DifferentialEquations, Macmillan and Co. Ltd., London.
4. **Ross, S. L**.: Differential Equations, John Willey & Sons, 2004.

Semester - IV
MAT-DSC4 (MAJ/MIN)-401: Calculus-II

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to analyse plane curves by looking at certain key parameters such as concavity, convexity, points of inflection, multiple points, and asymptotes. Students will also explore concepts of curvature and radius of curvature for various curves. The course will cover integration techniques for hyperbolic and inverse hyperbolic functions, along with reduction formulae for essential integrals.

Learning Outcomes: By the end of this course, students will be able to:

1. Analyse and classify plane curves based on parameters like concavity, convexity, points of inflection, multiple points, and asymptotes.
2. Understand and compute the curvature and radius of curvature for different curves.
3. Perform integration involving hyperbolic and inverse hyperbolic functions.
4. Apply reduction formulae to solve integrals of varying complexity.
5. Implement numerical methods for approximate computation of definite integrals.

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Unit-I

Concavity,convexity, pointsofinflexion,multiplepoints, double pointsand its types for curves in a plane. Tangents at origin, asymptote and its types, methods for finding asymptotes of rational algebraic curves. Special methods for finding oblique asymptotes of rational algebraic curves. Intersection of a curve and its asymptotes.

Unit--II

Introduction to the polar coordinate system, tracing of curves represented by equations in Cartesian coordinates, Polar coordinates, and in parametric forms.
Curvature and radiusofcurvature,atapoint,of curves in Cartesian andPolarCo-ordinates including parametric forms as well as curves represented by equation $f(x, y) = 0$ implicitly.

Unit-III

Integralcalculus: Integrationofhyperbolicandinversehyperbolicfunctions.ReductionFormulae for following integrals.

$\int x^n e^x dx$, $\int x^m (\log x)^n dx$, $x > 0$ *m and n natural* $numbers.$ $\int sin^n x\, dx$, $\int cos^n x\, dx$, $\int tan^n x\, dx$, $\int cot^n x\, dx$, $\int sec^n x\, dx$, $\int cosec^n x\, dx$, $\int sin^n x\ e^x\, dx$, $\int cos^n x\ e^x\, dx$, $\int x\ sin^n x\, dx$, $\int x\, cos^n x\, dx$, $\int x^m\ sin nx\, dx$, $\int x^m\ cos nx\, dx$, $\int x^m \cos nx\, dx$, $\int x^m \sin nx\, dx$, $\int cos^m x \cos nx\, dx$, $\int cos^m x \sin nx\, dx$, $\int cos^m x\ sin^n x\, dx$

$\int e^{ax} sin^n bx\, dxn \geq 2$ integer, $\int e^{ax} cos^n bx\, dxn \geq 2$ integer, $\int_0^{\frac{\pi}{2}} sin^n x\, dx\ \ n \in N$

$\int_0^{\frac{\pi}{2}} cos^n x\, dx\ \ n \in N$, $\int \frac{\sin nx}{\sin x}\, dx$, $\int x^m (a + bx^n)^p\, dx.$

Unit-IV

NumericalIntegration: Trapezoidal,PrismoidalandSimpsonRules.

Applicationofdefiniteintegral:SummationofSeries,Quadrature,rectification,volumesandsurfacesofsolidsofrevolution(Cartesianco-ordinatesonly)

References:

1. **G.B.Thomas& R.L.Finney: CalculusandAnalyticGeometry(Ninthedition),PearsonPublication.**
2. **GabrielKlambauer**:MathematicalAnalysis,MarcelDekkar,Inc.NewYork, 1975.
3. **N.Piskunov**:DifferentialandIntegralCalculus,PeacePublishers,Moscow.
4. **P. K.Jainand S.K.Kaushik:AnIntroductiontoReal Analysis,S.Chand&Co.NewDelhi,2000.**

Semester - IV

[1]MAT-DSC4 (MAJ)-402: DISCRETE MATHEMATICS AND GRAPH THEORY

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: To provide basic knowledge about mathematical logic, lattices, Boolean algebra, graph theory and coding theory.

Learning Outcomes: This course will develop logical reasoning and rational thinking of the students.

UNIT – I

Mathematical Logic: Introduction, Conjunction, Disjunction & negation, Propositions and truth table, Tautologies andcontradictions, Equivalence of formulas, Duality law, Normal forms: Disjunctive Normal form,Conjunctive Normal form,

Predicate Calculus: Predicates, the statement function, variables andquantifiers, predicate formulas. Methods of proof (Inference Theory).

UNIT – II

Ordered Sets, Latices, Boolean Algebra : Partially Ordered Sets, External elements of POSET, HASSE Diagrams of POSETS, Lattices, Finite Boolean Algebra, Functions on Boolean algebra, Circuit Designs.

UNIT – III

Graphs: Incidence and Degree, Isomorphism, Connectedness, Walk, Path and Circuits, Shortest Path algorithm between two Vertices, Eulerian graph, Directed graphs, Kuratowski's graphs; Detection of planarity.

Trees: Properties of Trees, Rooted and Binary Trees; Directed Tree, Spanning Tree and tree traversals (In-order, Pre-order and Post-order)

UNIT – IV

Functions: Composite functions & their range, domain, Functions for computer science like characteristicfunction, Hashing function, Growth of a function.

Coding Theory: Recursion, Recurrence Relation, Coding of binary Information & Error detection, Decoding &error correction.

[1]For students who haveopted mathematics as a major subject in table-1.

Suggested Readings:

1. Kolman, Busby, Ross: discrete Mathematical Structure, PEARSON Education, 5th Edition.
2. Schaum Series: Theory and Problems of Essential Computer Mathematics, McGraw Hill,New York, 3rd Edition.
3. Sengadir,T. : Discrete Mathematics and Combinatorics, PEARSON Education.
4. Kenneth.H.Rosen: Discrete Mathematics and its Applications, Mc GRAW Hill, International Edition.
5. Deo, N. : Graph Theory with Applications to Engineering and Computer Science, PHI.
6. Clark, John &Hetlan: A First Look at Graph Theory, Allied Publishers Limited.

Semester - IV

[2*]MAT-DSC4 (MAJ)-403: COORDINATE GEOMETRY -I

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The main goal of this course is to deliver basics of plane geometry. The course introduces the concept of pair of straight lines, circles and other conics. The objective is to understand the geometrical concepts related to standard conic sections and their properties.

Learning Outcomes:
The students will be able to determine the joint equation of a pair of lines and angle bisectors, equation of a circle in different forms, equation of tangents, normal and chord of contact. The students will also be able to identify a conic from a general second degree equation.

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Unit-I

Transformation of axes in two dimensions: Shifting of origin, rotation of axes, invariants. **Pair of Straight lines**: Joint equation of pair of straight lines and angle between them, Condition of parallelism and perpendicularity, Joint equation of the angle bisectors, Jointequation of lines joining origin to the intersection of a line and a curve.

Unit-II

Circle : General equation of circle, Circle through intersection of two lines, tangents, normals, chord of contact, pole and polar, pair of tangents from a point, equation of chord in terms of mid-point, angle of intersection and orthogonality.

Unit-III

Conic : General equation of conics in Cartesian and Polar coordinate system, parametric formsof conics, equations of tangents, normal, chords of contact, pole and polar, pair of tangents from a point, equation of chord in terms of mid-point.

Unit-IV

Diameter and conjugate diameters of the conics. Special properties of Parabola, Hyperbola and Ellipse. Conjugate hyperbola, asymptotes to the hyperbola, rectangular hyperbola. Identification of conic in general second degree equations.

*For students who haveopted mathematics as a major subject in table-1.

References:

1.S.L. Loney:*The Elements of Coordinate Geometry,* Macmillan andCompany,London, 2nd Edition 2007.

2.P.K. Jain and Khalil Ahmad:*A Text Book of Analytical Geometry of two dimensions,Wiley Eastern Ltd., 1999.*

3.Erwin Kreyszig:*Advanced Engineering Mathematics, John Wiley & Sons,1999.*

4. Gorakh Prasad and H.C. Gupta:*Text Book on Coordinate Geometry, Pothishala Pvt. Ltd., Allahabad, 1955.*

5. G. B. Thomas and R. L. Finney: Calculus and Analytic Geometry, Addison-Wesley Publishing Company.

Semester - IV
*MAT-DSC4 (MAJ)-404: NUMERICAL ANALYSIS

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)
Imp. Remark: Normal / Scientific calculator is allowed in the examination.

Note: **1.** The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to equip students with essential techniques for solving different types of mathematical problems that arise in various fields. It also introduces the fundamentals of calculus involving two and three variables, which are crucial for understanding and solving ordinary differential equations effectively.

Learning Outcomes: By the end of this course, students will be able to:

1. Apply appropriate methods to solve various types of differential equations.
2. To understand calculus concepts for functions of two and three variables.
3. Apply direct methods like Gauss Elimination, LU Decomposition, and Cholesky Decomposition, as well as iterative methods such as Jacobi, Gauss-Seidel, and relaxation methods, to solve systems of linear equations.

Unit-I

Solution of Equations: Bisection Methods, Regula Falsi Method, Secant Method, Newton-Raphson Method, Roots of Polynomials.
Interpolation: Lagrange and Hermite Interpolation, Newton's Divided Differences, Difference Schemes, Interpolation Formulas using Differences.

UNIT II.

Numerical Differentiation Numerical Quadrature: Newton-Cote's Formulas, Gauss Quadrature Formulas , Chebyshev's Formulas.

Unit-III

Linear Equations: Direct Methods for Solving Systems of Linear Equations (Gauss Elimination, LU Decomposition, Cholesky Decomposition), Iterative Methods (Jacobi, Gauss-Seidel, Relaxation Methods).

Unit-IV

The Algebraic Eigenvalue problem: Jacobi's Method, Givens' Method, Householder's Method, Power Method, QR Method, Lanczos' Method.

*For students who haveopted mathematics as a major subject in table-1.

OrdinaryDifferentialEquations:EulerMethod,Single-stepMethods,Runge-Kutta'sMethod,Multi-stepMethods.

References

1. **C.E.Froberg**:*IntroductiontoNumericalAnalysis*(SecondEdition),Addison-Wesley, 1979.
2. **MelvinJ.Maron**:*NumericalAnalysis:APracticalApproach,*MacmillanPublishingCo., New York, 1982.
3. **M.K.Jain,S.R.K.Iyengarand R.K.Jain**:*NumericalMethodsforScientificand Engineering Computation,* New Age International (P.) Ltd., 1999.
4. **R.Y.Rubistein**:*SimulationandtheMonteCarloMethods,*JohnWiley,1981.
5. **D.J.Yakowitz**:*ComputationalProbabilityandSimulation,*Addison-Wesley,1977.
6. **S.S.Sastry**:*IntroductoryMethodsofNumericalAnalysis,*3rdEdition(2000),Prentice Hall of India Pvt. Ltd., New Delhi.

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Semester - IV
*MAT-DSC4 (MAJ)-405: DIFFERENTIAL EQUATION-I

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to equip students with essential techniques for solving different types of differential equations. It also introduces the fundamentals of calculus involving two and three variables, which are crucial for understanding and solving differential equations effectively.

Learning Outcomes:By the end of this course, students will be able to:

1. Apply appropriate methods to solve various types of differential equations.
2. To understand calculus concepts for functions of two and three variables.

Unit-I

Limit and continuity of functions of two and three variables. Partial differentiation up to second order. Total differential, differentiation of composite and implicit functions. Euler's Theorem on homogeneous functions, differentiability of real-valued functions of two and three variables. Schwarz and Young's theorems (without proof).

Unit-II

Exact differential equations; Necessary and sufficient condition for a differential equation of the type M(x, y) dx + N(x ,y) dy = 0 to be exact. Integrating factor of a D.E. and methods to find it. First order and higher degree differential equations solvable for x, y, $p = \frac{dy}{dx}$. Clairaut's form and equations reducible to Clairaut's form. Singular solution as an envelope of general solutions.

Unit-III

Geometrical meaning of a differential equation, orthogonal trajectories. Homogeneous linear differential equations with constant coefficients and its solutions.

Theorems for finding particular integrals $\frac{1}{\mathrm{D}-a}f(x)$, $\frac{1}{f(D)}e^{ax}$, $\frac{1}{f(D)}e^{ax}\mathrm{V}(x)$, $\frac{1}{f(D^2)}sin\,x$, $\frac{1}{f(D^2)}\cos x$ **. Non-homogeneous linear differential equations with constant coefficients and its solutions.**

*Only for those students who have optedmath as a major subject in table-2.

Unit-IV

Solution of Homogenous and Non Homogeneous Linear differential equations with variable coefficients-Cauchy and Legendre Equations. Exactness of Linear differential equations of nth order and solution of linear differential equations that can be made exact using integrating factor.

References

2. **Erwin Kreyszig**: Advanced Engineering Mathematics, John Wiley &Sons Inc., New York, 1999.
3. **D.A. Murray**:Introductory Courseon Differential Equations, Orient Longmen, (India), 1967.
4. **A.R. Forsyth**: A Treatiseon DifferentialEquations, Macmillan and Co. Ltd., London.
5. **Ross, S. L**.: Differential Equations, John Willey & Sons, 2004.

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MICROBIOLOGY

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(MIC-DSC-3 MIN - 301)	Molecular Biology	3+1=4	24	III	L-200	(MIC-DSC-3 MIN - 301)	Molecular Biology	3+1=4	24
		(MIC-DSC-4 MIN - 302)	Bacteriology and Virology	3+1=4				(MIC-DSC-4 MIN - 302)	Bacteriology and Virology	3+1=4	
								DSC-MAJ	As opted by the student	3+1=4	
		DSC-MAJ	As opted by the student	3+1=4				DSC-MAJ	As opted by the student	3+1=4	
		DSC-MAJ	As opted by the student	3+1=4							
		(MIC-MDC-3-301)	Entrepreneurship in Microbial biotechnology	3				(MIC-MDC-3-301)	Entrepreneurship in Microbial biotechnology	3	
		(MIC-SEC-3-301)	Infield aspects and Entrepreneurship Microbiology	3				(MIC-SEC-3-301)	Infield aspects and Entrepreneurship Microbiology	3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(MIC-DSC-5 MIN - 401)	Immunology	3+1=4	24	IV	L-200	(MIC-DSC-5 MIN - 401)	Immunology	3+1=4	24
		DSC-MAJ	As opted by the student	3+1=4				DSC-MAJ	As opted by the student	3+1=4	
		DSC-MAJ	As opted by the student	3+1=4				DSC-MAJ	As opted by the student	3+1=4	
		DSC-MAJ	As opted by the	3+1=4				DSC-MAJ	As opted by the	3+1=4	

			student						student		
		DSC- MAJ	As opted by the student	3+1=4				DSC-MAJ	As opted by the student	3+1=4	
		(VAC-3-401)	To be opted by the student	2				(VAC-3-401)	To be opted by the student	2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Notes:

Table 1*						**Table 2***					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+ Pr+ Tut)**	**Total Credits**
V	L-300	DSC-MAJ	To be opted by the student	3+1=4	24	V	L-300	MIC-DSC-9 MIN - 501	Agro and Industrial Microbiology	3+1=4	24
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		*(INT/COM-506)	Internship/Community Work	4				*(INT/COM-506)	Internship/Community Work	4	
VI	L-300	(DSC-14 /Min-601)	To be opted by the student	3+1=4	24	VI	L-300	MIC-DSC-14 MIN - 601	Food Microbiology	3+1=4	24
		DSC-MAJ	To be opted by the student	3+1=4				MIC-DSC-15-MIN - 602	Microbial Pathogenesis	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	

		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4
72 (Maj)+32(Min)+9(MDC)+8(AEC)+6(VAC)+9(SEC)+4(INT/COM)+4(PBI/HCP)										
The student successfully securing 144 credits will be awarded B.Sc.degree										

***The internship will consist of a 4-credit course (100 marks) comprising 30 hours (2 hours per week) of practical or field training. The internship may be completed within the parent institute or through institute/industrial training for 2–4 weeks, with a minimum requirement of 30 hours. Evaluation will be based on the project report and a detailed presentation of the internship. The internal assessment will constitute 10% of the total marks, as per university and institutional guidelines.**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

SEMESTER- III

MIC DSC-3 MIN-301: MOLECULAR BIOLOGY (THEORY)

Total marks:75
Theory: 68
Internal assessment:07
Time: 3hours
Credits: 03

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objective and Learning Outcomes: Students should be able to learn the inheritance material and molecular mechanisms of genetic material. Students will be able to understand the structure and organization of genomic material in prokaryotes and eukaryotes; analyzing the mechanisms of DNA replication, transcription, translation, and gene regulation; and integrating theoretical knowledge with practical skills in DNA isolation, electrophoresis, plasmid analysis, and restriction digestion to interpret molecular processes underlying gene expression.

UNIT-I

DNA: **Chemical composition and structure of single and double stranded DNA.**

Types of biological significance of : A-DNA, B-DNA, Z-DNA, D-DNA, E-DNA.

Genome organization in prokaryotes: Composition and structure of prokaryotic DNA and RNA.

Genome organization in eukaryotes: Composition and structure of eukaryotic DNA. Nucleosome model and DNA packaging in eukaryotes.

Repetitive DNA: **Tandemly repetitive DNA and Mini and microsatellite DNA.**

Transposons: **Insertional elements, Composite transposons, retrotransposons, their role and importance.**

UNIT-II

DNA replication: **Prokaryotic DNA replication (Origin and site of replication, DNA DNA polymerases (I, II, III), replication factors, leading strand and lagging strand synthesis, Termination regions). Procesessivity and fidelity of DNA.**

Replication of single stranded DNA, M13 viral DNA.

Eukaryotic DNA replication: **Concept of replicon, Origin and site of replication, Pre- RC complex, licensing factor, replication factors, eukaryotic DNA polymerases and their characteristics, leading strand and lagging strand synthesis.**

End DNA replication: **Telomerase, mode of action and biological significance. Disassembly of chromatin components and reassembly during replication.**

UNIT-III

Gene Expression: Overview of central dogma

RNA: **rRNAs; Structural features of rRNAs (prokaryotic and eukaryotic).** tRNAs: **structural features, their anticodon feature.** mRNAs: **prokaryotic and eukaryotic mRNAs, structural features.**

Transcription: **RNA Polymerase (Structure and function), transcription factors, mechanism of transcription in prokaryotes and eukaryotes,**

UNIT IV

Genetic code: Characteristics of the genetic code, Wobble hypothesis

Translation: **Initiation factors, activation of amino acids and charging of tRNA Mechanism of translation process in prokaryotes and eukaryotes.**

Gene Regulation: **Regulation of gene expression in response to environmental conditions (Temperature, chemicals and light).**

Operon system**: Lactose operon, Tryptophan operon, catabolite repression, attenuation.**

REFERENCE BOOKS:

1. Lewin, B. (2004). Genes VIII. Oxford University Press, Noida, India.
2. Ingraham, J. L. (2004). Introduction to Microbiology (3rd ed.). Thomson Brooks/Cole, USA.
3. Malacinski, G. M. (2005). Freifelder's Essentials of Molecular Biology. Narosa Publishing House, New Delhi, India.
4. Karp, G. (2005). Cell and Molecular Biology. Wiley International, USA.
5. Snustad, D. P., & Simmons, M. J. (2006). Principles of Genetics. John Wiley & Sons, Inc., USA.
6. Brown, T. A. (2007). Genomes 3. Garland Science, USA.
7. Weaver, R. F. (2005). Molecular Biology. McGraw-Hill, USA.
8. Molecular cloning: A laboratory manual Vol. I-III Sambrook J. and Russel, David N. 2001 Cold Spring Harbor.

MIC DSC-3 MIN-301: MOLECULAR BIOLOGY -(PRACTICAL)

Total Credit:1
Max Marks 25
External Exam :22
Internal assessment 3
Time: 3 hours

1. Preparation of Reagents for DNA isolation.
2. Isolation of Plasmid DNA from bacteria.
3. Isolation of genomic DNA from Bacteria.
4. Agarose gel electrophoresis of DNA.
5. Spectrophotometeric analysis of DNA.
6. Isolation of DNA fragment from agarose gel.

MIC-DSC-4 MIN-302: BACTERIOLOGY AND VIROLOGY (THEORY)

Total marks:75
Theory: 68
Internal assessment:07
Time:3 hours
Credits: 03

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objective And Learning outcome: students should be able to learn the Organization and Fundamental concepts of Bacteria and viruses, their origin, isolation, cultivation, characterization of viruses.

Unit-I

Bacterial Systematics

Aim and principles of classification, systematics and taxonomy, concept of species, taxa, strain; Characters used in bacterial systematic. Differences between eubacteria and archaebacteria

Important archaeal and eubacterial groups

Archaebacteria: General characteristics of genera belonging to Nanoarchaeota (Nanoarchaeum), Crenarchaeota (Sulfolobus, Thermoproteus) and Euryarchaeota [Methanogens (Methanobacterium, Methanocaldococcus), thermophiles (Thermococcus, Pyrococcus, Thermoplasma), and Halophiles (Halobacterium, Halococcus)]

Eubacteria: Morphology, metabolism, ecological significance and economic importance of following groups: Gram Negative: General characteristics:Non proteobacteria, Alpha proteobacteria: Beta proteobacteria, Delta proteobacteria, Epsilon proteobacteria, Zeta proteobacteria. Gram Positive: Low G+ C (Firmicutes), High G+C (Actinobacteria). Cyanobacteria: An Introduction.

UNIT-II

Basics in Medical microbiology : Human diseases caused by Gram postive Bacteria (*Staphylococcus aureus*, *Mycobacterium tuberculosis*) Gram negative Bacteria (*Escherichiae coli*) , other bacteria Chlamydiae, Rickettsiaceae, Mycoplasma -Etiology, Epidemiology ,route of transmission, Pathogenicity and virulence, laboratory diagnosis and Treatments .

Human diseases caused by fungi and protozoa: *Candida albicans*, *Plasmodium falciparum* Etiology, Epidemiology, route of transmission, Pathogenicity and virulence, laboratory diagnosis and Treatments.

Unit-III

Discovery of viruses, nature and definition of viruses, general properties, classification of viruses. Concept of viroids, virusoids, satellite viruses and Prions. Structure of Viruses: Capsid symmetry, enveloped and non-enveloped viruses Isolation, purification and cultivation of viruses. Lytic and lysogenic cycle of T4/T2 phage.

UNIT-IV

Viruses and Cancer Introduction to oncogenic viruses Types of oncogenic DNA and RNA viruses: Concepts of oncogenes and proto-oncogenes.

Cultivation and purification of viruses: Different in vivo, in vitro and in ovo growth systems for bacterial, plant and animal viruses.

SUGGESTED READINGS

1. Ananthanarayan And Paniker (2017). Textbook Of Microbiology 10th Edition.
2. Dimmock, NJ, Easton, AL, Leppard, KN (2016). Introduction to Modern Virology. 7th edition, Blackwell Publishing Ltd.
3. Murray PR, Rosenthal KS, Kobayashi GS, Pfaller MA.(2020) Medical Microbiology. 9th edition, Mosby, Inc.
4. Carter J and Saunders V (2007).Virology: Principles and Applications. John Wiley and Sons.
5. Flint SJ, Enquist, LW, Krug, RM, Racaniello, VR, Skalka, AM (2004).
6. Flint, S. Jane, et al. Principles of virology,pathogenesis and control. Vol. 2. John Wiley & Sons, 2020.
7. Willey JM, Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
8. Gillespie, S. H., & Hawkey, P. M. (Eds.). (2006). *Principles and practice of clinical bacteriology*. John Wiley & Sons. 2nd edition.
9. Cappucino J and Sherman N. (2010). Microbiology: A Laboratory Manual. 9th edition. Pearson Education Limited.

MIC-DSC-4 MIN-302: BACTERIOLOGY AND VIROLOGY

(PRACTICAL)

Total marks:25
Theory: 22
Internal assessment:3
Time: 3hours
Credits: 01

1. Isolation of phages from water.
2. Antibiotic sensitivity testing of bacteria (disc diffusion method).
3. IMViC test.
4. Study of bacterial growth curve.
5. Enumeration of bacteriophage by plate assay.
6. Serological test for viral diagnosis.

SEMESTER IV

MIC-DSC-5 MIN 401: IMMUNOLOGY (THEORY)

Total marks:75
Theory: 68
Internal assessment:07
Time: 3 hours
Credits: 03

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objective And Learning outcome: Students will understand the fundamental principles of innate and adaptive immunity and the roles of immune cells and lymphoid organs. They will be able to explain antigen–antibody interactions, immunogenicity, and the molecular basis of humoral and cell-mediated immune responses. Learners will gain knowledge of immunoglobulins, MHC, and T-cell and B-cell activation and regulation. They will also acquire basic laboratory skills used in immunological analysis and diagnosis.

UNIT-I

Introduction **Overview of immune system, Historical perspectives, Innate and acquired immunity, Clonal nature of immune response.**
Cells of the immune system : **Hematopoiesis and differentiation, lymphocyte trafficking, B-lymphocytes, T-lymphocytes, macrophages, dendritic cells, Natural killer cells, eosinophils, neutrophils & mast cells.**

UNIT-II

Organs of the immune system: Primary and secondary lymphoid organs, systemic function of immune system.

Antigen**: Immunogenicity Vs. antigenicity, factors effecting immunogeniticity, nature of immunogen, epitopes, heptans and antigenicity, pattern recognition receptors.**

UNIT-III

B-Cell Activation, Differentiation: B-Cell Activation and Proliferation, In Vivo Sites for Induction of Humoral Responses, T-dependent and T-independent antigens.
Immunoglobulins: **Structure of antibody, antibody effector function, antibody classes and biological activities,antigenic determinants on Immunoglobulins, Immunoglobulins superfamilies, generation of antibody diversity.**

UNIT-IV

Major histocompatibility complex: **General organization and inheritance, MHC molecules and genes, genetic map, cellular distribution, regulation of MHC expression and disease susceptibility.**

T-Cell Maturation, Activation, and Differentiation:T-Cell Receptors: Structure and Roles, T- Cell Receptor Complex, Thymic Selection of the T-Cell Repertoire, T_H-Cell Activation, T-Cell Differentiation, Cell Death and T-Cell populations, Peripheral-T-Cells.

Reference Books:

1. *Kuby Immunology* (2006) by Thomas J. Kindt, Richard A. Goldsby, Barbara A. Osborne, Janis Kuby (W.H. Freeman).
2. *Immunology – A Short Course* (2009) by Richard Coico, Geoffrey Sunshine (Wiley-Blackwell).
3. *Fundamentals of Immunology* (1999) by William Paul (Lippincott Williams & Wilkins).
4. *Immunology* (2001) by Ivan Maurice Roitt, Jonathan Brostoff, David K. Male (Mosby).
5. Kindt, T.J., Goldsby, R.A., Osborne, B.A., Kuby, J. *Immunology* (6th Ed.). W.H. Freeman & Co., New York, 2006.
6. Delves, P.J., Martin, S.J., Burton, D.R., Roitt, I.M. *Roitt's Essential Immunology*.

MIC-DSC-5 MIN 401: IMMUNOLOGY (PRACTICAL)

Total marks:25
Theory:22
Internal assessment:3
Time: 3 hours
Credits: 01

1. Lymphoid organs and their microscopic organization.
2. Differential leucocytes count.
3. Separation of serum from blood.
4. Separation of plasma from blood.
5. Agglutination (Blood group testing).
6. Isolation of mononuclear cells from peripheral blood and viability test by dye exclusion methods.

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PHYSICS

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(PHY-DSC-3 Maj/Min-301)	Optics and Lasers	3+1=4 (x2)	24	III	L-200	(PHY-DSC-3 Maj/Min-301)	Optics and Lasers	3+1=4 (x3)	24
		(PHY-DSC-4 Maj/Min-302)	Vibrations, Waves & E.M. Theory	3+1=4(x2)				(PHY-DSC-4 Min-302)	Vibrations, Waves & E.M. Theory	3+1=4	
		(PHY-MDC-3-301)	Basic Computational Tools	3							
		(PHY-SEC-3-301)	Physics instrumentation Skills-III	3				(PHY-MDC-3-301)	Basic Computational Tools	3	
		(AEC-3-301)	Eng-3/MIL-1	2				(PHY-SEC-3-301)	Physics instrumentation Skills-III	3	
								(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(PHY-DSC-5 Maj/Min-401)	Quantum Mechanics	3+1=4 (x2)	24	IV	L-200	(PHY-DSC-5 Maj/Min-401)	Quantum Mechanics	3+1=4 (x3)	24
		(PHY-DSC-6 Maj-402)	Statistical Physics and Thermodynamics	3+1=4				(PHY-DSC-6 Maj-402)	Vibrations, Waves & E.M. Theory (Same Syllabus as PHY-DSC-4 Min-302 in Sem III)	3+1=4 (x2)	
		(PHY-DSC-7 Maj-403)	Condensed Matter Physics - I	3+1=4							

								(PHY-VAC-3-401)	Basics of Communication System	2	
		(PHY-DSC-8 Maj-404)	Electronics and Solid-State Devices - I	3+1=4							
		(PHY-VAC-3-401)	Basics of Communication System	2							
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

B.Sc. Semester 3
PHY-DSC-3 Maj/Min-301
(Optics and Lasers)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-3 Hrs per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- To understand the principles and conditions required for interference phenomena, including spatial and temporal coherence, and explore applications like Young's double-slit experiment, Newton's rings, and interferometry.
- To study the theory and applications of diffraction, including Huygens-Fresnel principles, resolving power, and diffraction gratings. To gain insight into polarization phenomena, double refraction, and the tools for producing and analyzing polarized light, such as Nicol prisms and retardation plates.
- To explore the fundamentals of lasers, their working principles, types, and applications, with a focus on coherence, population inversion, and different laser systems.
- To understand the construction, properties, and applications of optical fibers, including their design, numerical aperture, and light transmission losses.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Interference: Concept of coherence, spatial and temporal coherence, coherence time, coherence length, area of coherence. Conditions for observing interference fringes, Interference by wavefront division and amplitude division, Young's double slit experiment, Lloyd's mirror and Fresnel's biprism, phase change on reflection, Newton's rings, Michelson interferometer—working, principle and nature of fringes, Interference in thin films, Role of interference in anti-reflection, Multiple beam interference, Fabry-Perot interferometer.

Diffraction: Huygen-Fresnel theory half period zones, zone plates, resolving power of microscope and telescope, diffraction grating, resolving power,

Polarization: Concept and analytical treatment of unpolarised, plane polarized and elliptically polarized light, Double refraction, Nicol prism, sheet polarisers, retardation plates, Production and analysis of polarized light (quarter and half wave plates).

UNIT-II

Laser Fundamentals: Interaction of light with matter: Absorption, spontaneous emission, stimulated emission, Wave mechanical explanation, Properties of Spectral Lines, Temporal and spatial coherence, Characteristics of stimulated emission, Einstein coefficients and their relations, Light amplification and threshold condition, Population inversion, Doppler broadening & Natural broadening (qualitative account only), Principal pumping schemes, Three level and four level laser schemes.

Laser Systems: Types of lasers, Construction and working of Ruby, Nd: YAG, He–Ne, Dye and CO_2 lasers. Semiconductor laser (Qualitative idea), Applications of lasers—a general outline, Holography: Principle, recording of hologram and reconstruction of image. (qualitative only).

Fiber Optics: Photonics, Optical fibre, Construction, Numerical aperture, acceptance angle, skip distance, Step index fibre – single mode and multimode, Graded index fibre, Losses in optical fibre, Applications of optical fibre.

Books Suggested:

Essential Readings:

1. Optics, Jenkins and White, McGraw Hill.
2. Optics, Ajoy Ghatak, McMillan India.
3. Physics for Degree Students, C. L. Arora and P .S. Hemne, S. Chand & Co., 2014.
4. Laser Fundamentals, W.T. Silfvast, Foundation Books.
5. Lasers and Non-linear Optics, B.B. Laud, Wiley Eastern Limited.
6. A text book of Optics, Subtahmanyam N., Lal B. and Avadhanulu, M. N., S. Chand & Co., 2012.
7. Optical Fiber Communication, Keiser, MH.

Further Readings:

1. Optics, Born and Wolf, Pergamon.
2. Lasers, SveltoPergmon.

PHYSICS PRACTICALS
(PHY-DSC-3 Maj/Min-301)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 2Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal ssessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To find the angle of prism by using a spectrometer.
2. To measure the height of inaccessible objects using sextant.
3. To determine the refractive index of the material of the given prism using a spectrometer.
4. To determine the refractive index of given transparent liquid using spectrometer.
5. To set up a spectrometer and study variation of the angle of deviation with angle of incidence (i) for a prism. Measure the angle of minimum deviation (δ) directly and compare it with the value from i-δ curve.
6. To study the variation of refractive index with the wavelength of the spectral lines of the mercury source and hence calculate the Cauchy's constants.
7. To determine the refractive index of doubly refracting prism.
8. To determine the wavelength of sodium light by using a plane diffraction grating.
9. To determine the wavelength of sodium light by Newton's ring method.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried & Jitender Chhabra.

B.Sc. Semester 3

(PHY-DSC-4 Min-302) for Table 2
(PHY-DSC-4 Maj/Min-302) for Table 1

(Vibrations, Waves & E.M. Theory)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- To understand SHM and analyze damped and forced oscillations in mechanical and electrical systems.
- To apply differential equations to study oscillatory motion and interpret damping, Q-factor, and resonance.
- To explain wave propagation in media, including energy transfer, reflection, and group velocity.
- To use Maxwell's equations to describe EM waves, energy flow, and wave behavior in various media.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Simple harmonic motion, energy of a SHM, Compound Pendulum, Torsional Pendulum, Electrical Oscillations, Transverse Vibrations of a mass on a string, Decay of free vibrations due to damping, differential equation of motion, types of damping, determination of damping co-efficient; Logarithmic decrement, relaxation time and Q- Factor, Electromagnetic damping (Electrical oscillator).

Differential equation for forced mechanical and electrical oscillators, Transient and steady state behaviour, Displacement and velocity variation with driving force frequency, variation of phase with frequency, resonance. Power supplied to an oscillator and its variation with frequency, Q-value and band width

UNIT-II

Waves in physical media, Wave equation and its solution, Types of waves, particle velocity, acceleration and energy in progressive waves, Transverse waves on a string, characteristic impedance of a string, Reflection and Transmission of transverse waves on a string at discontinuity, Reflection and transmission

of energy, Reflection and transmission of longitudinal waves at a boundary. Impedance matching, Energy of vibrating string, Wave and group velocity

Maxwell's equations, E.M. waves and wave equation in a medium having finite permeability, permittivity and conductivity, Energy flow due to EM wave - Poynting vector, Impedance of a dielectric to EM waves, EM waves in a conducting medium and skin depth, Impedance and Refractive index of a dielectric and a conductor.

Books Suggested:

Essential Readings:

1. Text Book of Vibrations and Waves by S.P. Puri (Macmillan India Ltd.).
2. Physics of Vibrations and Waves by H.J. Pain, ELBS & John Wiley, London.

Further Readings:

1. Vibrations and Waves by A.P. French (Arnold Heinemann India, New Delhi).
2. The Mathematics of Waves and Vibrations by P.K. Ghosh (Mcmillan India).
3. Waves and Oscillations by N. Subrahmanayam & B. Lal (Vikas Pub., Delhi).

PHYSICS PRACTICALS
(PHY-DSC-4 Min-302) for Table 2
(PHY-DSC-4 Maj/Min-302) for Table 1

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal ssessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To determine time period of a simple pendulum.
2. To compare the moment of inertia of solid sphere and hollow sphere (or a solid disc and annular disc) of same mass using torsion pendulum. Hence show that M.I. depends upon distribution of masses.
3. To measure the time periods of oscillations using objects of various geometrical shapes but the same mass using Torsional pendulum.
4. To study the dependence of moment of inertia on distribution of mass by noting time periods of oscillations using objects of various geometrical shapes but of same mass.
5. Measure time period as a function of distance of centre of suspension (oscillation) from centre of mass, plot relevant graphs, determine radius of gyration and acceleration due to gravity.
6. To find the value of g by Kater's or Bar pendulum.
7. Measure time period of oscillation of a Maxwell needle and determine modulus of rigidity of the material of a given wire.
8. To measure/obtain logarithmic decrement, coefficient of damping, relaxation time, and quality factor of a damped simple pendulum.
9. To determine the frequency of electrically maintained tuning fork by means of Melde's Experiment.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal

B.Sc. Semester 4
PHY-DSC-5 Maj/Min-401
(Quantum Mechanics)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+5 marks (internal assessment 20%)]
Teaching hours: Theory-3Hrs per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- Understand the concept of wave-particle duality and its experimental demonstrations through phenomena like the photoelectric effect, Compton effect, and De Broglie waves.
- Learn the fundamentals of quantum mechanics, including the Schrödinger equation (time-dependent and time-independent), Born's interpretation of wave functions, and concepts such as eigenfunctions, eigenvalues, and operators.
- Explore applications of the Schrödinger equation to physical systems like particles in a box, potential steps, tunneling, harmonic oscillators, and the hydrogen atom's energy levels and spectra, Zeeman effects, spin-orbit coupling, and angular momentum.
- Examine the production, properties, and applications of X-rays, including X-ray spectra, Moseley's law, and the Auger effect, providing insights into their theoretical and practical implications.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Wave-particle duality – Photoelectric effect, Compton effect, De Brogile waves, wave packet, Phase velocity and Group velocity, Uncertainty principle and its applications, Time-dependent and Time independent Schrodinger wave equation, Born's interpretation of Wave function, normalization, orthogonality condition, Eigen function and eigen value, Operator formalism, Position, momentum and energy operators, Hermitian operator expectation values, Ehrenfest theorem.

Solution of Schrodinger wave equation for particle in one-dimension box, Potential step (E<V and E> V), Tunnel effect (Qualitative), Linear harmonic oscillator, SchrÖdinger equation for spherically symmetric potential, spherical harmonics,

UNIT-II

Spectrum of hydrogen atom, Normal Zeeman effect, Degeneracy of H-atom energy levels, fine structure, Electron orbital and spin angular momentum and magnetic moments, Larmor's frequency, spin-orbit

coupling, total angular momentum, examples of one electron systems, Anomalous Zeeman effect and its application to sodium D-lines.

Symmetric and Antisymmetric wave functions, exclusion principle, Many electron atoms, electronic configurations, Hund's rule, Spin-Orbit coupling, L-S coupling, J-J couplings, term symbols. Atomic spectra of Na and He

Production and properties of X-rays, Types of X-rays spectra, Absorption of X-rays, Mosley law, Auger effect

Books Suggested:

Essential Readings:

1. Concept of Modern Physics, A. Beiser, S. Mahajan and S. R. Choudhury, Tata McGraw Hill, 6th Edition.
2. Quantum Physics of Atoms, Molecular, R. Eisberg &R.Resnick, Second Edition, John Wiley, 2002.
3. Physics for Degree Students, C.L. Arora and P.S. Hemne, S. Chand & Co., 2014
4. Elements of Modern Physics, S.H. Patil, McGraw Hill, 1998.

Further Readings:

1. Modern Physics, J. Bernstein, P.M. Fishbane, S.G. Gasiorowicz, Pearson, 2000.
2. Elements of Modern Physics, S.H. Patil, McGraw Hill, 1998.
3. Quantum Mechanics, E. Merzbacaher, II Ed., John Wiley.
4. Fundamentals of Molecular Spectroscopy, C.N. Banwell, T.M.H., 1972
5. Atomic and Molecular Spectra, Rajkumar (Kedarnath Ramnath Prakashan, Meerut).
6. Atomic Spectra, H.G. Kuhn, Academic Press, N.Y.

PHYSICS PRACTICALS
(PHY-DSC-5 Maj/Min-401)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**
Teaching hours: 2Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal ssessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals

1. To study the Photoelectric effect and determine the value of Planck's constant.
2. To study the variation of photoelectric current with anode potential in case of photo cell.
3. To study the gas discharge spectrum of hydrogen.
4. To study the absorption spectra of iodine vapours.
5. To determine the ionization potential of mercury.
6. Study of variation of light intensity using photovoltaic cell/inverse square law.
7. To determine the wavelength of a given He-Ne laser.
8. To verify uncertainty principle using laser.
9. To determine the resolving power of a telescope.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried & Jitender Chhabra.

B.Sc. Semester 4
(PHY-DSC-6 Maj-402)
(Vibrations, Waves & E.M. Theory)
(for Table-2)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- To understand SHM and analyze damped and forced oscillations in mechanical and electrical systems.
- To apply differential equations to study oscillatory motion and interpret damping, Q-factor, and resonance.
- To explain wave propagation in media, including energy transfer, reflection, and group velocity.
- To use Maxwell's equations to describe EM waves, energy flow, and wave behavior in various media.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Simple harmonic motion, energy of a SHM, Compound Pendulum, Torsional Pendulum, Electrical Oscillations, Transverse Vibrations of a mass on a string, Decay of free vibrations due to damping, differential equation of motion, types of damping, determination of damping co-efficient; Logarithmic decrement, relaxation time and Q- Factor, Electromagnetic damping (Electrical oscillator).

Differential equation for forced mechanical and electrical oscillators, Transient and steady state behaviour, Displacement and velocity variation with driving force frequency, variation of phase with frequency, resonance. Power supplied to an oscillator and its variation with frequency, Q-value and band width

UNIT-II

Waves in physical media, Wave equation and its solution, Types of waves, particle velocity, acceleration and energy in progressive waves, Transverse waves on a string, characteristic impedance of a string, Reflection and Transmission of transverse waves on a string at discontinuity, Reflection and transmission of energy, Reflection and transmission of longitudinal waves at a boundary. Impedance matching, Energy of vibrating string, Wave and group velocity

Maxwell's equations, E.M. waves and wave equation in a medium having finite permeability, permittivity and conductivity, Energy flow due to EM wave - Poynting vector, Impedance of a dielectric to EM waves, EM waves in a conducting medium and skin depth, Impedance and Refractive index of a dielectric and a conductor.

Books Suggested:

Essential Readings:

1. Text Book of Vibrations and Waves by S.P. Puri (Macmillan India Ltd.).
2. Physics of Vibrations and Waves by H.J. Pain, ELBS & John Wiley, London.

Further Readings:

1. Vibrations and Waves by A.P. French (Arnold Heinemann India, New Delhi).
2. The Mathematics of Waves and Vibrations by P.K. Ghosh (Mcmillan India).
3. Waves and Oscillations by N. Subrahmanayam & B. Lal (Vikas Pub., Delhi).

PHYSICS PRACTICALS
(PHY-DSC-6 Maj-402)
(for Table-2)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal ssessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To determine time period of a simple pendulum.
2. To compare the moment of inertia of solid sphere and hollow sphere (or a solid disc and annular disc) of same mass using torsion pendulum. Hence show that M.I. depends upon distribution of masses.
3. To measure the time periods of oscillations using objects of various geometrical shapes but the same mass using Torsional pendulum.
4. To study the dependence of moment of inertia on distribution of mass by noting time periods of oscillations using objects of various geometrical shapes but of same mass.
5. Measure time period as a function of distance of centre of suspension (oscillation) from centre of mass, plot relevant graphs, determine radius of gyration and acceleration due to gravity.
6. To find the value of g by Kater's or Bar pendulum.
7. Measure time period of oscillation of a Maxwell needle and determine modulus of rigidity of the material of a given wire.
8. To measure/obtain logarithmic decrement, coefficient of damping, relaxation time, and quality factor of a damped simple pendulum.
9. To determine the frequency of electrically maintained tuning fork by means of Melde's Experiment.

Texts and Reference Books:

1. B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 4
(PHY-DSC-6 Maj-402)
(Statistical Physics and Thermodynamics)
(for Table-1)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory)+1 (Practical)

Learning Outcomes

- To understand the basic ideas of statistical physics, distribution of four distinguishable particles in two compartments, deviation from the state of maximum probability
- To study phase space and its division into elementary cells, B.E. statistics and F.D. statistics.
- To explore the statistical definition of entropy, change of entropy of a system, brief review of the terms
- To understand the Derivation of Maxwell's thermodynamical relations, Clausius Clayperon Equation and Joule-Thomson effect

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Basic ideas of Statistical Physics, Scope of Statistical Physics, basic ideas about probability, distribution of four distinguishable particles in two compartments of equal size. Concept of macrostates, microstates, thermodynamic probability, effects of constraints on the system, distribution of n particles in two compartments, deviation from the state of maximum probability, equilibrium state of dynamic system, Phase space and its division into elementary cells, three kinds of statistics. The basic approach in the three statistics, Maxwell-Boltzmann statistics applied to an ideal gas in equilibrium, Need of quantum statistics-B.E. statistics, derivation of Planck's law of radiation, F.D. statistics, Comparison of M.B., B.E. and F.D. statistics.

UNIT-II

Statistical definition of entropy, change of entropy of a system, additive nature of entropy, law of increase of entropy, reversible and irreversible processes with examples, Work done in a reversible process, Examples of increase of entropy in natural processes, Entropy and disorder, Entropy of a perfect gas, Equation of state of ideal gas from simple statistical consideration. Heat death of the universe, Derivation of Maxwell's thermodynamical relations, cooling produced by adiabatic stretching, Expression for (C_p-C_v), change of state and Clausius Clayperon Equation, Thermodynamical treatment of Joule-Thomson effect and applications (Qualitative).

Books Suggested:

Essential Readings:

1. "Statistical Physics and Thermodynamics", V.S. Bhatia, (Shoban Lal Nagin Chand, Jalandhar).
2. "A Treatise on Heat" Saha and Srivastava (Indian Press, Ahmedabad, 1972).

Further Readings:

1. Thermal Physics by C. Kittel & H. Kroemer, CBS Pub., 1987.
2. Thermal Physics, S.C. Garg, R.M. Bansal, and C.K. Ghosh, TMH, 2000.

PHYSICS PRACTICALS
(PHY-DSC-6 Maj-402)
(for Table-1)

Max. Marks: 25 (20+5 internal assessment)
Total Credit: 1
Teaching hours: 30 Hrs. / 2 Hrs. per week
General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal ssessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals

1. To study adiabatic expansion of a gas.
2. To measure thermal expansion of crystal using interference fringes.
3. To measure thermal conductivity in poor conductor by Lee's method.
4. To measure the thermo e.m.f.
5. To study Probability distribution using coloured dice, coins.
6. To determine the heating efficiency of an electric kettle with varying voltages.
7. Total radiation law, temperature dependence of radiation.
8. To determine principal points of a lens system.
9. To determine the wavelength of the spectral lines of mercury using diffraction grating and mercury source.
10. To find the thermal conductivity of copper using Searle's conductivity apparatus.

Texts and Reference Books :

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried&JitenderChhabra.

B.Sc. Semester 4
(PHY-DSC-7 Maj-403)
(Condensed Matter Physics - I)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory)+1 (Practical)

Learning Outcomes

- **Understand Crystal Structures**: Describe symmetry operations, Bravais lattices, Miller indices, and basic crystal structures like diamond and NaCl.
- **Apply Crystal Diffraction Concepts**: Use Bragg's Law, Laue equations, and reciprocal lattices to analyze crystal structures and Brillouin zones.
- **Explain Band Theory**: Understand energy band formation, Bloch's theorem, and distinguish between conductors, semiconductors, and insulators.
- **Analyze Electrical Properties**: Apply free electron theory, effective mass, carrier mobility, and study temperature effects, Fermi levels, and Hall effect.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Crystal structure: Symmetry operations for a two-dimensional crystal. Two dimensional Bravais lattices, Three dimensional Bravais lattices, Basic primitive cells, Crystal planes and Miller indices, Diamond and NaCl structure. Crystal diffraction: Bragg's Law, Determination of crystal structure, Laue equations, Reciprocal lattices of SC, BCC and FCC, Bragg's law in reciprocal lattice, Brillouin zones and its derivation in two dimensions, structure factor and atomic form factor.

UNIT-II

Band Theory of solids, periodic potential and Bloch theorem, Kronig-Penney model, band gaps, band structures in conductors, direct and indirect semiconductors and insulators. Free electron theory of metals, effective mass, drift current, mobility and conductivity (carrier concentration and mobility of carriers) and their variation with temperature in semi-conductors, Fermi level positions in intrinsic and extrinsic semiconductors, Wiedemann-Franz law, Hall effect in metals and semiconductors.

Recommended Books:

Essential Readings:

` 1. Introduction to Solid State Physics, C. Kittel, Wiley Eastern
2. Elements of Modern Physics, S.H. Patil, Tata McGraw Hill, 1985.
3. Solid State Physics, 6th Edition, S.O. Pillai, New Age International Publishers.
4. Physics for Degree Students, C.L. Arora and P.S. Hemne, S. Chand & Co., 2014.

Further Readings:

1. Elements of Solid State Physics, 2nd Edition, J.P. Srivastava, Prentice Hall.
2. Elementary Solid State Physics, M. Ali Omar, Pearson.
3. Crystallography for Solid State Physics, A.R. Verma, O.N. Srivastava, Wiley Eastern.

PHYSICS PRACTICALS (PHY-DSC-7 Maj-403)

Max. Marks: 25 (20+5 internal assessment)
Total Credit: 1
Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error). 09
(ii) One exercise based on experiment 04
(iii) Viva-Voce 04
(iv) Record (Practical file) 03
(v) Internal Assessment 05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals

1. Measurement of reverse saturation current in p-n junction diode at various temperatures and to find the approximate value of energy gap.
2. To find the conductivity of a given semi-conductor crystal using four probe method.
3. To determine the Hall coefficient for a given semiconductor.
4. To study the characteristics of a thermistor and find its parameters.
5. To draw the characteristics of a Light emitting diode.
6. To study the characteristics of photo voltaic cell
7. To determine the resistivity of wires of different materials.
8. To determine the dielectric constant of a given sample.
9. To study dependence of intensity of radiative transitions in LED as a function of temperature.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried&JitenderChhabra.

B.Sc. Semester 4
(PHY-DSC-8 Maj-404)
(Electronics and Solid-State Devices – I)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory)+1 (Practical)

Learning Outcomes

- **Apply Network Theorems and CRO Usage**: Understand and apply Thevenin's/Norton's theorems, source conversions, and use CRO for waveform analysis, frequency, and phase measurements.
- **Explain Semiconductor Device Behavior**: Interpret energy band diagrams, Fermi level positions, and diode characteristics including Zener, LED, photodiode, and solar cell.
- **Analyze Diode-Based Circuits**: Design and evaluate rectifiers, filter circuits, clipping circuits, and voltage regulators using Zener diodes.
- **Understand and Use BJT Configurations**: Explain BJT operation in CB, CE, CC modes, perform biasing, load-line analysis, and evaluate CE amplifier performance.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Concepts of current and voltage sources, Thevenin's theorem, Norton's theorem, Source conversion. CRO, Block diagram, construction and principle of working, Use of CRO for frequency, time period, special features of dual trace, phase measurements. Energy band diagrams in semiconductors, Direct and indirect semiconductors, Formula to calculate Position of Fermi level in p and n semiconductors, Barrier formation, energy band diagram of p-n junction, Formula for Depletion width, Qualitative ides of current flow mechanism in forward and reverse biased diode, v-i characteristics, static and dynamic resistance, Depletion and diffusion capacitance, zener diode, LED, photodiode and solar cell.

UNIT-II

Diode circuits, Clipping circuits, Rectification: half wave, full wave and bridge rectifiers, filter circuits (C, LC and π filters), rectification efficiency and ripple factor in LC filter, voltage regulation circuit using zener diode, voltage multiplier circuits. Bipolar Junction transistors: Structure and working, different currents in transistor, switching action. Characteristics of CB, CE and CC configurations, Active, cutoff and saturation regions. Load line analysis of transistors, Q-point, Transistor biasing and stabilization of operating point, fixed bias, collector to base bias, bias circuit with emitter resistor, voltage divider biasing circuit. Working analysis of CE amplifier using h-parameters, current, voltage and power gain, input and output impedance.

Recommended Books:

Essential Readings:

1. Electronic Devices and Circuit Theory, 7th Ed., R. Boylestad, L. Nashelsky, Prentice Hall Inc.
2. Electronic Principles, A.P. Malvino, and D.J. Bates, 7th ed. McGraw Hill
3. Solid State Electronic Devices, 6th Ed., Ben G. Streetman and S. Banerjee, Eastern Economy Edition.

Further Readings:

1. Basic Electronics, 5 th Edition, B.L. Thareja, S. Chand.
2. Basic Electronics and Linear Circuits, N.N. Bhargave, D.C. Kulshreshtha, and S.C. Gupta,Tata Mc Graw Hill.
3. Foundations of Electronics, D. Chatopadhyay, P.C. Rakshit, B. Saha, and N.N. Purkit, New Age International

PHYSICS PRACTICALS (PHY-DSC-8 Maj-404)

Max. Marks: 25 (20+5 internal assessment)
Total Credit: 1
Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii) One exercise based on experiment	04
(iii) Viva-Voce	04
(iv) Record (Practical file)	03
(v) Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals

1. To draw the characteristics of a P-N Junction diode.
2. To draw the characteristics of a Zener diode.
3. To study the stabilization of output voltage of a power supply with Zener diode.
4. To measure and plot Common Emitter Characteristics of a transistor (pnp or npn).
5. To plot Common Base Characteristics and determine h-parameters of a given transistor.
6. To measure the efficiency and ripple factors for (a) Half-wave, (b) Full wave, and (c) Bridge rectifier circuits.
7. To study the reduction in the ripples in the rectified output with RC, LC and π-filters.
8. To study the function of diode as a clipping element.
9. To obtain the waveform of a given oscillator using a CRO/DSO.

Texts and Reference Books :

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

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STATISTICS

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*) B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						**Table 2***					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
III	L-200	(STAT-DSC-3 Maj/Min-301)	Descriptive Statistics-II	3+1=4 (x2)	24	III	L-200	(STAT-DSC-3 Maj/Min-301)	Descriptive Statistics-II	3+1=4 (x3)	24
		(STAT-DSC-4 Maj/Min-302)	Probability Distributions	3+1=4 (x2)				(STAT-DSC-4 Min-302)	Probability Distributions	3+1=4	
		(STAT-MDC-3-301)	Applied Statistics	3				(STAT-MDC-3-301)	Applied Statistics	3	
		(STAT-SEC-3-301)	Statistics for Business Analytics with Microsoft Excel	3				(STAT-SEC-3-301)	Statistics for Business Analytics with Microsoft Excel	3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(STAT-DSC-5 Maj/Min-401)	Statistical Inference	3+1=4 (x2)	24	IV	L-200	(STAT-DSC-4 Maj-402)	Probability Distributions	3+1=4 (x2)	24
		(STAT-DSC-6 Maj-402)	Sampling Distributions	4				(STAT-DSC-5 Maj/Min-401)	Statistical Inference	3+1=4(X3)	
		(STAT-DSC-7 Maj-403)	Applied Statistics-I	4							
		(STAT-DSC-8 Maj-404)	Practical	4							
		(VAC-3-401)*	---	2				(VAC-3-401)*	----	2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											
***Value Added Courses are not being offered by the Deaprtment of Statistics, Panjab University, Chandigarh**											

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

SEMESTER-III

Course Code: (STAT-DSC-3-Maj/Min-301)
Course Name: Descriptive Statistics-II

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Descriptive Statistics-II (STAT-DSC-3-Maj/Min-301)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

DESCRIPTIVE STATISTICS-II

Objectives: The main objective of this paper isto provide students with a comprehensive understanding of correlation analysis, curve fitting techniques, linear regression, and multivariate data analysis. Additionally, students will learn about multivariate data analysis techniques such as multiple and partial correlation, and gain the ability to interpret and analyze relationships among multiple variables.

Learning Outcomes: On completion of this course students will be able to:

1. Understand the concepts and types of correlation, regression, multivariate data and their utilization in real life.
2. Utilize the principle of least squares for curve fitting, including fitting straight lines, second degree parabolas, power curves, and exponential curves.
3. Analyze multivariate data by understanding the plane of regression, properties of residuals, and variance of the residual.
4. Apply multiple and partial correlation techniques for three variables, including understanding the coefficient of multiple correlation, coefficient of partial correlation, and coefficient of determination.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks except the compulsory question, the distribution of marks for which will be as follows:-

UNIT-I

Correlation: Concept and types of correlation, methods of finding correlation - scatter diagram, Karl Pearson's Coefficient of correlation (*r*), its properties, coefficient of correlation for a bivariate frequency distribution. Rank correlation with its derivation, its merits and demerits, limits of rank correlation coefficient, tied or repeated ranks.

UNIT-II

Curve Fitting: Principle of least squares, fitting of straight line, second degree parabola, power curves of the type $Y=aX^b$, exponential curves of the types $Y=ab^X$ and $Y=ae^{bX}$, regression curves.

UNIT-III

Linear Regression: Two lines of regression, regression coefficients, properties of regression coefficients, angle between two regression lines, standard error of estimate obtained from regression line, correlation coefficient between observed and estimated values, distinction between correlation and regression.

UNIT-IV

Multivariate Data: Plane of regression, properties of residuals, variance of the residual. Multiple and partial correlation for three variables: coefficient of multiple correlation and its properties, coefficient of partial correlation and its properties, multiple correlation in terms of total and partial correlations and coefficient of determination.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Introduction to Theory of Statistics	Mood A.M., Graybill F.A. &Boes D.C.	McGraw Hill
2.	Applied General Statistics	Croxton F.E., Cowden D.J. &Kelin S.	Prentice Hall
3.	Introduction to Mathematical Statistics	Kapoor&Saxena.	S.Chand
4.	Statistical Methods	Snedecor G.W. & Cochran W.G.	Iowa State Uni. Press
5.	Fundamentals of Mathematical Statistics	Gupta S.C.& Kapoor V.K.	Sultan Chand & Sons

(STAT-DSC-3-Maj/Min-301): **DESCRIPTIVE STATISTICS-II (PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-3-Maj/Min-301) will be based on the following topics:

1. Problems based on Karl Pearson's Coefficient of correlation.
2. Problems based on Spearman's Rank Correlation coefficient without ties.
3. Problems based on Spearman's Rank Correlation coefficient with ties.
4. Problems based on fitting of straight line.
5. Problems based on second degree parabola curve.
6. Problems based on exponential curve.
7. Problem based on two lines of regression.
8. Problem based on regression coefficients.
9. Problem based on multiple correlation coefficients.
10. Problem based on partial correlation coefficients.

Course Code: (STAT-DSC-4 Maj/Min-302)
Course Name: Probability Distributions

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/Credit	Practical(P)/ Credit	Marks Distribution	
Probability Distributions (STAT-DSC-4 Maj/Min-302)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

PROBABILITY DISTRIBUTIONS

Objectives: The main objective of the paper is to provide students with a comprehensive understanding of various probability distributions including Bernoulli, Binomial, Poisson, Negative Binomial, Geometric, Continuous Uniform, Gamma, Exponential, and Normal distributions. This includes learning their key characteristics, moments, moment generating functions, mean deviation, mode, characteristic functions, and their applications in real-world scenarios such as modeling and inference.

Learning Outcomes: On completion of this course students will be able to:

1. Understand about various discrete and continuous distributions and their properties.
2. Compute moments, moment generating functions, and characteristic functions for different probability distributions.
3. Analyze various types of numerical problems with the help of probability distributions.
4. Apply properties of discrete as well as continuous probability distributions to solve the real life problems.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 7 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Bernoulli distribution and its moments, Binomial distribution: Moments, recurrence relation for the moments, mean deviation about mean, mode, moment generating function (m.g.f), characteristic function, additive property and recurrence relation for the probabilities of Binomial distribution.

UNIT-II

Poisson distribution: Poisson distribution as a limiting case of Binomial distribution, moments, mode, recurrence relation for moments, m.g.f., additive property of independent Poisson variates. Negative Binomial distribution: m.g.f., deduction of moments of negative binomial distribution from those of binomial distribution. Geometric distribution: moments, m.g.f,and lack of memory.

UNIT-III

Continuous uniform distribution: Moments, m.g.f.,characteristic function and mean deviation. Gamma distribution: m.g.f., and additive property. Exponential distribution: m.g.f., moments and lack of memory.

UNIT-IV

Normal distribution as a limiting form of binomial distribution, chief characteristics of Normal distribution; mode, median, m.g.f.,and moments of Normal Distribution, A linear combination of independent normal variates, points of inflexion, mean deviation about mean, area property of Normal distribution, importance and fitting of normal distribution.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Statistics:A Beginner's Text Vol. II	Bhat B.R., Srivenkatramana T. & RaoMadhava K.S.	New Age International
2.	Fundamentals of Mathematical Statistics	Gupta S.C. & Kapoor V.K.	Sultan chand & Sons
3.	Introduction to Mathematical Statistics	Kapoor&Saxena.	S.Chand
4.	Statistics	Johnson R.	Wiley Publishers
5.	Mathematical Statistics With Applications	Freund's J.E.	Prentice Hall

(STAT-DSC-4 Maj/Min-302): **PROBABILITY DISTRIBUTIONS (PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-4 Maj/Min-302) will be based on the following topics:

1. Practical problem based on binomial distribution.
2. To fit binomial distribution of the given data.
3. Practical problem based on Poisson distribution.
4. To fit Poisson distribution of the given data.
5. Practical problem based on negative binomial distribution.
6. Practical problem based on geometric distribution.
7. Practical problem based on continuous uniform distribution.
8. Practical problem based on exponential distribution.
9. Practical problem based on area property of normal distribution.
10. To fit normal distribution of the given data.

B.Sc. (Hons) STATISTICS
Semester-IV

Course Code: (STAT-DSC-4 Maj-402)
Course Name: Probability Distributions

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L) /Credit	Practical(P)/ Credit	Marks Distribution	
Probability Distributions (STAT-DSC-4 Maj-402)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

PROBABILITY DISTRIBUTIONS

Objectives: The main objective of the paper is to provide students with a comprehensive understanding of various probability distributions including Bernoulli, Binomial, Poisson, Negative Binomial, Geometric, Continuous Uniform, Gamma, Exponential, and Normal distributions. This includes learning their key characteristics, moments, moment generating functions, mean deviation, mode, characteristic functions, and their applications in real-world scenarios such as modeling and inference.

Learning Outcomes: On completion of this course students will be able to:

1. Understand about various discrete and continuous distributions and their properties.
2. Compute moments, moment generating functions, and characteristic functions for different probability distributions.
3. Analyze various types of numerical problems with the help of probability distributions.
4. Apply properties of discrete as well as continuous probability distributions to solve the real life problems.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 7 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Bernoulli distribution and its moments, Binomial distribution: Moments, recurrence relation for the moments, mean deviation about mean, mode, moment generating function (m.g.f), characteristic function, additive property and recurrence relation for the probabilities of Binomial distribution.

UNIT-II

Poisson distribution: Poisson distribution as a limiting case of Binomial distribution, moments, mode, recurrence relation for moments, m.g.f., additive property of independent Poisson variates. Negative Binomial distribution: m.g.f., deduction of moments of negative binomial distribution from those of binomial distribution. Geometric distribution: moments, m.g.f,and lack of memory.

UNIT-III

Continuous uniform distribution: Moments, m.g.f.,characteristic function and mean deviation. Gamma distribution: m.g.f., and additive property. Exponential distribution: m.g.f., moments and lack of memory.

UNIT-IV

Normal distribution as a limiting form of binomial distribution, chief characteristics of Normal distribution; mode, median, m.g.f.,and moments of Normal Distribution, A linear combination of independent normal variates, points of inflexion, mean deviation about mean, area property of Normal distribution, importance and fitting of normal distribution.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Statistics:A Beginner's Text Vol. II	Bhat B.R., Srivenkatramana T. & RaoMadhava K.S.	New Age International
2.	Fundamentals of Mathematical Statistics	Gupta S.C. & Kapoor V.K.	Sultan chand & Sons
3.	Introduction to Mathematical Statistics	Kapoor&Saxena.	S.Chand
4.	Statistics	Johnson R.	Wiley Publishers
5.	Mathematical Statistics With Applications	Freund's J.E.	Prentice Hall

(STAT-DSC-4 Maj-402) **PROBABILITY DISTRIBUTIONS (PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-4 Maj-402) will be based on the following topics:

1. Practical problem based on binomial distribution.
2. To fit binomial distribution of the given data.
3. Practical problem based on Poisson distribution.
4. To fit Poisson distribution of the given data.
5. Practical problem based on negative binomial distribution.
6. Practical problem based on geometric distribution.
7. Practical problem based on continuous uniform distribution.
8. Practical problem based on exponential distribution.
9. Practical problem based on area property of normal distribution.
10. To fit normal distribution of the given data.

B.Sc. (Hons) STATISTICS
Semester-IV
Course Code: (STAT-DSC-5 Maj/Min-401)
Course Name: Statistical Inference

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L) /Credit	Practical(P) / Credit	Marks Distribution	
Statistical Inference (STAT-DSC-5 Maj/Min-401)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

STATISTICAL INFERENCE

Objectives: The learning objectives of this course are as follows:

1. To understand the concept of estimation theory and testing of hypothesis.
2. To draw inferences about the unknown population parameters based on random samples.
3. To validate the estimation/ inference about the population parameters.

Learning Outcomes: After successful completion of this course, students should be able to:

1. Understanding of estimation theory, Point and interval estimations.
2. Get knowledge about various characteristics of a good estimator and different methods of estimation.
3. Develop the best/most powerful statistical tests to test the hypotheses regarding unknown population parameters by using Neyman-Pearson theory.
4. Apply the likelihood ratio test and Sequential probability ratio test (SPRT) to test the different problems.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks except the compulsory question, the distribution of marks for which will be as follows:-

UNIT I

Estimation Theory: Parameter space, sample space, point estimation, requirement of a good estimator, consistency, unbiasedness, efficiency, sufficiency, Minimum variance unbiased estimators, Factorization theorem, Fisher- Neyman Criterion: statement and applications, Cramer- Rao inequality: statement and application, MVB estimators and their applications, Statement of Rao-Blackwell theorem.

UNIT II

Method of Estimation: Maximum likelihood, method of moments, least squares and minimum variance, Properties of maximum likelihood estimators (illustration), Interval Estimation: confidence interval and confidence limits for the parameters of normal distribution, confidence intervals for large samples.

UNIT III

Test of Significance: Principles of test of significance: Null and alternative hypotheses, simple and composite, Type-land Type-II errors, critical region, level of significance, power of the test, best critical region, most powerful test, uniformly most powerful test, uniformly most powerful unbiased (UMPU)critical region, Neyman- Pearson Lemma: statement and its applications to construct most powerful test.

UNIT-IV

Likelihood Ratio Test: Introduction of likelihood ratio test, its properties, test for the mean of a normal population, test for the equality of means of two normal populations, test for the variance of a normal population, test for equality of variances of two normal populations.
Sequential Analysis: Sequential probability ratio test (SPRT), operating characteristic (OC) Function of SPRT, average sample number (ASN).

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	A First Course on Parametric Inference	Kale B.K.	Narosa
2.	Introduction to Theory of Statistics	Mood A.M., Graybill F.A. &Boes D.C.	McGraw Hill
3.	Mathematical Statistics With Applications	Freund's J.E.	Prentice Hall
4.	Fundamentals of Mathematical Statistics	Gupta S.C. & Kapoor V.K.	Sultan chand & Sons

(STAT-DSC-5 Maj/Min-401): **STATISTICAL INFERENCE (PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-5 Maj/Min-401)will be based on the following topics:

1. Problem based on unbiased estimators.
2. Problem based on consistent estimators.
3. Problem based on efficient estimators.
4. Problem based on sufficiency.
5. Problem based on maximum likelihood estimators of standard distributions.
6. Problem based on method of moments.
7. Problem based on least square method of estimation.
8. Problem based on Best critical region or most powerful test.
9. Problem based on uniformly most powerful (UMP) test.
10. Problem based on likelihood ratio test.
    Testing problem based on SPRT and find.

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ZOOLOGY

STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table1*)
AND STUDENTS OPTING FOR MULTI-DISCIPLINARY/INTER-DISCIPLINARY (Table 2*)

B.Sc.,B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN ZOOLOGY

Table1*						Table2*					
Sem	**Level**	**CourseCode**	**CourseName**	**Credits Th(L) + Pr(P)**	**Total Credits**	**Sem**	**Level**	**CourseCode**	**CourseName**	**Credits Th(L) + Pr(P)**	**Total Credits**
III	L-200	(ZOO-DSC-3 Maj/Min-301)	Animal Physiology	3+1=4 (x2)	24	III	L-200	(ZOO-DSC-3 Maj/Min-301)	Animal Physiology	3+1=4(x3)	24
		(ZOO-DSC-4 Maj/Min-302)	Developmental Biology	3+1=4 (x2)				(ZOO-DSC-4 Min-302)	Developmental Biology	3+1=4	
		(ZOO-MDC-3-301)	First Aid and Emergency Management	2+1=3				(ZOO-MDC-3-301)	First Aid and Emergency Management	2+1=3	
		(ZOO-SEC-3-301)	Poultry Farming	1+2=3				(ZOO-SEC-3-301)	Poultry Farming	1+2=3	
		(AEC-3-301)	Eng-3/MIL-1	2				(AEC-3-301)	Eng-3/MIL-1	2	
IV	L-200	(ZOO-DSC-5 Maj/Min-401)	Biochemistry	3+1=4 (x2)	24	IV	L-200	(ZOO-DSC-5 Maj/Min-401)	Biochemistry	3+1=4(x3)	24
		(ZOO-DSC-6 Maj-402)	Biomolecules and Nutrition	3+1=4				(ZOO-DSC-4 Maj-402)	Developmental Biology	3+1=4(x2)	
		(ZOO-DSC-7 Maj-403)	Basic Biochemical Techniques	3+1=4							
		(ZOO-DSC-8 Maj-404)	Endocrinology	3+1=4							
		(ZOO-VAC-3-401)	Reproductive Health	1+1=2				(ZOO-VAC-3-401)	Reproductive Health	1+1=2	
		(AEC-4-401)	Eng-4/MIL-2	2				(AEC-4-401)	Eng-4/MIL-2	2	
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship											

*Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre

5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Semester- III (L-200)

Course Code: ZOO-DSC-3-Maj/Min-301
Course Title: Animal Physiology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) +08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will be able to explore basic physiological principles common to animals.*
- *Students will understand the fundamental concepts of digestion, respiration, blood circulation and excretion.*
- *Students will learn about the structure and functioning of nerves and muscles, andthe role of hormones in physiological processes.*

UNIT- 1
Digestion: Extra and Intra Cellular Digestion, Absorption of carbohydrates, proteins and fats.
Respiration: Exchange and transport of respiratory gases, Oxygen dissociation curve, Bohr Effect, Chloride shift and Haldane effect.

UNIT- 2
Blood: Composition and functions of blood and lymph,Blood Clotting, Blood groups
Heart: Origin and conduction of heart beat, cardiac cycle, cardiac output, electrocardiogram and blood pressure.

UNIT- 3
Excretion: Structure of nephron, urine formation, composition of urine, osmoregulation and counter current mechanism.
Muscles: Types of muscles and their structure. Skeletal Muscle Contraction.

UNIT- 4
Neural Integration: Structure of neuron, resting membrane potential, origin and propagation of nerve impulse.
Endocrine System: Hormones and functions of Endocrine glands: Hypothalamus, pituitary, thyroid, parathyroid, adrenal, pancreas and gonads.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-3-Maj/Min-301: Animal Physiology

Practical

1. Study of blood smear of a mammal.
2. Demonstration of estimation of haemoglobin content in blood.
3. Recording of blood pressure of man
4. Determination of heart rate and pulse rate
5. Study of waveforms in electrocardiogram.
6. Study of stained sections of different mammalian tissues and organs
7. An idea of location of endocrine glands in mammals through charts / models/ video clippings

SUGGESTED READINGS

1. Taneja, S.K.: Biochemistry & Animal Physiology, Trueman Book Co.,1997.
2. Guyton, A.S.: Text Book of Medical Physiology, 14th edition, W.B. Saunders Company, 2020.
3. H.R Singh &Neeraj Kumar, Animal Physiology & Biochemistry Edition. 2005,Vishal Publication Co. Jalandhar.
4. VasantikaKashyap: Animal Physiology & Biochemistry, KedarNath Ram Nath Publications, Meerut.
5. Berry, A.K. Text Book of Animal Physiology 2005 Edition, Emkay Publications Delhi.

Course Code: ZOO-DSC-4 Min-302
Course Title: Developmental Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will get detailed knowledge about the formation and structure of gametes.*
- *Students will understand development patterns in different animals ranging from proto chordates to mammals.*
- *Students will be able to apply their knowledge and pursue research in the field of Developmental biology.*

UNIT- 1
Gametogenesis: Spermatogenesis, spermiogenesis, Oogenesis &Vitellogenesis
Types of eggs, egg membranes and polarity of egg.

UNIT- 2
Fertilization & parthenogenesis.
Cleavage: Types of cleavage patterns depending upon amount and distribution of yolk and position of spindle.
Blastulation and types of blastula.

UNIT- 3
Gastrulation: Definition, Morphogenetic movements & Fate of three germ layers
Development up to three germ layers in *Herdmania*, frog, chick and rabbit.

UNIT- 4
Foetal membranes, their formation and role.
Mammalian placenta – its formation, types and functions.
Metamorphosis in *Herdmania* and *Rana* (frog)

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-4 Min-302

Practical

1. Study of the following permanent slides for gametogenesis- T.S. Testis and T.S. ovary of mammals
2. Study of the development of frog from permanent slides.
3. Study of the development of chick embryo from permanent slides up to 96 hours.
4. Study of metamorphosis of *Herdmania* and Frog through charts/video/ models

SUGGESTED READINGS

1. Ballinsky, B.I. An Introduction to Embryology, 5th Edition, Saunders Co., 1981.
2. Gilbert, S.F. Developmental Biology, 10th Edition, Sinauer Ass., Inc. Publ., 2014.
3. Urspaung, H. Major Problems in Developmental Biology, Academic Press, New York, 1966.
4. Wolport, Lewis Principles of development, Oxford University press, Oxford, 2007
5. Veer BalaRastogi, Chordate Embryology (Developmental Biology) KedarNath Ram Nath, Meerut/ Delhi 2011

Semester- IV (L-200)

Course Code: ZOO-DSC-5 Maj/Min-401
Course Title: Biochemistry

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) +08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will gain knowledge about biomolecules, the basic building blocks of all living organisms.*
- *Students will get to know about the nature and mode of action of enzymes, the biological catalysts.*
- *Students will be able to understand the metabolic pathways of carbohydrates, lipids and proteins.*

UNIT- 1
Biomolecules: Carbohydrates, proteins, lipids and nucleic acids: their classifications and functions.

UNIT- 2
Enzymes: Nature, nomenclature, mode of action, their classification, coenzymes and cofactors.
Carbohydrate metabolism: EmbdenMeyerhof Parnas pathway (glycolysis), Tricarboxylic Acid Cycle, Glycogenesis.

UNIT- 3
Lipid metabolism: Beta-oxidation of fatty acids, gluconeogenesis, lipogenesis and ketosis.

UNIT- 4
Protein metabolism: Metabolism of amino acids: Oxidative deamination, Transamination Decarboxylation and Ornithine cycle.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-5 Maj/Min-401: Biochemistry

Practical

1. Preparation of Molar, Normal Solutions and Buffers
2. Determination of pH
3. Study the action of salivary amylase under optimum conditions.
4. Study the effect of pH and temperature on the action of salivary amylase.
5. Qualitative analysis of carbohydrates, proteins and lipids.
6. Analysis of urine for urea, uric acid, chloride and glucose.

SUGGESTED READINGS

1. Taneja, S.K.: Biochemistry & Animal Physiology, Trueman Book Co.,1997.
2. Robert, K., Murray, R.K.: Harper's Biochemistry, 22nd edition, Daryl, M., Granner, K., Prentice – Hall International, Inc.,1990 Victor, W. and Woodwell.
3. Nelson, D.L. and Cox, M.M.: Lehninger Principles of Biochemistry, 5th edition, W.H., Freeman and Company, New York, 2008.
4. Murray, R.K., Bender, D.A., Botham, K.M., Kennelly, P.J., Rodwell, V.W. and Well, P.A. (2009). Harper's Illustrated Biochemistry. XXVIII Edition, International Edition, The McGraw- Hill Companies Inc. Suggestive readings
5. Stryer, L., Berg, J., Tymoczko, J., Gatto, G. (2019). Biochemistry (9th Ed.). New York, WH: Freeman. 4. Voet, D., Voet. J. G. (2013). Biochemistry (4th ed.). New Jersey, John Wiley & Sons, Asia Pvt. Ltd.
6. H.R Singh &Neeraj Kumar, Animal Physiology & Biochemistry, Vishal Publication Co. Jalandhar Ed. 2005
7. VasantikaKashyap: Animal Physiology & Biochemistry, KedarNath Ram Nath Publications, Meerut

Course Code: ZOO-DSC-4 Maj-402
Course Title: Developmental Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will get detailed knowledge about the formation and structure of gametes.*
- *Students will understand development patterns in different animals ranging from proto chordates to mammals.*
- *Students will be able to apply their knowledge and pursue research in the field of Developmental biology.*

UNIT- 1
Gametogenesis: Spermatogenesis, spermiogenesis, Oogenesis &Vitellogenesis
Types of eggs, egg membranes and polarity of egg.

UNIT- 2
Fertilization & parthenogenesis.
Cleavage: Types of cleavage patterns depending upon amount and distribution of yolk and position of spindle.
Blastulation and types of blastula.

UNIT- 3
Gastrulation: Definition, Morphogenetic movements & Fate of three germ layers
Development up to three germ layers in *Herdmania*, frog, chick and rabbit.

UNIT- 4
Foetal membranes, their formation and role.
Mammalian placenta – its formation, types and functions.
Metamorphosis in *Herdmania* and *Rana* (frog)

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-4 Maj-402: Developmental Biology

Practical

1. Study of the following permanent slides for gametogenesis- T.S. Testis and T.S. ovary of mammals.
2. Study of the development of frog from permanent slides.
3. Study of the development of chick embryo from permanent slides up to 96 hours.
4. Study of metamorphosis of *Herdmania* and Frog through charts/video/ models

SUGGESTED READINGS

1. Ballinsky, B.I. An Introduction to Embryology, 5th Edition, Saunders Co., 1981.
2. Gilbert, S.F. Developmental Biology, 10th Edition, Sinauer Ass., Inc. Publ., 2014.
3. Urspaung, H. Major Problems in Developmental Biology, Academic Press, New York, 1966.
4. Wolport, Lewis Principles of development, Oxford University press, Oxford, 2007
5. Veer BalaRastogi, Chordate Embryology (Developmental Biology) KedarNath Ram Nath, Meerut/ Delhi 2011

Course Code: ZOO-DSC-6 Maj-402
Course Title: Biomolecules and Nutrition

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory= 75 + Practical= 25)
Theory Marks= 75 {67 (Theory examination) + 8 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 3 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning outcomes:
Students will be able to:

- *Acquire scientific knowledge about the basic biomolecules and their types*
- *Understand the importance of Nutrition and balanced diet for a healthy body*

UNIT- 1
Carbohydrates: Types (Monosaccharides, Disaccharides & Polysaccharides) andtheir sources
Lipids: **Types (triglycerides, phospholipids, and sterols) andtheir**sources

UNIT- 2
Proteins: Types (Simple, Conjugated, Fibrous, Globular) andtheir sources
Nucleic acids: Types (DNA & RNA) and their sources

UNIT- 3
Vitamins: Sources and functions of Fat-solubleVitamins(A,D,E and K) and Water-soluble Vitamins (BI, B2, B3, B6, B12 and C).
Minerals:Sources and functions of Macrominerals(sodium, potassium, calcium, magnesium, chloride) and Micro/Trace minerals (iodine, iron, zinc).

UNIT- 4
Balanced diet, Recommended Dietary Allowance (RDA), Undernutrition (Kwashiorkor, Marasmus), Overnutrition (Obesity, diabetes, hypertension), Micronutrient deficiency (Anemia, Goitre), Key objectives of the National Nutrition Policy of India (1993).

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-6 Maj-402: Biomolecules and Nutrition

Practical

- Qualitative analysis of carbohydrates (Molisch's Test, Benedict's test, Iodine test)
- Qualitative analysis of lipids (Solubility test, Spotting test)
- Qualitative analysis of proteins (Biuret test, Xanthoproteic test)
- Preparation of Balanced Diet Chart for different age groups- Children (1-9 years), Adolescents (10-18 years), Adults (19-59 years) and elderly (60+ years)
- Project on calculation of Body Mass Index (BMI) of individuals and assessment of their nutritional status (Underweight, Normal, Overweight and Obese) based on BMI

SUGGESTED READINGS

- Swaminathan, M. (2004). Handbook of food and nutrition. The Bangalore Printing and Publishing Co. Ltd.
- Joshi, S. (2023). Nutrition and dietetics (2nd ed.). Tata McGraw-Hill Publishing Company Limited. ISBN: 978-93-907278-27.
- Joshi, V. D. (1999). Handbook of nutrition and dietetics. Vora Medical Publications.
- Arora, M. P. (2004). Biomolecules. The Bangalore Printing and Publishing Co. Ltd. ISBN: 978-93-5024-579-8.
- Sihag, M. K. (2023). Biochemistry and human nutrition. Tata McGraw-Hill Publishing Company Limited. ISBN: 978-93-87973-54-1.
- Chahande, S. J. (2022). Textbook of biomolecules and nutritional biochemistry. ISBN: 978-81-976705-7-2.
- Gopalan, C., Rama Sastri, B. V., &Balasubramanian, S. C. (2003). Nutritive value of Indian foods (5th ed.). National Institute of Nutrition. ISBN: 978-81-85038-27-5.

Course Code: ZOO-DSC-7 Maj-403
Course Title: Basic Biochemical Techniques

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory= 75 + Practical= 25)
Theory Marks= 75 {67 (Theory examination) + 8 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +3 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Familiarize themselves with the fundamentals of key biochemical techniques.*
- *Understand the principles and practical applications of chromatography, spectroscopy, centrifugation, and electrophoresis.*

Unit- 1
Chromatographic Techniques
General methodology of chromatography, Retention Factor (Rf value), Retention Time (Rt)
Principle and applications of Paper Chromatography, Thin Layer Chromatography, Column Chromatography (Adsorption, Partition, Gel/Size exclusion, Ion exchange)

Unit- 2
Spectroscopic Techniques
Beer-Lambert Law,Principle and applications of Colorimetry, UV-Visible Spectroscopy, Fluorescence Spectroscopy, Infrared (IR) Spectroscopy and Nuclear Magnetic Resonance (NMR) Spectroscopy

Unit- 3
Centrifugation Techniques
Principle of sedimentation, Types of centrifuges(Benchtop, High speed, Ultracentrifuge), Types of rotors (Fixed-angle, Swinging-bucket, Vertical)
Principle and applications of Differential Centrifugation, Density Gradient Centrifugationand Ultracentrifugation

Unit- 4
Electrophoretic Techniques
Principle and applications of Agarose Gel Electrophoresis, Polyacrylamide Gel Electrophoresis (SDS-PAGE, Native PAGE), Isoelectric focusing, Capillary Electrophoresis

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-7 Maj-403: Basic Biochemical Techniques

Practical

1. Separation of amino acids/plant pigments by paper chromatography.
2. Separation of sugars/lipids/leaf pigments/dyes by Thin Layer Chromatography.
3. Estimation of proteins by Lowry/Bradford method(Demonstration or Virtual simulation).
4. Operation and Calibration of Benchtop and High-Speed Centrifuges.
5. Separation of milk components using centrifuge.
6. Agarose gel electrophoresis for the separation of DNA fragments (Demonstration or Virtual simulation).
7. SDS-PAGE for protein separation (Demonstration or Virtual simulation).

SUGGESTED READINGS

- **Upadhyay A., Upadhyay K &Nath N. –*Biophysical Chemistry and Biochemical Techniques***, 4th edition (2023), Himalaya Publishing House, Mumbai. ISBN: 978-93-5142-227-3.
- **Pattabiraman T.N. –*Laboratory Manual and Practical Biochemistry,*** 4th edition**, All India Publishers & Distributors**, New Delhi. **ISBN:**978-81-8004-002-3.
- **Gupta R.C. –*Practical Biochemistry,*** 5th edition (2018),CBS Publishers**& Distributors Pvt. Ltd.,** New Delhi. **ISBN:**978-81-239-2230-0.
- **Wilson K & Walker**J. **–*Principles and Techniques of Biochemistry and Molecular Biology, 8th*** edition (2018), Cambridge University Press. ISBN: 978-1-107-16227-3.
- **Plummer, D.T. –*An Introduction to Practical Biochemistry, 3rd***edition (2017), Tata McGraw-Hill Education, New Delhi**. ISBN:**978-0-07-099487-4.
- **Robyt J.F. & White B.J. –*Biochemical Techniques: Theory and Practice, 3rd edition, Pearson Publishers.*** **ISBN:**978-0-321-53657-2.

Course Code: ZOO-DSC-8 Maj-404
Course Title: Endocrinology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory= 75 + Practical= 25)
Theory Marks= 75 {67 (Theory examination) + 8 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) +3 (Internal assessment)}
Duration for final examination: 3 hours

Theory

***Learning Outcome*:**
Students will be able to:

- *Understand the hormones and functions of endocrine glands*
- *Evaluate the mechanisms of hormone action*
- *Identify major endocrine disorders*

UNIT- 1
Introduction to Endocrinology: Classification of hormones, Mechanism of action of Hormones, Feedback mechanisms
Hormones of Gastrointestinal tract, and their functions

UNIT- 2
Hypothalamo- hypophysial portal system
Hormones of hypothalamus and pituitary gland, and their functions
Disorders of pituitary gland: Acromegaly, Cushing's Syndrome, Diabetes Insipidus
.
UNIT- 3
Hormones of Pancreas, Thyroid gland and Parathyroid glands, and their functions
Disorders: Diabetes mellitus, Goitre, Grave's disease

UNIT- 4
Hormones of Adrenal gland and their role in stress management
Disorders: Addison's disease, Cushing Syndrome
Hormones of Ovary and Testis, and their functions

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-8 Maj-404:Endocrinology

Practical

1. Demonstration of location of Endocrine glands in Rat/Human with the help of Charts/Models/Videos.
2. Study of the permanent slides of endocrine glands.
3. Demonstration of plasma level of any hormone using ELISA.
4. Demonstration of hormonal disorders mentioned in the theory with the help of Charts/Models/Videos.
5. Assignment on causes, symptoms and treatment of hormonal disorders.

SUGGESTED READINGS

- Turner, C. D. (1976). General Endocrinology, 6th edition. Pub- Saunders Toppan. ISBN-0721689337.
- Nussey, S. and Whitehead, S. (2001). Endocrinology: An Integrated Approach. Oxford: BIOS Scientific Publishers. ISBN- 9781859962527.
- Hadley, M.E. and Levine J.E. (2007). Endocrinology, 6th Edition. Pearson Prentice-Hall, Pearson Education Inc., New Jersey. ISBN- 9780131876064.
- Norris, D. O. and Carr. J. (2013). Vertebrate Endocrinology, 5th edition. Academic Press. ISBN-9780123948151.
- Ramamurthy, V. and Durgadevi, K. (2020). Endocrinology, AkiNik Publications. **ISBN-**978-93-90420-39-1.

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